SEVEN OPEN UNIT EQUILATERAL TRIANGLES DO NOT COVER A REGULAR UNIT HEXAGON

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ABSTRACT. We prove that the regular hexagon of side length one cannot be covered by seven open equilateral triangles of side length one. The center and the six vertices canonically label the seven covering triangles. A finite classification routes every hypothetical cover to one of three strategies: trace-length deficit, area loss, or a finite equilateral-enclosure obstruction. The finite-enclosure strategy includes the explicit zero-gap nine-point theorem and the nonzero-gap residual witness sets. Appendices collect the exact local reach calculations certifying that the relevant finite sets do not fit in a unit C triangle. The paper is organized as one proof, and the two mixed support-arc intersections are verified in a self-contained exact certificate appendix over $\mathbb{Q}(\sqrt{3})$.

CONTENTS

1. INTRODUCTION AND PROOF OUTLINE

1.1. **The theorem and the canonical labeling.** Let

$$O = (0,0), \qquad V_i = \left(\cos\frac{i\pi}{3}, \sin\frac{i\pi}{3}\right), \qquad H = \operatorname{conv}\{V_0, \dots, V_5\},$$

where indices lie in $\mathbb{Z}/6\mathbb{Z}$. Write

$$e_{i,i+1} = [V_i, V_{i+1}], \qquad r_i = [O, V_i], \qquad M_i = \frac{1}{2}V_i,$$

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and put

$$S = \partial H \cup \bigcup_{i=0}^{5} r_i.$$

Then

$$\mathcal{H}^1(\partial H) = 6,$$
$$\mathcal{H}^1\left(\bigcup_{i=0}^{5} r_i\right) = 6, \tag{1}$$
$$\mathcal{H}^1(S) = 12.$$

Theorem 1.1 (Main theorem)**.** *The hexagon H cannot be covered by seven open equilateral triangles of side length one.*

Corollary 1.2 (Soifer's Hexagon Problem)**.** *For every $L > 1$, the regular hexagon $H_L = LH$ cannot be covered by seven closed unit equilateral triangles.*

The seven points $O, V_0, \ldots, V_5$ are pairwise at distance at least one, whereas any two points in an open unit equilateral triangle are at distance strictly less than one. Hence the seven covering triangles admit a unique labeling

$$U_C, U_0, \ldots, U_5$$

such that

$$O \in U_C, \qquad V_i \in U_i.$$

We call U_C the *C triangle* and U_i the *V triangle at* V_i. Set

$$T_C = \overline{U_C}, \qquad T_i = \overline{U_i}.$$

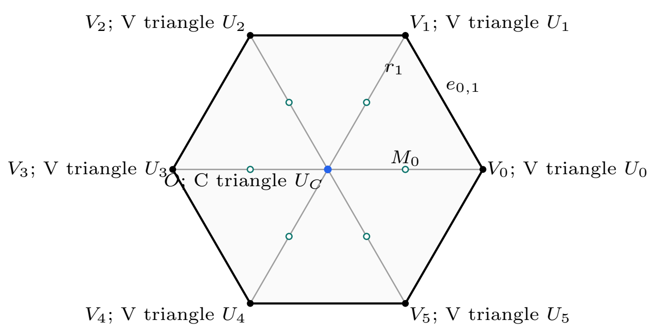


Figure 1. The center, vertices, radial segments, and radial midpoints of H.

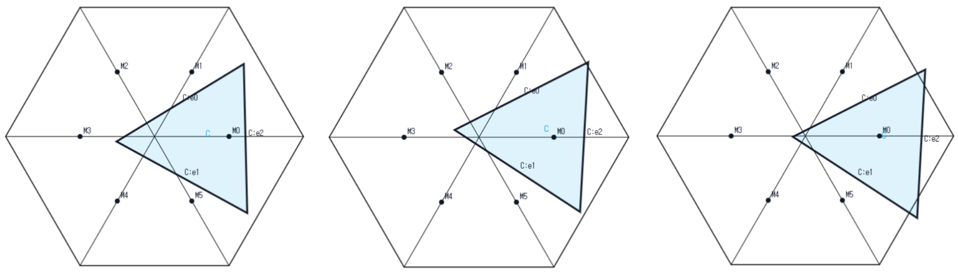
1.2. **Finite classifications.**

Set

$$N_E(T_C) := \left|\left\{i \in \mathbb{Z}/6\mathbb{Z} : \mathcal{H}^1(T_C \cap e_{i,i+1}) > 0\right\}\right|.$$

The only possible values are $0, 1, 2$. We call the corresponding C triangle types CE0, CE1, CE2; thus

$$T_C \text{ has type CE0} \iff N_E(T_C) = 0,$$
$$T_C \text{ has type CE1} \iff N_E(T_C) = 1,$$
$$T_C \text{ has type CE2} \iff N_E(T_C) = 2.$$



(a) CE0 (b) CE1 (c) CE2

FIGURE 2. Examples of the three C triangle classes. From left to right, the C triangle has positive-length trace on zero, one, and two edges of ∂H; point contacts do not count. The screenshots are illustrative and not to scale; no proof depends on visual inspection.

For the V triangle T_i, let $o(T_i)$ be the number of its triangle vertices outside H and put

$$n(T_i) = \left|\left\{j \in \{i-1, i+1\} : \mathcal{H}^1(T_i \cap r_j) > 0\right\}\right|,$$

with indices modulo 6. The raw pair $(3, 0)$ says that all three vertices of T_i lie outside H; it does not say that T_i is disjoint from H, since $V_i \in \operatorname{int}(T_i) \cap H$. Proposition 2.5 replaces every such triangle by a same-orientation translate with exactly the same trace in H, still containing V_i in its interior, and with at most two vertices outside H. We make this exact-trace normalization before defining any reach. The normalized exhaustive types used below are

$(o(T_i), n(T_i))$	established type
$(1, 0)$	Vd0
$(2, 0)$	Vd0
$(1, 1)$	Vd1
$(1, 2)$	Vd2
$(2, 1)$	T3-like.

The actual maximal reaches are

$$A_i := \max\{x \in [0, 1] : V_i + x(V_{i-1} - V_i) \in T_i\},$$
$$B_i := \max\{x \in [0, 1] : V_i + x(V_{i+1} - V_i) \in T_i\},$$
$$C_i := \max\{x \in [0, 1] : V_i + x(O - V_i) \in T_i\}.$$

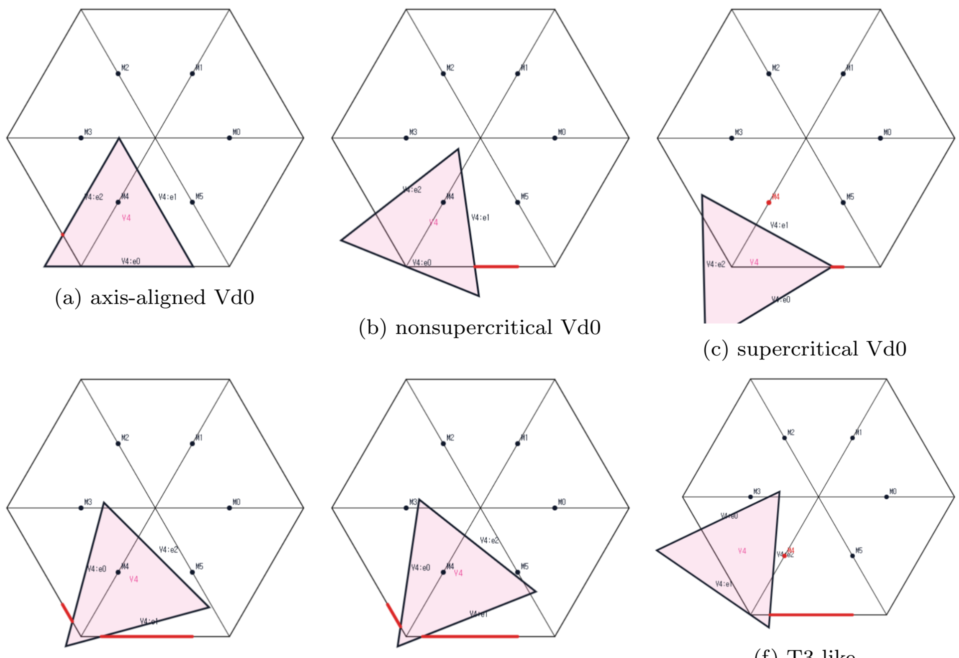
The same numbers are the suprema of the corresponding reaches in the open triangle U_i. The V triangle T_i is *supercritical* when $A_i + B_i > 1$, and

$$N_+ := |\{i \in \mathbb{Z}/6\mathbb{Z} : A_i + B_i > 1\}|. \tag{2}$$

Lowercase reaches are used only after a selection satisfying

$$0 \le a_i \le A_i,$$
$$0 \le b_i \le B_i,$$
$$0 \le c_i \le C_i.$$

They never define N_+.



(a) axis-aligned Vd0

(b) nonsupercritical Vd0

(c) supercritical Vd0

(d) Vd1

(e) Vd2

(f) T3-like

FIGURE 3. Examples of the normalized V triangle types. The top row shows an axis-aligned Vd0, a nonsupercritical Vd0 with $A_i + B_i \le 1$, and a supercritical Vd0 with $A_i + B_i > 1$. The bottom row shows Vd1, Vd2, and T3-like. The screenshots are illustrative and not to scale; no proof depends on visual inspection.

Parametrize $e_{i,i+1}$ from V_i to V_{i+1} by

$$X_i(x) := V_i + x(V_{i+1} - V_i), \qquad 0 \le x \le 1.$$

The incident open traces are

$$U_i \cap e_{i,i+1} = X_i([0, B_i)),$$
$$U_{i+1} \cap e_{i,i+1} = X_i((1 - A_{i+1}, 1]).$$

Define the *boundary gap*

$$J_i := e_{i,i+1} \setminus \big((U_i \cap e_{i,i+1}) \cup (U_{i+1} \cap e_{i,i+1})\big).$$

The displayed trace identities give

$$J_i \ne \varnothing \iff B_i + A_{i+1} \le 1.$$

Equality gives a singleton gap, which remains uncovered. Put

$$N_{\mathrm{gap}} := |\{i \in \mathbb{Z}/6\mathbb{Z} : J_i \ne \varnothing\}|.$$

For the case tables, put

$$d := |\{i \in \mathbb{Z}/6\mathbb{Z} : (o(T_i), n(T_i)) \in \{(1,1), (1,2)\}\}|,$$
$$t := |\{i \in \mathbb{Z}/6\mathbb{Z} : (o(T_i), n(T_i)) = (2,1)\}|,$$

and

$$N_{\mathrm{sp}} := |\{i \in \mathbb{Z}/6\mathbb{Z} : n(T_i) > 0\}| = d + t.$$

The common trace calculation proves that the number of V triangles that are either supercritical or meet an adjacent radial segment in positive length is

$$N_+ + N_{\mathrm{sp}}.$$

In particular, every CE1/CE2 configuration with $N_+ + N_{\mathrm{sp}} \geq 3$ is impossible by a trace-length bound on S.

1.3. **Exhaustive routing.** Lemma 2.9 proves the fundamental equivalence

$$N_{\mathrm{gap}} = 0 \quad \Longleftrightarrow \quad U_0, \ldots, U_5 \text{ cover } \partial H.$$

Thus Table 1 first dispatches every zero-gap case without reference to the center class. The nonzero-gap part is read from top to bottom; the first applicable row is the unique route for a hypothetical cover. The full classification proof is Proposition 2.14.

N_{gap}	N_+	d	t	$N_E(T_C)$	mechanism
0	0	$d + t \leq 6$		$N_E(T_C) \in \{0, 1, 2\}$	Trace length
0	1	$d \geq 1$	$t \geq 0$	$N_E(T_C) \in \{0, 1, 2\}$	Trace length
0	1	$d = 0$	$t \geq 1$	$N_E(T_C) \in \{0, 1, 2\}$	Area loss
0	1	$d = 0$	$t = 0$	$N_E(T_C) \in \{0, 1, 2\}$	Finite enclosure
0	$N_+ \geq 2$	$d + t \leq 6$		$N_E(T_C) \in \{0, 1, 2\}$	Area loss
$N_{\mathrm{gap}} \geq 1$	$N_+ \geq 2$	$d + t \leq 6$		$N_E(T_C) \in \{1, 2\}$	Trace length
$N_{\mathrm{gap}} \geq 1$	$N_+ \in \{0, 1\}$	$N_+ + d + t \geq 3$		$N_E(T_C) \in \{1, 2\}$	Trace length
$N_{\mathrm{gap}} \geq 1$	0	$d = 0$	$t = 0$	$N_E(T_C) \in \{1, 2\}$	Finite enclosure
$N_{\mathrm{gap}} \geq 1$	0	$d \geq 1$	$d + t \leq 2$	$N_E(T_C) \in \{1, 2\}$	Trace length
$N_{\mathrm{gap}} \geq 1$	0	$d = 0$	$t \in \{1, 2\}$	$N_E(T_C) \in \{1, 2\}$	Finite enclosure
$N_{\mathrm{gap}} \geq 1$	1	$d = 0$	$t = 0$	$N_E(T_C) \in \{1, 2\}$	Finite enclosure
$N_{\mathrm{gap}} \geq 1$	1	$d = 0$	$t = 1$	$N_E(T_C) \in \{1, 2\}$	Finite enclosure
$N_{\mathrm{gap}} \geq 1$	1	$d = 1$	$t = 0$	$N_E(T_C) = 1$	Trace length
$N_{\mathrm{gap}} \geq 1$	1	$d = 1$	$t = 0$	$N_E(T_C) = 2$	Trace and finite enclosure

TABLE 1. The exhaustive routing. Strategy 1 is a trace-length argument, Strategy 2 sums area loss, and Strategy 3 forces a finite residual set into the C triangle and proves an equilateral-enclosure obstruction. Nonzero gap count forces center class CE1 or CE2. In that part the high-count rows are applied first; the final two rows separate the CE1 length route from the detailed CE2 finite-enclosure placement assembly.

1.4. **Organization of the proof.** Section 2 proves the structural classification, strict handoff selection, and midpoint facts. Appendix A proves the signed-center normal form used by the later arguments. Section 3 gives the complete trace-length argument. Section 4 proves the local and cyclic area-loss estimates. Section 5 gives the finite residual-hull principle and the nonzero-gap and zero-gap enclosure obstructions. Section 6 then applies the zero-gap matrix before the nonzero-gap rows of Table 1 and closes the proof. Appendices A–E supply the corresponding solved optimizations; Appendix F is the exact polynomial certificate for the two mixed zero-gap intersections.

2. Common Geometry and Structural Reductions

The introduction fixes the exact C triangle and V triangle classifications, the actual maximal reaches A_i, B_i, C_i, the selected lower bounds a_i, b_i, c_i, and the boundary gaps J_i. This section proves that those definitions give the finite structural alternatives used by all three mechanisms. Coordinate evaluations needed in the proofs are collected in Appendix A.

2.1. **Open and closed formulations.**

Proposition 2.1 (Compactness and scaling equivalence)**.** *The following assertions are equivalent.*

1. *H is covered by seven open equilateral triangles of side* 1*.*
2. *For some $\varepsilon > 0$, H is covered by seven closed equilateral triangles of side $1-\varepsilon$.*
3. *For some $L > 1$, $H_L := LH$ is covered by seven closed equilateral triangles of side* 1*.*

Proof. The uniform inward displacement for (1)⇒(2) is the compactness-margin calculation theorem A.1. Enlarging each closed triangle about its incenter gives (2)⇒(1), and dilation by $(1-\varepsilon)^{-1}$ identifies (2) with (3). □

Lemma 2.2 (Distinct triangles)**.** *The seven triangles can be labelled so that*

$$O \in U_C, \qquad V_i \in U_i \quad (0 \le i \le 5),$$

and these seven triangles are distinct. Consequently

$$O \in \mathrm{int}(T_C), \qquad V_i \in \mathrm{int}(T_i).$$

Proof. The seven points $O, V_0, \ldots, V_5$ are pairwise at distance at least 1, whereas two points of an open unit equilateral triangle are at distance strictly less than its diameter 1. Selecting a covering triangle for each point therefore gives a bijection. □

2.2. **C triangle and V triangle classifications.**

Point contacts have zero trace length, so they do not create an additional positive boundary trace in the definition of $N_E(T_C)$.

Proposition 2.3 (Exhaustive center types)**.** *The closed C triangle T_C has exactly one of the types* CE0, CE1, CE2*; equivalently,*

$$N_E(T_C) \le 2.$$

Proof. If $N_E(T_C) \ge 3$, two of the traced edges are nonadjacent in the boundary six-cycle. Choose a relative-interior point from each positive-length trace. By theorem A.6, those points are more than unit distance apart, so a unit equilateral triangle cannot contain both. □

Lemma 2.4 (Local wedge)**.** *Every original V triangle has $1 \le o(T_i) \le 3$. If $\mathcal{H}^1(T_i \cap r_j) > 0$ for some $j \in \{i-1, i+1\}$, at least one vertex of T_i belongs to H; if both adjacent radial traces have positive length, at least two vertices belong to H.*

Proof. This is the wedge-incidence calculation in theorem A.7. It is stated for the closed T_i because closure may add a tangency, although it adds no trace length. □

Raw roles with $(o(T_i), n(T_i)) = (3, 0)$ admit an exact-trace normalization.

Proposition 2.5 (Exact-trace normalization of raw $(3,0)$ roles)**.** *Let U be an original open V triangle at V_i, put $T = \overline{U}$, and suppose $(o(T), n(T)) = (3, 0)$. There is a same-orientation translate $\widehat{U}$, with closure $\widehat{T}$, such that*

$$V_i \in \widehat{U}, \qquad (o(\widehat{T}), n(\widehat{T})) = (2, 0), \tag{3}$$

and

$$T \cap H = \widehat{T} \cap H, \qquad U \cap H = \widehat{U} \cap H. \tag{4}$$

In particular, all three actual maximal reaches and the inequality $A_i + B_i > 1$ are unchanged.

Proof. The two-orientation affine calculation and the trace-preserving translation are given in theorem A.8. □

We apply this translation simultaneously and reuse U_i, T_i for the normalized triangles.

Proposition 2.6 (Exhaustive normalized vertex types)**.** *Every normalized V triangle has exactly one of the four types* Vd0, Vd1, Vd2, T3-like *in the exact dictionary of the introduction.*

Proof. Lemma 2.4 gives $o \le 2$ when $n \ge 1$ and $o \le 1$ when $n = 2$, while $o \ge 1$. Proposition 2.5 replaces the only additional raw pair, $(3, 0)$, by $(2, 0)$. □

The next construction is a trace majorant, not a new open covering triangle.

Proposition 2.7 (T3-like closed-trace domination)**.** *Let T be the closure of an original* T3-like *V triangle at V_i. There is a same-orientation translate $\widehat{T}$ such that*

$$V_i \in \mathrm{relint}(\textit{a side of } \widehat{T}), \qquad T \cap H \subsetneq \widehat{T} \cap H,$$

and $(o(\widehat{T}), n(\widehat{T})) = (2, 1)$, while

$$\mathrm{int}(T) \cap (H \setminus \{V_i\}) \subseteq \mathrm{int}(\widehat{T}).$$

Since V_i becomes a boundary point, $\widehat{T}$ is only a closed-trace majorant.

Proof. See the two-orientation feasibility calculation in theorem A.2. □

2.3. **Initial reaches and boundary gaps.** The exact reach definitions are those in the introduction. The following elementary fact records how their closed maxima and open suprema are used.

Lemma 2.8 (Initial reach segments)**.** *On the three segments from V_i toward V_{i-1}, V_{i+1}, O, the closed parameter sets are respectively*

$$[0, A_i], \qquad [0, B_i], \qquad [0, C_i],$$

and the open parameter sets are

$$[0, A_i), \qquad [0, B_i), \qquad [0, C_i).$$

Thus the closed maxima equal the corresponding open-triangle suprema.

Proof. Each intersection with a segment issuing from the interior point V_i is convex. Closedness of T_i, openness of U_i, and $T_i = \overline{U_i}$ give the stated endpoints. $\square$

The gap J_i is the exact set difference defined in the introduction. The name includes the equality case $B_i + A_{i+1} = 1$: then J_i is a singleton missed by both incident open V triangles.

Lemma 2.9 (Exhaustiveness of the gap split)**.** *Every J_i is empty or one closed interval, with singleton intervals retained. Under a hypothetical cover,*

$$N_{\text{gap}} = \left|\{i \in \mathbb{Z}/6\mathbb{Z} : J_i \neq \varnothing\}\right|,$$

and every nonempty J_i lies in a positive-length C triangle trace. Consequently a closed C triangle T_C of type CE0, CE1, CE2 *permits respectively at most* 0, 1, 2 *gaps, and*

$$N_{\text{gap}} = 0 \iff U_0, \dots, U_5 \text{ cover } \partial H.$$

Proof. The two incident open traces are initial intervals from opposite endpoints, so their complement is exactly the interval displayed in the introduction. A nonincident V triangle cannot contain a relative-interior point of the edge: that point is more than unit distance from its distinguished vertex. Hence a gap lies in U_C; openness turns even a singleton V-gap into a positive-length C trace. Proposition 2.3 gives the count and the final equivalence. $\square$

Lemma 2.10 (T3-like V triangles are nonsupercritical)**.** *Every original* T3-like *V triangle satisfies*

$$A_i + B_i \leq 1. \tag{5}$$

It is therefore not counted by N_+.

Proof. This is the second assertion of theorem A.2. $\square$

2.4. **Strict handoffs and midpoint forcing.**

Proposition 2.11 (Strict boundary handoffs)**.** *Assume that $U_0, \dots, U_5$ cover ∂H. There are choices*

$$\xi_i \in (1 - A_{i+1}, B_i), \qquad a_i := 1 - \xi_{i-1}, \qquad b_i := \xi_i,$$

such that

$$0 < a_i < A_i, \qquad 0 < b_i < B_i, \qquad a_i^2 + a_i b_i + b_i^2 < 1.$$

The selected anchors lie in U_i. If exactly one actual V triangle is supercritical, every strict selection has that same unique selected supercritical index. If at least two actual V triangles are supercritical, some simultaneous strict selection retains at least two.

Proof. The overlap intervals and the cyclic selector are proved in theorem A.3. Geometrically, each ξ_i lies in the overlap of two adjacent open edge traces, which makes both associated reach demands strict. $\square$

Lemma 2.12 (Self-midpoint obstruction)**.** *Let u, v be unit vectors meeting at* 120°. *If a closed unit equilateral triangle contains*

$$0, \qquad au, \qquad bv, \qquad \frac{u+v}{2},$$

then $a + b \le 1$*. Consequently*

$$M_i \in T_i \Longrightarrow A_i + B_i \le 1.$$

If the boundary points and midpoint are contained with positive margin, the conclusion is strict.

Proof. This is the four-point caliper optimization theorem A.10. □

Proposition 2.13 (CE1/CE2 exactly-one-midpoint property)**.** *If* $O \in \mathrm{int}(T_C)$ *and* T_C *is* CE1 *or* CE2*, then*

$$|T_C \cap \{M_0, \ldots, M_5\}| = 1.$$

Proof. Normalize one positive boundary trace. The signed-center optimization theorem A.11 gives the exact midpoint set $\{M_0\}$; undoing the symmetry proves the assertion. □

The interior hypothesis is essential: $\mathrm{conv}\{O, V_0, V_1\}$ has a boundary edge overlap and contains both M_0 and M_1, but O is not interior to it.

2.5. **Exhaustive structural routing.**

Proposition 2.14 (Exhaustive structural reduction)**.** *Every hypothetical cover admits the distinct C triangle and V triangles used in Table 1, and its actual maximal reaches determine exactly one row of that table.*

Proof. Lemma 2.2 and Propositions 2.3 and 2.6 give the exhaustive, mutually exclusive roles. The actual reaches determine N_+, while the open traces determine N_{gap}.

If $N_{\mathrm{gap}} = 0$, Lemma 2.9 makes the six V triangles boundary-complete, independently of the center class. The cases $N_+ = 0$ and $N_+ \ge 2$ are single routes. For $N_+ = 1$, exactly one of $d \ge 1$, $d = 0$ with $t \ge 1$, and $(d, t) = (0, 0)$ holds.

If $N_{\mathrm{gap}} \ge 1$, the closed C triangle T_C is CE1 or CE2 and has one radial midpoint. When $N_+ \ge 2$, Lemma 3.5 and Theorem 3.6 close the high-count route. The same theorem closes $N_+ + d + t \ge 3$ for $N_+ \in \{0, 1\}$. The survivors have $N_+ = 0, d + t \le 2$ or $N_+ = 1, d + t \le 1$. Their exact alternatives are

$$(d, t) = (0, 0), \qquad d \ge 1, \qquad d = 0 \ \wedge \ t \in \{1, 2\}$$

and

$$(d, t) = (0, 0), \qquad (d, t) = (0, 1), \qquad (d, t) = (1, 0),$$

respectively. The final alternative splits by CE1 and CE2. These predicates are disjoint and exhaust Table 1. □

3. Trace-Length Mechanism

For a closed triangle T and a rectifiable target E, define

$$L_E(T) := \mathcal{H}^1(T \cap E).$$

Passing from an original open triangle to its closure can only enlarge its trace. Hence an open cover of E implies

$$\mathcal{H}^1(E) \le L_E(T_C) + \sum_{i=0}^{5} L_E(T_i). \tag{6}$$

The mechanism applies this inequality only to the perimeter ∂H or the full skeleton S, under the hypotheses stated below.

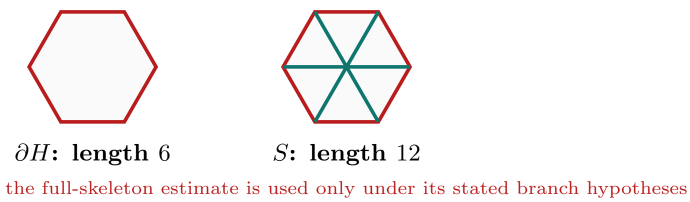


FIGURE 4. The perimeter and full-skeleton targets. The skeleton is used only under the hypotheses of the corresponding trace budget.

3.1. Skeleton interfaces.

Lemma 3.1 (Center skeleton cap). *If $N_E(T_C) \in \{1, 2\}$ and*

$$|T_C \cap \{M_0, \dots, M_5\}| = 1,$$

then

$$L_S(T_C) < \frac{3}{2}.$$

Proof. This is the signed-center trace optimization theorem B.1. □

Lemma 3.2 (Positive-support skeleton cap). *Let T_i be the closure of an original open V triangle. If*

$$\exists j \in \{i-1, i+1\} \quad \mathcal{H}^1(T_i \cap r_j) > 0,$$

then

$$L_S(T_i) < \frac{3}{2}.$$

Proof. Normalize to $i = 0$ and translate the hexagon by V_0. The five components of S on which a unit triangle containing V_0 can have positive trace become the corresponding local components of the translated skeleton. The positive adjacent-radial trace becomes a positive boundary trace there. Proposition 2.13 and Lemma 3.1 give the bound; every nonlocal component is more than unit distance from V_0, apart from zero-measure endpoints. □

Lemma 3.3 (No-support skeleton cap). *If*

$$A_i + B_i \le 1,$$
$$\mathcal{H}^1(T_i \cap r_{i-1}) = 0, \qquad \mathcal{H}^1(T_i \cap r_{i+1}) = 0,$$

then

$$L_S(T_i) \le 2.$$

Proof. See theorem B.2; geometrically, the two incident boundary traces and the own radial trace are the only components that can contribute positive length. □

Lemma 3.4 (Supercritical skeleton cap). *If $A_i + B_i > 1$, then*

$$L_S(T_i) < \frac{3}{2}.$$

Proof. The exact admissible-cell and polynomial-sign calculation is theorem B.3. □

Lemma 3.5 (Positive-support rescuer)**.** *Suppose $N_E(T_C) \in \{1,2\}$ and, after symmetry,*

$$T_C \cap \{M_0,\dots,M_5\} = \{M_0\}.$$

If $N_+ \geq 2$, then for a supercritical index $s \neq 0$ there is $j \in \{s-1, s+1\}$ such that

$$M_s \in U_j, \qquad \mathcal{H}^1(T_j \cap r_s) > 0.$$

Proof. Choose a supercritical $s \neq 0$. Lemma 2.12 gives $M_s \notin T_s$, and the displayed midpoint identity gives $M_s \notin T_C$. The open cover therefore puts M_s in another U_j. Diameter locality restricts j to $s-1$ or $s+1$, and openness makes its trace on r_s have positive length. □

Theorem 3.6 (Direct CE1/CE2 skeleton count)**.** *If*

$$N_E(T_C) \in \{1,2\},$$
$$|T_C \cap \{M_0,\dots,M_5\}| = 1,$$

and

$$N_+ + N_{\mathrm{sp}} \geq 3, \tag{7}$$

then the seven triangles do not cover S.

Proof. The three geometric caps above make the total trace strictly smaller than $\mathcal{H}^1(S) = 12$; the exact budget is theorem B.4. This contradicts (6). □

3.2. **Boundary trace register.** The exact evaluation is in Appendix B; the following table is the complete interface used by the routing argument.

Theorem 3.7 (Boundary trace table)**.** *For closures of original open triangles, the following bounds hold.*

(8)

exact hypothesis	*established class*	*exact bound*
$N_E(T_C) = 0$	CE0	$L_{\partial H}(T_C) = 0$
$N_E(T_C) = 1$	CE1	$L_{\partial H}(T_C) \leq \dfrac{\sqrt{3}}{2} - \dfrac{3}{4}$
$N_E(T_C) = 2$	CE2	$L_{\partial H}(T_C) < \dfrac{1}{2}$
$(o(T_i), n(T_i)) \in \{(1,0),(2,0)\}$ $A_i + B_i \leq 1$	Vd0	$L_{\partial H}(T_i) \leq 1$
$(o(T_i), n(T_i)) \in \{(1,0),(2,0)\}$ $A_i + B_i > 1$	Vd0	$L_{\partial H}(T_i) \leq \dfrac{2}{\sqrt{3}}$
$(o(T_i), n(T_i)) \in \{(1,1),(1,2)\}$	Vd1 *or* Vd2	$L_{\partial H}(T_i) < \dfrac{1}{2}$
$(o(T_i), n(T_i)) = (2,1)$	T3-like	$L_{\partial H}(T_i) < 1.$

For every V triangle,

$$L_{\partial H}(T_i) = A_i + B_i. \tag{9}$$

Proof. The locality identity and every row are proved in theorem B.7. □

Corollary 3.8 (Branches closed by the common budgets)**.** *The following exact alternatives are impossible.*

(1) $N_E(T_C) = 0$, $N_+ = 1$, *and* $d \geq 1$;
(2) $N_E(T_C) \in \{1,2\}$, $N_+ = 0$, *and* $d \geq 1$;

(3) $N_E(T_C) = 1$, $N_+ = 1$, *and* $d \geq 1$*;*
(4) $N_E(T_C) = 2$, $N_+ = 1$, *and* $d \geq 2$*;*
(5) $N_E(T_C) = 2$, $N_+ = 1$, *and there is an index* i *such that*

$$(o(T_i), n(T_i)) = (1, 2)$$

and

$$\{M_{i-1}, M_{i+1}\} \cap T_i \neq \varnothing;$$

(6) $N_E(T_C) \in \{1, 2\}$ *and* $N_+ + N_{\mathrm{sp}} \geq 3$*;*
(7) $N_E(T_C) \in \{1, 2\}$ *and* $N_+ \geq 2$.

Proof. The perimeter sums for items (1)–(5) are theorem B.11. Item (6) is Theorem 3.6. For item (7), normalize the unique C triangle midpoint and apply Lemma 3.5; its rescuer is distinct from the two supercritical triangles, so item (6) applies. □

Proposition 3.9 (Vd2 adjacent-midpoint branch)**.** *Assume*

$$N_E(T_C) = 2, \qquad N_+ = 1, \qquad (d, t) = (1, 0).$$

*If the unique non-*Vd0 *triangle* T_i *satisfies*

$$(o(T_i), n(T_i)) = (1, 2)$$

and

$$\{M_{i-1}, M_{i+1}\} \cap T_i \neq \varnothing,$$

then ∂H *cannot be covered.*

Proof. The strict one-third cap is theorem B.6; item (5) of Corollary 3.8 then applies. □

3.3. **Routing conclusion.**

Proposition 3.10 (Branches closed by the trace-length mechanism)**.** *The following routing entries and hybrid subcase are impossible.*

(1) $N_{\mathrm{gap}} = 0$ *and* $N_+ = 0$*;*
(2) $N_{\mathrm{gap}} = 0$, $N_+ = 1$, *and* $d \geq 1$*;*
(3) $N_E(T_C) \in \{1, 2\}$ *and* $N_+ + N_{\mathrm{sp}} \geq 3$*;*
(4) $N_E(T_C) \in \{1, 2\}$, $N_+ = 0$, *and* $d \geq 1$*;*
(5) $N_E(T_C) = 1$, $N_+ = 1$, *and* $(d, t) = (1, 0)$*;*
(6) *the subcase of* $N_E(T_C) = 2$, $N_+ = 1$, *and* $(d, t) = (1, 0)$ *in which the unique non-*Vd0 *triangle* T_i *satisfies*

$$(o(T_i), n(T_i)) = (1, 2)$$

and

$$\{M_{i-1}, M_{i+1}\} \cap T_i \neq \varnothing.$$

Proof. For item (1), Lemma 2.9 makes the incident open V traces overlap on every edge, so

$$B_i + A_{i+1} > 1 \qquad (0 \leq i \leq 5).$$

Summing contradicts $N_+ = 0$. Items (2)–(5) are clauses (1), (6), (2), and (3), respectively, of Corollary 3.8; their numerical budgets are evaluated in Appendix B. Item (6) is Proposition 3.9. □

4. Area-Loss Estimate

The area-loss method records how much of each V triangle must lie outside H. The local estimate applies to every V triangle; the sharper T3-like estimate uses its classified geometry. A final cyclic argument adds these losses without imposing a center type.

Put $A_\triangle = \sqrt{3}/4$ and normalize all areas by $A_\triangle$. At V_0 define

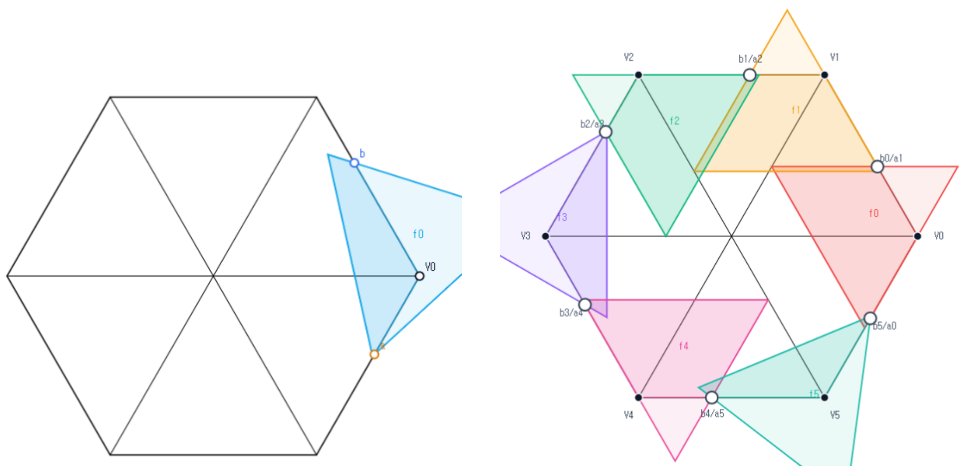
$$\mathcal{T}(a,b) := \left\{ T : \begin{array}{l} T \text{ is a closed unit equilateral triangle,} \\ \{V_0, V_0 + a(V_5 - V_0), V_0 + b(V_1 - V_0)\} \subseteq T \end{array} \right\},$$

$$\mathcal{L}(T) := \frac{\operatorname{area}(T \setminus H)}{A_\triangle}, \qquad \mathcal{L}(a,b) := \inf_{T \in \mathcal{T}(a,b)} \mathcal{L}(T),$$

where $a, b \in [0,1]$ and $\mathcal{T}(a,b) \neq \varnothing$. Hexagon symmetry gives the corresponding family at every V_i. In particular, if

$$0 \le a_i \le A_i, \qquad 0 \le b_i \le B_i,$$

then the rotated copy of T_i belongs to $\mathcal{T}(a_i, b_i)$.



(a) local retained area (b) cyclic retained areas

Figure 5. The area-loss method. The dark regions indicate the portions retained in H; the exact quantities are $\mathcal{L}(T)$ and $\mathcal{L}(a,b)$.

4.1. **Local area loss.**

Theorem 4.1 (Local square loss)**.** *For every feasible* (a,b)*,*

$$\mathcal{L}(a,b) \ge \min\{a,b\}^2. \tag{10}$$

If $a + b > 1$*, then*

$$\mathcal{L}(a,b) \ge \max\{a,b\}^2. \tag{11}$$

The same bounds hold for every $T \in \mathcal{T}(a,b)$*.*

Proof. The two exhaustive orientation families give these minima. Their polygon areas and the factorizations proving both inequalities are computed in theorem C.2. $\square$

4.2. T3-like **loss.**

Theorem 4.2 (Direct T3-like loss)**.** *Let T_i be a V triangle with*

$$(o(T_i), n(T_i)) = (2, 1),$$

and let

$$0 \le a_i \le A_i, \qquad 0 \le b_i \le B_i.$$

Then

$$a_i + b_i \le 1. \tag{12}$$

Moreover, if

$$0 \le \mu \le \frac{1}{2}, \qquad a_i, b_i \ge \mu,$$

then

$$\mathcal{L}(T_i) \ge 2\mu - 4\mu^2. \tag{13}$$

Proof. The first assertion is theorem 2.10. The translated closed-trace majorant from theorem 2.7 reduces the second assertion to a one-variable loss calculation, carried out in theorem C.3. □

4.3. **Cyclic accumulation.** Use the strict handoffs from theorem 2.11:

$$\xi_i \in (1 - A_{i+1}, B_i), \qquad a_i := 1 - \xi_{i-1}, \qquad b_i := \xi_i.$$

They satisfy

$$0 < a_i < A_i, \qquad 0 < b_i < B_i, \qquad a_i^2 + a_i b_i + b_i^2 < 1.$$

Lemma 4.3 (Cyclic area-loss certificate)**.** *The six actual V triangles satisfy*

$$|\{i \in \mathbb{Z}/6\mathbb{Z} : a_i + b_i > 1\}| \ge 2 \quad \Longrightarrow \quad \sum_{i=0}^{5} \mathcal{L}(T_i) > 1.$$

They also satisfy

$$\left.\begin{array}{r} |\{i \in \mathbb{Z}/6\mathbb{Z} : a_i + b_i > 1\}| = 1, \\ \forall i \in \mathbb{Z}/6\mathbb{Z}, \quad (o(T_i), n(T_i)) \in \{(1,0), (2,0), (2,1)\}, \\ \exists j \in \mathbb{Z}/6\mathbb{Z}, \quad (o(T_j), n(T_j)) = (2,1) \end{array}\right\} \quad \Longrightarrow \quad \sum_{i=0}^{5} \mathcal{L}(T_i) > 1.$$

Proof. Apply the local bounds around the six-cycle. The two exact sums, including the strict contribution from the second ascent, are given in theorem C.4. □

Proposition 4.4 (Branches closed by area)**.** *The following branches are impossible:*

(1) $N_{\mathrm{gap}} = 0$ *and* $N_+ \ge 2$*;*
(2) $N_{\mathrm{gap}} = 0$*,* $N_+ = 1$*,* $d = 0$*, and* $t \ge 1$*.*

Proof. By theorem 2.9, the six V triangles cover ∂H. The appropriate clause of theorem 2.11 supplies the handoffs required by theorem 4.3. Thus the V triangles have total normalized inside area strictly below five. The C triangle contributes at most one, so seven triangles cannot cover the normalized area six of H. □

5. Finite-Enclosure Obstructions

The finite-enclosure method finds compact sets that the V triangles miss. Under a hypothetical cover those sets lie in the open C triangle, while their least equilateral enclosure has side at least one.

Put

$$h := \frac{\sqrt{3}}{2}.$$

For $X \in \mathbb{R}^2$ and $r \geq 0$, put

$$\mathbb{B}(X, r) := \{Y \in \mathbb{R}^2 : \|Y - X\| \leq r\}.$$

For every nonempty compact $K \subseteq \mathbb{R}^2$, define

$$\Lambda(K) := \inf \left\{ s > 0 : \begin{array}{l} \exists T\ [T \text{ is a closed equilateral triangle of side } s, \\ K \subseteq T] \end{array} \right\}.$$

Thus $\Lambda(K)$ is the least equilateral enclosure side for K.

Lemma 5.1 (Compact set in an open unit triangle)**.** *If K is compact, U is an open unit equilateral triangle, and $K \subset U$, then*

$$\Lambda(K) < 1.$$

Proof. The distance from K to each side line of U is positive. Moving all three lines inward by a smaller common distance gives a closed equilateral triangle of side below one that still contains K. $\square$

5.1. **Four-anchor and neighboring-arm capacities.** Normalize at V_0 and define the four-anchor hull

$$K(a, b, c) := \operatorname{conv}\{V_0, X_5(1-a), X_0(b), (1-c)V_0\},$$
$$\mathcal{A} := \{(a, b, c) \in [0, 1]^3 : \Lambda(K(a, b, c)) \leq 1\}.$$

For

$$0 \leq a, b \leq 1, \qquad a^2 + ab + b^2 \leq 1,$$

put

$$c_{\max}(a, b) := \max\{c \in [0, 1] : (a, b, c) \in \mathcal{A}\}.$$

For $0 \leq a, c \leq 1$, also put

$$M_c(a) := \max\{b \in [0, 1] : (a, b, c) \in \mathcal{A}\}.$$

The exact admissible set, attainment, and the evaluation of these extrema are proved in theorem A.9.

The neighboring-arm capacities are the variational quantities

$$C_+(a, b) := \max \left\{ c \in [0, 1] : \begin{array}{l} \exists T\ [T \text{ is a closed unit equilateral triangle}, \\ \{V_0, X_5(1-a), X_0(b), (1-c)V_1\} \subseteq T] \end{array} \right\},$$
$$C_-(a, b) := C_+(b, a), \qquad 0 \leq a, b \leq 1, \qquad a + b \leq 1.$$

Their coordinate evaluation is deferred to theorem D.6.

Let selected lower bounds satisfy

$$0 \leq a_j \leq A_j, \qquad 0 \leq b_j \leq B_j \qquad (0 \leq j \leq 5),$$
$$a_j^2 + a_j b_j + b_j^2 \leq 1 \qquad (0 \leq j \leq 5).$$

A neighboring term is used only when the actual V triangle has the corresponding positive radial trace:

$$\mathcal{H}^1(T_{i-1} \cap r_i) > 0 \Longrightarrow a_{i-1} + b_{i-1} \leq 1,$$
$$\mathcal{H}^1(T_{i+1} \cap r_i) > 0 \Longrightarrow a_{i+1} + b_{i+1} \leq 1.$$

Define

$$\Gamma_i := \max\Big(\{c_{\max}(a_i, b_i)\} \cup \{C_+(a_{i-1}, b_{i-1}) : \mathcal{H}^1(T_{i-1} \cap r_i) > 0\} \cup \{C_-(a_{i+1}, b_{i+1}) : \mathcal{H}^1(T_{i+1} \cap r_i) > 0\}\Big), \qquad D_i := (1-\Gamma_i)V_i.$$

The points D_i are the radial witnesses used below.

Lemma 5.2 (Type-aware radial forcing)**.** *For every i,*

$$D_i \in r_i \subseteq S, \qquad D_i \notin \bigcup_{j=0}^{5} U_j.$$

Consequently,

$$S \subseteq U_C \cup \bigcup_{j=0}^{5} U_j \Longrightarrow D_i \in U_C.$$

Proof. The own role is excluded by $c_{\max}$, and an adjacent role is excluded only by the capacity allowed by its actual V type. The other roles are more than unit distance from D_i. The full calculation and the endpoint strictness are in theorem D.7. □

Lemma 5.3 (Common-pair domination)**.** *If*

$$p, q \geq 0, \qquad p + q \leq 1,$$

then

$$C_+(p, q), C_-(p, q) \leq 1 - \min\{p, q\} \leq c_{\max}(p, q).$$

Proof. This is the branchwise comparison in theorem D.8. □

Lemma 5.4 (Open trace endpoint)**.** *Let U be an open equilateral triangle and $T = \overline{U}$. If $T \cap [V, O]$ is the closed interval from V to P, then $P \notin U$. If a noncentral point $Q \in [V, O]$ lies in U, then $\mathcal{H}^1(U \cap [V, O]) > 0$.*

Proof. The first point is an endpoint of the closed convex intersection. For the second assertion, intersect a small ball about Q contained in U with the radial arm. □

5.2. **Actual gaps and the case register.** For an actual gap on $e_{i,i+1}$, the definition from theorem 2.9 gives exactly

$$J_i = X_i([B_i, 1 - A_{i+1}]).$$

This remains a closed singleton when $B_i = 1 - A_{i+1}$. Equivalently,

$$U_i \cap e_{i,i+1} = X_i([0, B_i)), \qquad U_{i+1} \cap e_{i,i+1} = X_i((1 - A_{i+1}, 1]).$$

Thus every point of J_i, including its endpoints, is center-forced under the skeleton-cover hypothesis.

For the one-Vd rows define the normalized indices by

$$\{\sigma\} := \{i \in \mathbb{Z}/6\mathbb{Z} : A_i + B_i > 1\},$$
$$\{\tau\} := \{i \in \mathbb{Z}/6\mathbb{Z} : (o(T_i), n(T_i)) \in \{(1,1), (1,2)\}\}.$$

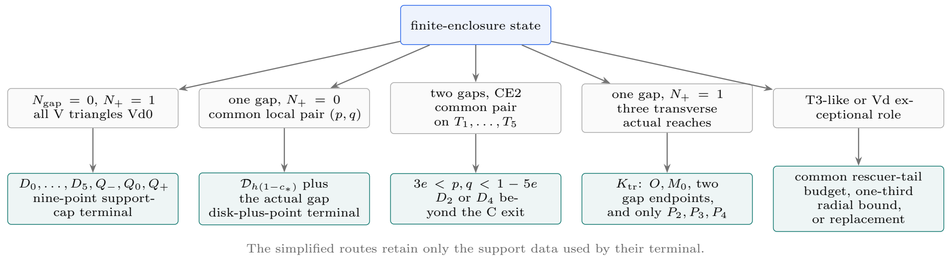


FIGURE 6. Roadmap of the simplified finite-enclosure chapter. The one-gap all-Vd0 witness uses only three transverse radial endpoints; the two-gap terminal uses the interval $3e < p, q < 1 - 5e$; and the T3-like and Vd1 paths share one rescuer-tail lemma.

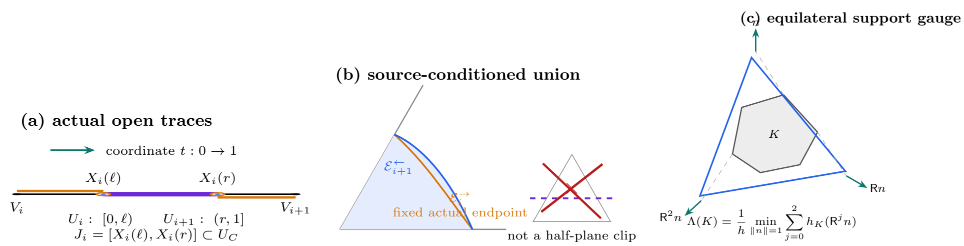


FIGURE 7. Three conventions used throughout the finite-enclosure argument. (a) A hollow outer marker records nonmembership in an open V trace, while the filled center marker records selection of the same endpoint as a center-forced witness. (b) A trace-exact envelope is obtained by restricting the source family to the prescribed actual maximal reach and then taking its union; it is not an affine half-plane clipping. (c) Equilateral enclosure is measured by three support directions separated by 120°. Incidences and interval orders are exact; lengths are schematic.

Rows A–E and the replacement router assume $S \subseteq U_C \cup \bigcup_{i=0}^{5} U_i$; row F assumes the hypothetical cover of H.

Table 2: Exact finite-enclosure case register. The symbolic columns determine the row; the final column gives its concise geometric route.

gap conditions	actual reaches	role and placement	route
$N_{\text{gap}} = 1$ $J = J_0$ $J_0 = X_0([\ell, r])$ $0 < \ell \le r < 1$	$N_+ = 0$	$d = 0$ $t \in \{0, 1, 2\}$ $N_E(T_C) \in \{1, 2\}$	**A.** Complementary gap.
$N_{\text{gap}} = 2$	$N_+ \in \{0, 1\}$ $N_+ + t \le 2$ $N_+ t = 0$	$d = 0$ $N_E(T_C) = 2$ $T_C \cap \{M_i\}_{i=0}^{5} = \{M_0\}$	**B.** CE2 short ray.

gap conditions	actual reaches	role and placement	route
$N_{\text{gap}} = 1$ $J = J_0$ $J_0 = X_0([\ell, r])$ $0 < \ell \leq r < 1$	$N_+ = 1$ $A_0 + B_0 > 1$	$(d,t) = (0,0)$ $N_E(T_C) \in \{1,2\}$ $T_C \cap \{M_i\}_{i=0}^5 = \{M_0\}$	**C.** Transverse seven-point witness.
$N_{\text{gap}} \in \{1,2\}$	$N_+ = 1$	$(d,t) = (0,1)$ $(o(T_0), n(T_0)) = (2,1)$ $N_E(T_C) \in \{1,2\}$ $T_C \cap \{M_i\}_{i=0}^5 = \{M_0\}$	$\mathbf{D}_T$**.** T3-like supported tail.
$N_{\text{gap}} \in \{1,2\}$	$N_+ = 1$ $A_0 + B_0 < 1/2$	$(d,t) = (1,0)$ $(o(T_0), n(T_0)) = (1,1)$ $N_E(T_C) = 2$ $T_C \cap \{M_i\}_{i=0}^5 = \{M_0\}$	$\mathbf{D}_V$**.** Vd1 supported tail.
$N_{\text{gap}} \in \{1,2\}$	$N_+ = 1$	$(d,t) = (1,0)$ $\sigma = 0$ $\tau \in \{1,5\}$ $N_E(T_C) = 2$ $T_C \cap \{M_i\}_{i=0}^5 = \{M_0\}$	$\mathbf{E}_a$**.** Adjacent radial separation.
$N_{\text{gap}} \in \{1,2\}$	$N_+ = 1$	$(d,t) = (1,0)$ $\sigma = 0$ $\tau \in \{2,3,4\}$ $N_E(T_C) = 2$ $T_C \cap \{M_i\}_{i=0}^5 = \{M_0\}$	$\mathbf{E}_n$**.** Nonadjacent radial separation.
$N_{\text{gap}} = 0$	$N_+ = 1$	$(d,t) = (0,0)$ $H \subseteq U_C \cup \bigcup_{i=0}^{5} U_i$	**F.** Zero-gap nine-point obstruction.
$N_{\text{gap}} \in \{1,2\}$	$N_+ = 1$	$(d,t) = (1,0)$ $\sigma \in \{1,2,3,4,5\}$ $\tau \in \{1,2,3,4,5\}$ $\sigma \neq \tau$ $\sigma - \tau \equiv \pm 1 \pmod 6$ $\big(o(T_\tau), n(T_\tau)\big) = (1,1)$ $M_\sigma \in T_\tau$ $N_E(T_C) = 2$ $T_C \cap \{M_i\}_{i=0}^5 = \{M_0\}$	**R.** Vd1 two-chart replacement; route by N'_{gap}.

The short names in the last column are navigation aids. The exact hypotheses and witnesses below are the mathematical interfaces.

5.3. **Complementary gap and CE2 short ray.**

Lemma 5.5 (One-gap common-pair adapter)**.** *Under the exact row-A hypotheses, rotate the actual gap to*

$$J_0 = X_0([\ell, r]), \qquad \ell = B_0, \qquad r = 1 - A_1,$$

and put

$$p := 1 - r, \qquad q := \ell.$$

Then

$$\forall i \in \mathbb{Z}/6\mathbb{Z}, \quad (p \le A_i \ \wedge \ q \le B_i) \ \vee (q \le A_i \ \wedge \ p \le B_i).$$

Proof. The exact five-handoff calculation is theorem D.2. □

Let

$$0 < p, q < 1, \qquad p + q \le 1, \qquad c_* := c_{\max}(p, q),$$

and put

$$J(p, q) := X_0([q, 1 - p]), \qquad \mathcal{D}(p, q) := \mathbb{B}(O, h(1 - c_*)).$$

Theorem 5.6 (Complementary-gap enclosure)**.**

$$\Lambda\big(\mathcal{D}(p, q) \cup J(p, q)\big) \ge 1.$$

The conclusion includes $p + q = 1$, when $J(p, q)$ is a singleton.

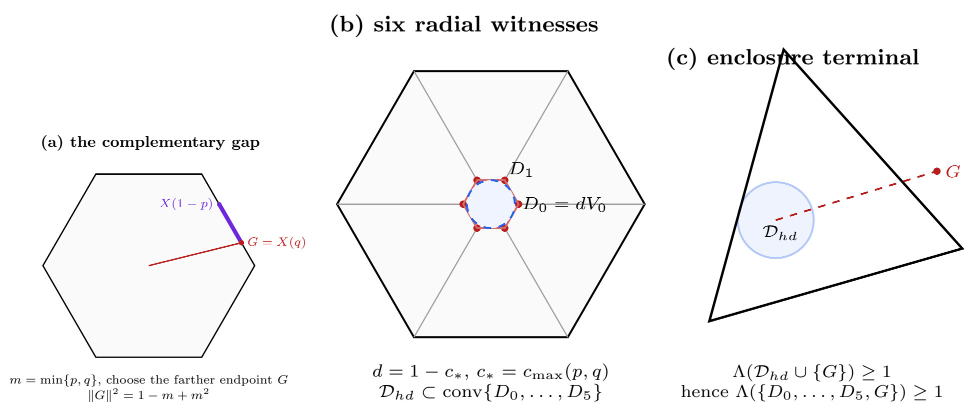
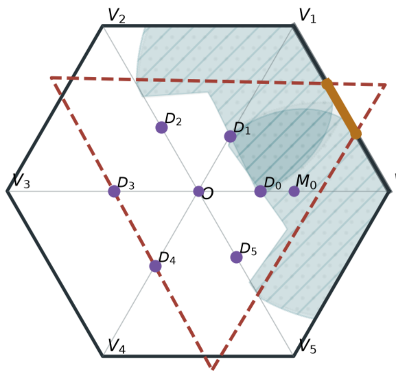


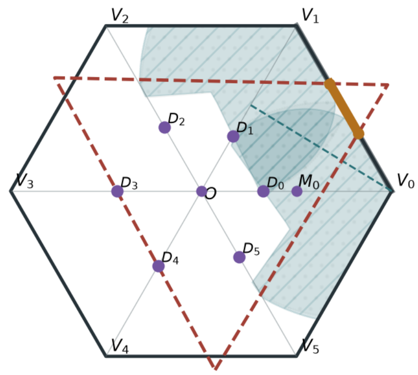
FIGURE 8. The complementary-gap enclosure mechanism. The farther endpoint G of $J(p, q)$ is center-forced. Common-pair domination produces the six center-forced points $D_i = (1 - c_*)V_i$, whose convex hull contains the disk $\mathcal{D}_{h(1-c_*)}$. The disk-plus-point formula then forces enclosing side length at least one. The disk is not an additional witness; it is contained in the convex hull of the six finite radial witnesses.

Proof. The enclosure optimization, including the disk–point transition, is theorems D.9 and D.10. □

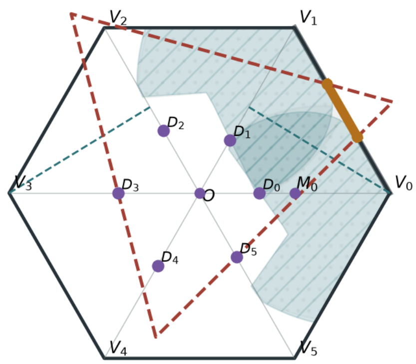
In the six case figures below, orange marks actual gaps, purple marks sampled finite-enclosure points, and the dashed red line is an enclosing triangle. The shading and the displayed triangle are illustrative, not proof objects.



All Vd0.
$N_{\mathrm{gap}} = 1$
$N_+ = 0$
$(d, t) = (0, 0)$



One T3-like **role.**
$N_{\mathrm{gap}} = 1$
$N_+ = 0$
$(d, t) = (0, 1)$



Two T3-like **roles.**
$N_{\mathrm{gap}} = 1$
$N_+ = 0$
$(d, t) = (0, 2)$

FIGURE 9. Case A: sampled complementary-gap data.

In the CE2 two-gap normalization, define the two positive C-trace endpoints geometrically by

$$p := \min\{x \in [0, 1] : X_5(x) \in T_C\},$$
$$q := 1 - \max\{x \in [0, 1] : X_0(x) \in T_C\},$$

and put

$$c_* := c_{\max}(p, q), \qquad D_2 := (1 - c_*)V_2, \qquad D_4 := (1 - c_*)V_4.$$

Lemma 5.7 (Two-gap common-pair adapter)**.** *Under the exact row-B hypotheses,*

$$\forall i \in \{1, 2, 3, 4, 5\}, \quad (p \le A_i \ \wedge \ q \le B_i) \ \vee (q \le A_i \ \wedge \ p \le B_i).$$

For $j \in \mathbb{Z}/6\mathbb{Z}$, *if* $(o(T_j), n(T_j)) = (2,1)$ *and* $\mathcal{H}^1\big(T_j \cap (r_2 \cup r_4)\big) > 0$, *then*

$$A_j + B_j \le 1, \qquad \max\{C_+(A_j, B_j), C_-(A_j, B_j)\} \le c_*.$$

Proof. The exact two-gap handoff calculation and the T3-like capacity check are theorems D.3 and 5.3. □

Theorem 5.8 (CE2 short-ray obstruction)**.** *Assume*

$$N_E(T_C) = 2, \qquad N_{\text{gap}} = 2, \qquad N_+ \in \{0,1\}, \qquad d = 0, \qquad N_+ + t \le 2.$$

If every $i \in \{1,2,3,4,5\}$ *has either*

$$p \le A_i, \qquad q \le B_i,$$

or

$$q \le A_i, \qquad p \le B_i,$$

then

$$D_2, D_4 \in U_C, \qquad \{D_2, D_4\} \nsubseteq T_C.$$

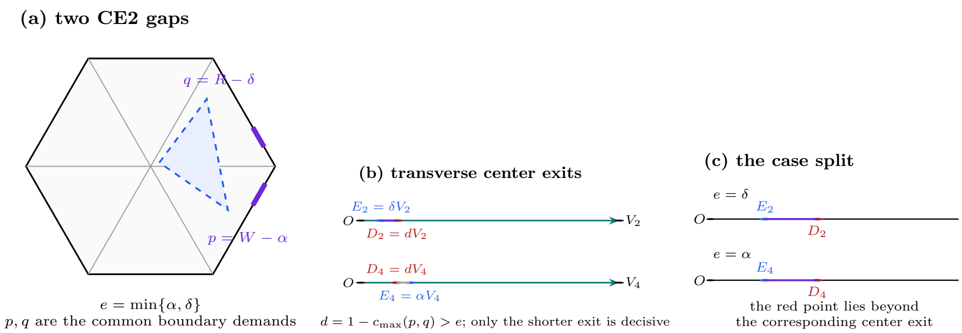
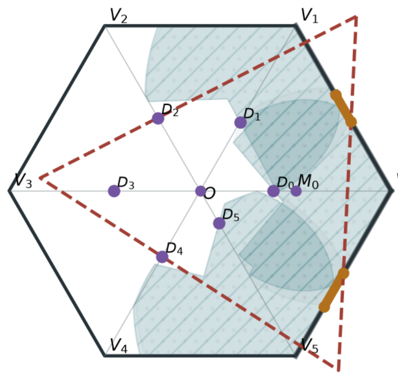
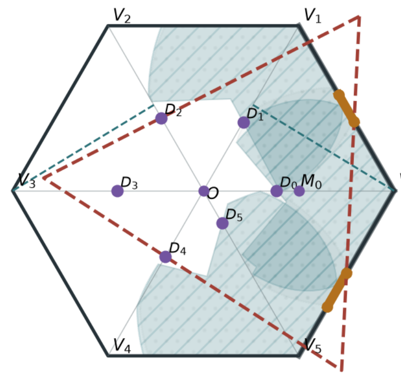


FIGURE 10. The CE2 short-ray terminal. The two boundary gaps determine the common pair (p, q) and the shorter center residual $e = \min\{\alpha, \delta\}$. The theorem proves $d = 1 - c_{\max}(p, q) > e$. Consequently the radial witness on the shorter of r_2 and r_4 lies strictly beyond the corresponding center exit and is uncovered. The displayed numerical ordering is illustrative; the labelled case split is exact.

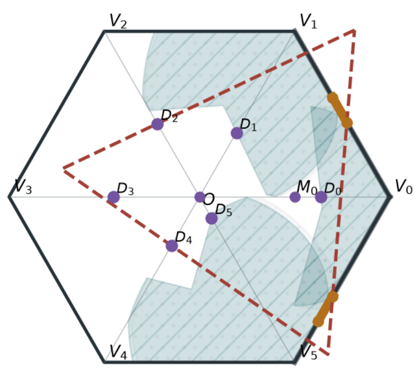
Proof. The first inclusion is type-aware radial forcing. The signed CE2 optimization proving the final strict noncontainment is theorem D.11. □



All Vd0**.**
$N_{\mathrm{gap}} = 2$
$N_E(T_C) = 2$
$N_+ = 0$
$(d, t) = (0, 0)$



Two T3-like **roles.**
$N_{\mathrm{gap}} = 2$
$N_E(T_C) = 2$
$N_+ = 0$
$(d, t) = (0, 2)$



One supercritical role.
$N_{\mathrm{gap}} = 2$
$N_E(T_C) = 2$
$N_+ = 1$
$(d, t) = (0, 0)$

FIGURE 11. Case B: sampled CE2 short-ray data.

Proposition 5.9 (The common-pair nonzero-gap closures)**.** *Every row A or B of table 2 is impossible.*

Proof. In row A, theorem 5.5, type-aware radial forcing, and common-pair domination put the six points $(1 - c_*)V_i$ in U_C. Their convex hull contains $\mathcal{D}(p, q)$, while the actual gap is $J(p, q) \subset U_C$. Now theorems 5.1 and 5.6 contradict each other.

In row B, theorem 5.7 supplies every hypothesis of theorem 5.8. That theorem puts both D_2, D_4 in $U_C \subseteq T_C$ and at least one outside T_C, a contradiction. This includes $N_+ = 1$. □

5.4. **The transverse seven-point witness.** Normalize row C so that

$$J = J_0 = X_0([\ell, r]), \qquad \ell := B_0, \qquad r := 1 - A_1, \qquad 0 < \ell \le r < 1.$$

For the actual radial reaches define

$$P_i := (1 - C_i)V_i,$$

and define exactly

$$K_{\mathrm{tr}} := \{O, M_0, X_0(\ell), X_0(r), P_2, P_3, P_4\}.$$

Lemma 5.10 (The transverse witness is center-forced)**.** *Under the exact row-C hypotheses,*

$$K_{\mathrm{tr}} \subset U_C.$$

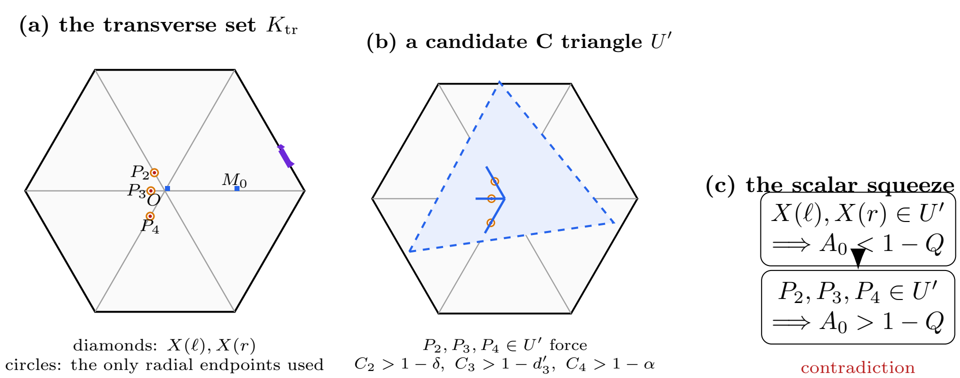


FIGURE 12. The simplified one-gap all-Vd0 enclosure certificate. Only the actual gap endpoints and the three transverse radial endpoints are needed. The gap gives the diameter upper squeeze at the supercritical role; the three radial endpoints supply the CE1/CE2 transverse return. The former points P_0, P_1, P_5 are unnecessary. Lengths are schematic.

Proof. The gap endpoints are missed by both incident open V triangles. Each P_i is the excluded endpoint of the actual closed radial trace; the Vd0 restriction excludes adjacent support, and diameter excludes nonlocal roles. The midpoint belongs only to the C triangle. □

Proposition 5.11 (CE2 transverse threshold terminal)**.** *For an open unit equilateral triangle U_C, put $T_C := \overline{U_C}$. Then*

$$N_E(T_C) = 2 \Longrightarrow K_{\mathrm{tr}} \not\subseteq U_C.$$

Proof. The exact endpoint squeeze and the two transverse thresholds are proved in theorems D.16 and D.17. □

Proposition 5.12 (CE1 three-transverse return)**.** *For an open unit equilateral triangle U_C, put $T_C := \overline{U_C}$. Then*

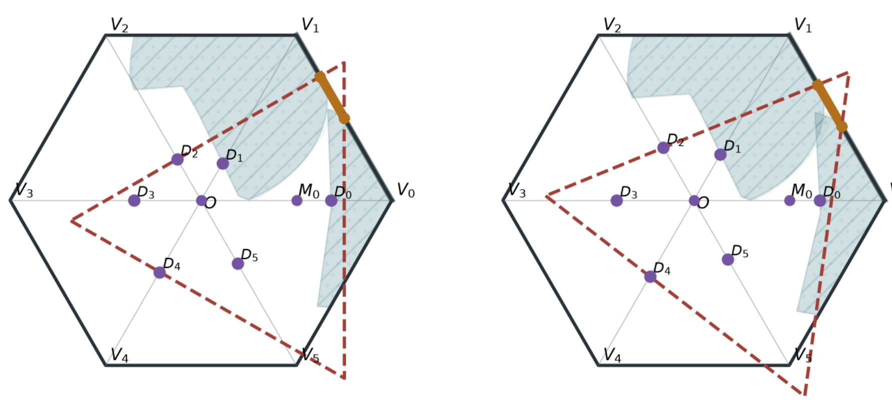
$$N_E(T_C) = 1 \Longrightarrow K_{\mathrm{tr}} \nsubseteq U_C.$$

Proof. The three actual endpoints P_2, P_3, P_4 give the CE1 radial hypotheses; the reverse-path calculation is theorem D.15. No radial point on r_0 is used. ☐

Proposition 5.13 (Transverse seven-point enclosure)**.** *Under the exact row-C hypotheses,*

$$\Lambda(K_{\mathrm{tr}}) \geq 1.$$

Proof. By theorem 2.3, the closure of a candidate C triangle has type CE1 or CE2. The preceding two propositions exclude both possibilities, and theorem 5.1 gives the enclosure conclusion. ☐



CE1 **terminal context.**

$N_{\mathrm{gap}} = 1$

$N_{+} = 1$

$(d, t) = (0, 0)$

$N_E(T_C) = 1$

CE2 **terminal context.**

$N_{\mathrm{gap}} = 1$

$N_{+} = 1$

$(d, t) = (0, 0)$

$N_E(T_C) = 2$

FIGURE 13. Case C: sampled transverse-witness data.

5.5. **Supported tails and radial separation.** For $0 \leq c < 1/2$, define

$$M_c^{\sup} := \sup\{M_c(a) : 0 \leq a \leq 1,\ M_c(a) > 1 - a\},$$

and set

$$M_{1/2}^{\sup} := \frac{1}{2}.$$

The endpoint value is the continuous extension, not the supremum of an empty strict feasible set.

Both supported-tail adapters use an actual interval

$$T_0 \cap r_1 = \{V_1 + x(O - V_1) : c \leq x \leq u\},$$

$$0 < c \leq \frac{1}{2} \leq u < 1, \qquad \varepsilon := 1 - u, \qquad C_1 \geq c.$$

With $M := M_c^{\sup}$, their common exact conclusion is

$$\sum_{i=2}^{5}(A_i + B_i) \geq 4 + M - B_1 > 4.$$

Proposition 5.14 (T3-like supported-tail terminal)**.** *Assume*

$$N_{\text{gap}} \in \{1,2\}, \qquad N_+ = 1, \qquad (d,t) = (0,1), \qquad N_E(T_C) \in \{1,2\},$$
$$T_C \cap \{M_0, \ldots, M_5\} = \{M_0\}, \qquad (o(T_0), n(T_0)) = (2,1),$$
$$(o(T_i), n(T_i)) \in \{(1,0),(2,0)\} \qquad (1 \leq i \leq 5).$$

Then the skeleton is not covered.

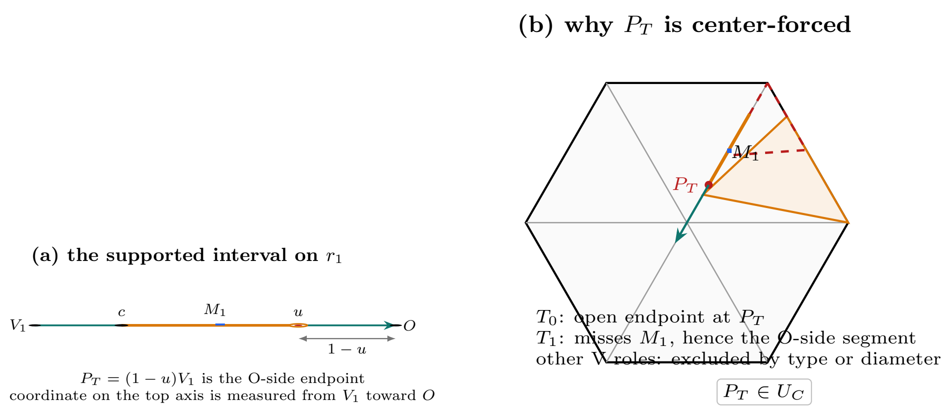
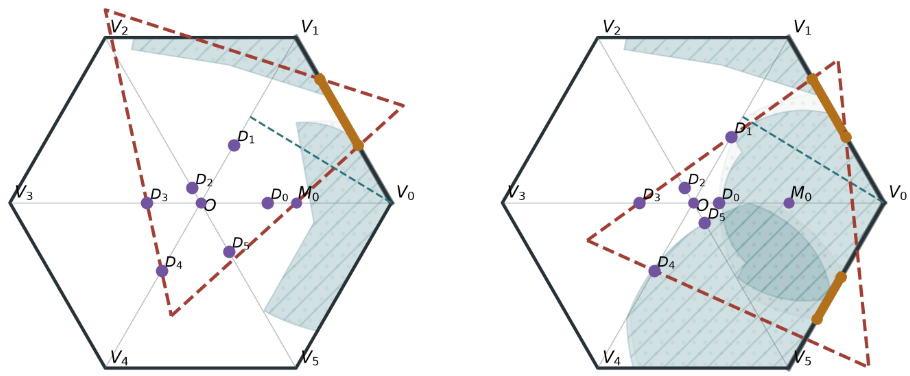


FIGURE 14. Construction of the T3-like rescuer endpoint. The interval $[c,u]$ is parametrized from V_1 toward O, so its O-side endpoint is $P_T = (1-u)V_1$. The T3-like open role omits its endpoint, the adjacent supercritical role contains V_1 but misses M_1 and therefore misses the O-side segment by convexity, and all remaining V roles are excluded. Thus P_T is forced into the C triangle.

Proof. Apply the T3-like supported-tail adapter and its tail budget, theorems D.20 and D.21. $\square$



One gap.
$N_{\text{gap}} = 1$
$N_+ = 1$
$(d, t) = (0, 1)$

Two gaps.
$N_{\text{gap}} = 2$
$N_+ = 1$
$(d, t) = (0, 1)$

FIGURE 15. Case D_T: sampled T3-like supported-tail data.

For radial separation along r_i, define

$$d_i^C := \max\{x \in [0,1] : xV_i \in T_C\},$$
$$c_{\text{loc}} := \sup\{s \in [0,1] : V_i + s(O - V_i) \in U_{i-1} \cup U_i \cup U_{i+1}\}.$$

Then

$$T_C \cap r_i = \{xV_i : 0 \le x \le d_i^C\},$$
$$(U_{i-1} \cup U_i \cup U_{i+1}) \cap r_i \subseteq \{xV_i : 1 - c_{\text{loc}} \le x \le 1\}.$$

Thus the exact strict inequality

$$c_{\text{loc}} < 1 - d_i^C$$

leaves the nonempty interval

$$\{xV_i : d_i^C < x < 1 - c_{\text{loc}}\} \subseteq S \setminus \left(U_C \cup \bigcup_{j=0}^{5} U_j \right).$$

For the nonadjacent adapter, if $I \subseteq [0,1]$ is empty or a closed interval, define

$$\mathcal{R}_I(p) := 1 - \sup\{x \in [0,1] : [0,x] \subseteq [0,p] \cup I\}.$$

Put

$$I_R := \{x \in [0,1] : X_0(x) \in T_C\}, \qquad I_L := \{x \in [0,1] : X_5(1-x) \in T_C\},$$
$$\rho_R := \mathcal{R}_{I_R}(B_0), \qquad \rho_L := \mathcal{R}_{I_L}(A_0), \qquad \rho := \min\{\rho_R, \rho_L\}, \qquad D_\tau := \rho V_\tau.$$

Appendix D proves the exact separation inequalities for both adapters.

5.6. **The Vd placement router.**

Proposition 5.15 (Direct one-Vd placement assembly). *Assume*

$$N_E(T_C) = 2, \qquad N_{\text{gap}} \in \{1, 2\}, \qquad N_+ = 1, \qquad (d, t) = (1, 0),$$
$$T_C \cap \{M_0, \dots, M_5\} = \{M_0\}.$$

Then the skeleton is not covered.

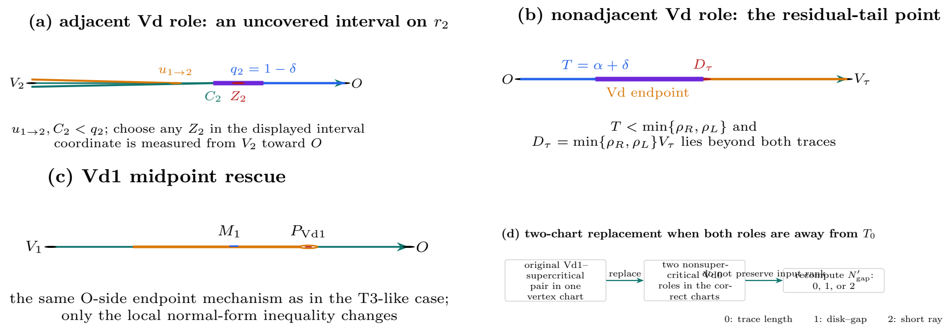


FIGURE 16. The four geometric terminals in the one-Vd placement assembly. In the adjacent placement the radial coordinate is measured from V_2 toward O; the two local V endpoints lie before the center interval entry $q_2 = 1 - \delta$, leaving a literal uncovered interval on r_2. The nonadjacent placement produces the point $D_\tau = \min\{\rho_R, \rho_L\} V_\tau$ beyond both the Vd and center traces; the Vd1 midpoint rescue uses the same O-side endpoint construction as the T3-like case; and the remaining placement is reduced by a two-chart replacement followed by recomputation of the output gap rank. Lengths are schematic; all labelled order relations are the proof relations.

Proof. If $\sigma = 0$ and $\tau \in \{1, 5\}$, the adjacent radial-separation calculation gives $c_{\text{loc}} < 1 - d_2^C$ on r_2. If $\sigma = 0$ and $\tau \in \{2, 3, 4\}$, the nonadjacent calculation gives

$$d_\tau^C < \rho < 1 - C_\tau, \qquad D_\tau \in S \setminus \left(U_C \cup \bigcup_{j=0}^{5} U_j \right).$$

If the Vd role is based at T_0, Vd2 closes by the trace bound and Vd1 closes by the supported-tail adapter. These three calculations are in theorems D.22 to D.24.

It remains that $\sigma, \tau \neq 0$. Midpoint rescue makes the two roles adjacent. Vd2 again closes by trace length. For Vd1, fresh cyclic renumbering and reflection give $(\tau, \sigma) = (0, 1)$, with

$$(o(T_0), n(T_0)) = (1, 1), \qquad A_0 + B_0 < \frac{1}{2}, \qquad M_1 \in T_0, \qquad A_1 + B_1 > 1,$$
$$A_i + B_i \le 1 \quad (i \neq 1),$$
$$(o(T_i), n(T_i)) \in \{(1, 0), (2, 0)\} \quad (1 \le i \le 5),$$
$$U_C \cap e_{0,1} = \varnothing, \qquad \mathcal{H}^1(T_j \cap r_k) = 0 \quad (2 \le j \le 5,\ k \in \{0, 1\}).$$

The two-chart construction in theorem D.5 produces $\varepsilon > 0$ and open unit triangles U_0', U_1', with $V_k \in U_k'$, such that, after

$$U_i' := U_i \quad (2 \le i \le 5), \qquad T_i' := \overline{U_i'} \quad (0 \le i \le 5),$$

one has

$$S \subseteq U_C \cup \bigcup_{i=0}^{5} U_i', \qquad (o(T_i'), n(T_i')) \in \{(1,0),(2,0)\} \quad (0 \le i \le 5),$$

$$A(T_i') + B(T_i') \le 1 \quad (0 \le i \le 5),$$
$$A(T_0') + B(T_0') = A(T_1') + B(T_1') = 1 - \varepsilon < 1,$$

where

$$A(T_i') := \max\{x \in [0,1] : V_i + x(V_{i-1} - V_i) \in T_i'\},$$
$$B(T_i') := \max\{x \in [0,1] : V_i + x(V_{i+1} - V_i) \in T_i'\}.$$
$$C(T_i') := \max\{x \in [0,1] : V_i + x(O - V_i) \in T_i'\}.$$

For the unchanged roles,

$$\big(A(T_i'), B(T_i'), C(T_i')\big) = (A_i, B_i, C_i) \qquad (2 \le i \le 5).$$

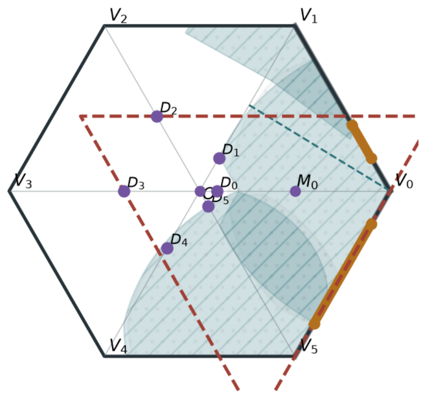
Moreover,

$$|\{i \in \mathbb{Z}/6\mathbb{Z} : A(T_i') + B(T_i') > 1\}| = 0.$$

Define, from the output open traces,

$$N_{\mathrm{gap}}' := \left|\left\{i \in \mathbb{Z}/6\mathbb{Z} : e_{i,i+1} \setminus \big((U_i' \cap e_{i,i+1}) \cup (U_{i+1}' \cap e_{i,i+1})\big) \neq \varnothing\right\}\right|.$$

Then $N_{\mathrm{gap}}' \in \{0,1,2\}$. If it is zero, theorem 3.10 applies. If it is one, apply theorem 5.5 to the primed actual reaches and then the row-A branch of theorem 5.9. If it is two, apply theorem 5.7 to the primed actual reaches and then the row-B branch of theorem 5.9. Thus the common-pair hypotheses are rederived after replacement; the input and output gap ranks are not asserted to be equal. $\square$



Vd1 **supported tail.**

$N_{\text{gap}} = 2$

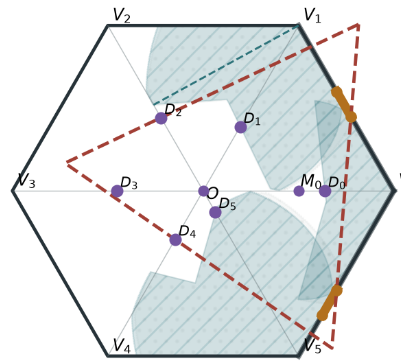
$N_+ = 1$

$(d, t) = (1, 0)$

$(\sigma, \tau) = (1, 0)$

$(o(T_0), n(T_0)) = (1, 1)$

$A_0 + B_0 < 1/2$



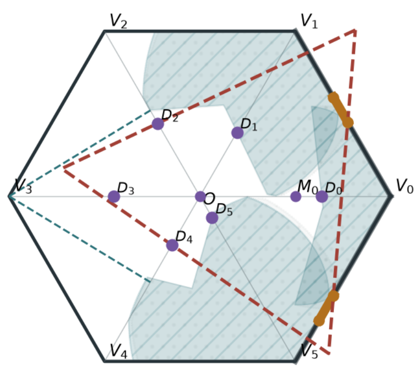
Adjacent separation.

$N_{\text{gap}} = 2$

$N_+ = 1$

$(d, t) = (1, 0)$

$(\sigma, \tau) = (0, 1)$



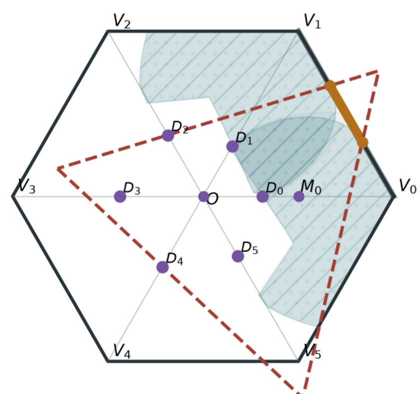
Nonadjacent separation.

$N_{\text{gap}} = 2$

$N_+ = 1$

$(d, t) = (1, 0)$

$(\sigma, \tau) = (0, 3)$



Replacement output $R \to A$**.**

$N'_{\text{gap}} = 1$

$A(T'_i) + B(T'_i) \le 1 \quad (i \in \mathbb{Z}/6\mathbb{Z})$

$(o(T'_i), n(T'_i)) \in \{(1, 0), (2, 0)\} \quad (i \in \mathbb{Z}/6\mathbb{Z})$

FIGURE 17. Cases D_V, E_a, E_n, and R: sampled router data.

Proposition 5.16 (All nonzero-gap finite-enclosure rows)**.** *Every nonzero-gap row of table 1 assigned to finite enclosure is impossible.*

Proof. Rows A and B are theorem 5.9; row C is theorem 5.13; and rows D, E, and R are theorems 5.14 and 5.15. □

5.7. **The zero-gap nine-point obstruction.** Assume

$$N_{\mathrm{gap}} = 0, \qquad N_+ = 1, \qquad (d,t) = (0,0).$$

Rotate the unique supercritical role to index 4. The exact-one handoff calculation theorem E.2 gives

$$\xi_3 < \xi_2 < \xi_1 < \xi_0 < \xi_5 < \xi_4. \tag{14}$$

Define

$$a := 1 - \xi_3, \qquad b := \xi_4, \qquad p := 1 - b, \qquad q := 1 - a. \tag{15}$$

Then

$$\begin{gathered} 0 < a, b < 1, \qquad a = a_4 < A_4, \qquad b = b_4 < B_4, \\ a + b > 1, \qquad a^2 + ab + b^2 < 1, \\ a_i \geq p, \qquad b_i \geq q \quad (0 \leq i \leq 5). \end{gathered} \tag{16}$$

Define the two boundary anchors

$$P_A := V_4 + a(V_3 - V_4), \qquad P_B := V_4 + b(V_5 - V_4),$$

and the feasible union

$$\mathcal{R}_{AB}(a,b) := \bigcup \left\{ T \cap H : \begin{array}{l} T \text{ is a closed unit equilateral triangle,} \\ V_4, P_A, P_B \in T \end{array} \right\}.$$

Put

$$c_* := c_{\max}(p,q), \qquad r_* := h(1 - c_*), \qquad P_i^{\mathrm{rad}} := (1 - c_*)V_i \quad (0 \leq i \leq 5).$$

The radial forcing calculation gives

$$\mathbb{B}(O, r_*) \subseteq \operatorname{conv}\{P_0^{\mathrm{rad}}, \ldots, P_5^{\mathrm{rad}}\} \subset U_C. \tag{17}$$

The two circles adjacent to the supercritical frontier are

$$\mathcal{C}_- := \partial\mathbb{B}(X_2(b), 1), \qquad \mathcal{C}_+ := \partial\mathbb{B}(X_5(1-a), 1).$$

Define ℓ_- and ℓ_+ as the corresponding maximal nondegenerate line segments of the interior frontier:

$$\{\ell_-\} := \left\{ [X,Y] : \begin{array}{l} X \neq Y, \quad [X,Y] \subseteq \partial\mathcal{R}_{AB}(a,b) \cap \operatorname{int}(H), \\ [X,Y] \cap \mathcal{C}_- \neq \varnothing, \\ \nexists X', Y' \ [X,Y] \subsetneq [X', Y'] \subseteq \partial\mathcal{R}_{AB}(a,b) \cap \operatorname{int}(H) \end{array} \right\},$$

$$\{\ell_+\} := \left\{ [X,Y] : \begin{array}{l} X \neq Y, \quad [X,Y] \subseteq \partial\mathcal{R}_{AB}(a,b) \cap \operatorname{int}(H), \\ [X,Y] \cap \mathcal{C}_+ \neq \varnothing, \\ \nexists X', Y' \ [X,Y] \subsetneq [X', Y'] \subseteq \partial\mathcal{R}_{AB}(a,b) \cap \operatorname{int}(H) \end{array} \right\}.$$

The three frontier points are the singleton intersections

$$\{Q_0\} := \ell_- \cap \ell_+, \qquad \{Q_-\} := \ell_- \cap \mathcal{C}_-, \qquad \{Q_+\} := \ell_+ \cap \mathcal{C}_+.$$

Their exact construction and exclusions in theorems E.1, E.4 and E.5 give

$$Q_-, Q_0, Q_+ \in U_C. \tag{18}$$

Define the public witness set

$$K_{\mathrm{wit}}(a,b) := \mathbb{B}(O, r_*) \cup \{Q_-, Q_0, Q_+\} \subset U_C. \tag{19}$$

Theorem 5.17 (Witness enclosure)**.** *Every pair (a,b) satisfying equation (16) satisfies*

$$\Lambda(K_{\mathrm{wit}}(a,b)) \geq 1.$$

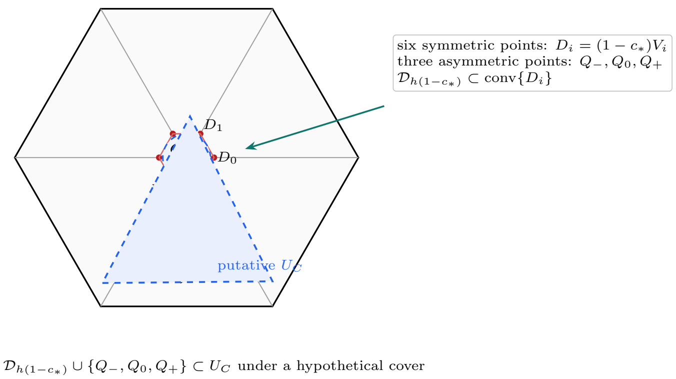


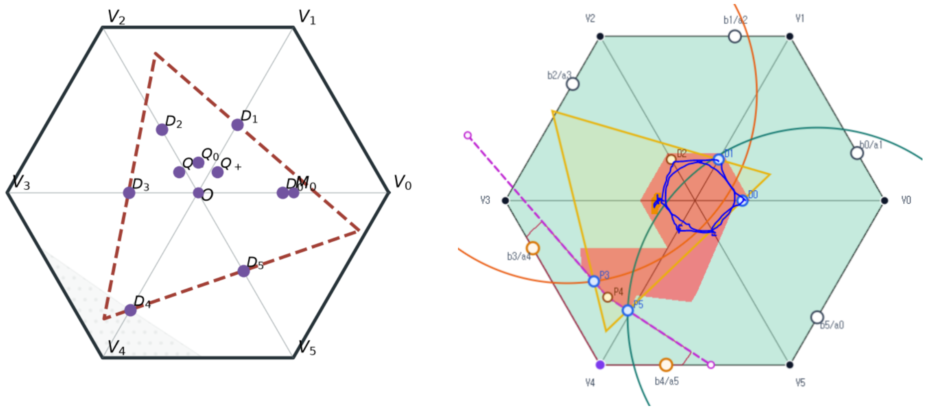
FIGURE 18. The zero-gap nine-point witness. The six symmetric radial points force a centered disk into the convex C triangle, while the three asymmetric frontier points supply the missing directional support. Marker shapes separate the two constructions: circles are radial witnesses and triangles are strict-supercritical AB-frontier witnesses. The displayed placement is illustrative; the point definitions are the exact ones in the preceding lemmas.

Proof. The disk closes the range $c_* \leq 2/3$. For $c_* > 2/3$, the Newton inner points and cyclic support arcs reduce the claim to the exact mixed-overlap certificate. The complete calculation is theorem E.9. □

Theorem 5.18 (Zero-gap obstruction)**.** *There is no hypothetical cover with*

$$N_{\mathrm{gap}} = 0, \qquad N_+ = 1, \qquad (d, t) = (0, 0).$$

Proof. The construction gives the compact containment $K_{\mathrm{wit}}(a, b) \subset U_C$, while theorem 5.17 gives $\Lambda(K_{\mathrm{wit}}(a, b)) \geq 1$. This contradicts theorem 5.1. □



Trace-exact witness.
$N_{\mathrm{gap}} = 0$
$N_+ = 1$
$(d, t) = (0, 0)$

Core configuration.
$N_{\mathrm{gap}} = 0$
$N_+ = 1$
$(d, t) = (0, 0)$

FIGURE 19. Case F: sampled zero-gap obstruction data.

Proposition 5.19 (Cases excluded by the nine-point obstruction)**.** *Every* $N_{\mathrm{gap}} = 0$*,* $N_+ = 1$*,* $(d, t) = (0, 0)$ *row of table 1 is impossible, independently of the center class.*

Proof. This is theorem 5.18; the equality $N_{\mathrm{gap}} = 0$ supplies full boundary coverage by theorem 2.9. □

6. Completion of the Proof

Proof of Theorem 1.1. Assume that seven open unit equilateral triangles cover H. By Proposition 2.14, the cover lies in exactly one row of Table 1.

First suppose $N_{\mathrm{gap}} = 0$. The $N_+ = 0$ row and the $N_+ = 1, d \geq 1$ row are excluded by Proposition 3.10. The $N_+ \geq 2$ row and the $N_+ = 1, d = 0, t \geq 1$ row are excluded by Proposition 4.4. The remaining $N_+ = 1, (d, t) = (0, 0)$ row is excluded by the explicit finite-enclosure theorem Proposition 5.19.

Now suppose $N_{\mathrm{gap}} \geq 1$. The rows assigned to trace length are excluded by Proposition 3.10. Every nonzero-gap row assigned to finite enclosure is excluded by the finite residual-hull theorem Proposition 5.16. In the final CE2 one-Vd1/Vd2 row, the trace-length complements and the finite-enclosure placements are assembled in Proposition 5.15; its replacement subcase recomputes the gap rank before selecting the closing strategy. Thus every zero- and nonzero-gap row is impossible, contradicting the assumed cover. □

Corollary 1.2 follows from Proposition 2.1.

Appendix A. Structural, Shared-Local, and Signed-Center Optimization

This appendix solves the coordinate and compactness problems used by the geometric interfaces in Section 2. It also proves the shared equilateral-enclosure calculus used by the later optimization appendices. Each calculation is attached to the body statement it supplies.

A.1. **Open–closed shrinking feasibility.**

Lemma A.1 (Uniform shrinking margin)**.** *If seven open unit equilateral triangles cover H, then, for some $\varepsilon > 0$, seven closed equilateral triangles of side $1-\varepsilon$ also cover H.*

Proof. For $p \in H$ put

$$m_\nu(p) := \operatorname{dist}(p, \mathbb{R}^2 \setminus U_\nu) \quad \big(\nu \in \{C, 0, \ldots, 5\}\big), \qquad m(p) := \max_{\nu \in \{C,0,\ldots,5\}} m_\nu(p).$$

The continuous function m is positive on compact H, hence

$$\eta := \min_{p \in H} m(p) > 0.$$

Choose

$$0 < \rho < \min\left\{\eta, \frac{\sqrt{3}}{6}\right\}.$$

Moving every side of U_ν inward by ρ gives a closed equilateral triangle K_ν of side $1 - 2\sqrt{3}\rho$. For each $p \in H$, some $m_\nu(p) \geq \eta > \rho$, so $p \in K_\nu$. Thus $\varepsilon = 2\sqrt{3}\rho$ is feasible. This supplies the only quantitative step in Proposition 2.1. □

A.2. **T3-like trace-majorant feasibility.**

The calculation uses the local wedge and the two orientation charts developed in the exact-trace module below. It is placed here beside the body warning that the output is a closed majorant, not a replacement open triangle.

Lemma A.2 (T3-like translation calculation)**.** *The closed-trace majorant in Proposition 2.7 exists. In the calculation chart, every original* T3-like *triangle also satisfies $A_i + B_i \leq 1$.*

Proof. Use the wedge chart of theorem A.7 and the orientation forms (27)– (28) in Subsection A.6. In Type I, direct inversion shows that when exactly two vertices lie outside W, the sole wedge vertex has both coordinates strictly below 1: the relevant bounds are $(1-t)/z$ and t/z, with strictness from $\alpha > 0$ or $\beta > 0$. The support argument in theorem A.7 then excludes positive-length contact with either adjacent radial line. Thus a T3-like triangle has Type II.

Reflect in $x = y$ if necessary so that $T_0 \cap r_1$ has positive length. The endpoint cases $t = 0, 1$ give no such interval, hence $0 < t < 1$. The exact axis reaches are

$$A_0 = \beta, \qquad B_0 = z - \alpha - \beta,$$

and the support condition is

$$Q_x > 1, \qquad Q_y \leq 1,$$

where Q is (30). In particular

$$A_0 + B_0 = z - \alpha < z \leq 1.$$

Define $\widehat{T}$ by replacing α with 0 and retaining t, β. This changes only the translation. At the origin, $\widehat{U} = 0$ and $0 < \widehat{V} = \beta < z$, so V_0 lies in the relative interior of a side. For $(x, y) \in W$,

$$\widehat{U} = (1-t)x + ty \geq 0, \qquad \widehat{V} = V, \qquad \widehat{U} + \widehat{V} = U + V - \alpha.$$

Thus $T \cap W \subset \widehat{T} \cap W$. The old x-axis reach is B_0 and the new reach is $B_0 + \alpha$, so the inclusion is strict in H.

The two outside vertices remain outside, while V_0 and the new wedge vertex lie in H. Moreover $\widehat{Q}_x > Q_x > 1$, and $Q_x > 1$ gives

$$tD_t(1 - \widehat{Q}_y) > D_t(1 - z) + (1 - t)\alpha > 0.$$

Hence $\widehat{Q}_y < 1$ and $(o(\widehat{T}), n(\widehat{T})) = (2, 1)$. At every nonzero point of W, $\widehat{U} > 0$; the other two displayed slack relations show that old interior points other than V_0 remain interior. Reflection proves the other supported-arm case. $\square$

A.3. **Strict-handoff feasibility.**

Lemma A.3 (Cyclic strict-handoff selection)**.** *Under the hypotheses of Proposition 2.11, all six overlap intervals are nonempty, and the selectors can preserve the stated one- or two-supercritical-index alternatives.*

Proof. Since V_i is interior to a diameter-one triangle, $0 < A_i < 1$ and $0 < B_i < 1$. No nonincident V triangle contains a relative-interior point of $e_{i,i+1}$. Thus the relatively open incident traces cover the edge and overlap, so

$$(1 - A_{i+1}, B_i) \neq \varnothing.$$

Choosing ξ_i in this interval places both selected anchors in U_i. Their squared distance is $a_i^2 + a_i b_i + b_i^2$, which is strictly below 1 because both are interior points of the same open unit triangle.

Put $\ell_i := 1 - A_{i+1}$ and $u_i := B_i$. Then

$$A_i + B_i > 1 \iff \ell_{i-1} < u_i, \qquad a_i + b_i > 1 \iff \xi_{i-1} < \xi_i. \tag{20}$$

If i is actually nonsupercritical, then

$$\xi_i < u_i \leq \ell_{i-1} < \xi_{i-1},$$

so no selection makes it supercritical. Also

$$\sum_{i=0}^{5}(a_i + b_i) = \sum_{i=0}^{5}(1 - \xi_{i-1} + \xi_i) = 6. \tag{21}$$

If p is the sole actual supercritical index, the other five selected sums are below 1, and (21) forces the pth sum above 1.

For two nonadjacent supercritical indices p, q, choose

$$\ell_{r-1} < z_r < u_r \qquad (r \in \{p, q\}),$$

and then

$$\xi_{r-1} \in (\ell_{r-1}, \min\{u_{r-1}, z_r\}), \qquad \xi_r \in (\max\{\ell_r, z_r\}, u_r).$$

The pairs are disjoint and create ascents at p, q. Any three indices on a six-cycle contain two nonadjacent indices. If the only two supercritical indices are adjacent, say $p, p+1$, the remaining inequalities give

$$\ell_{p-1} < \ell_{p+1}, \qquad \max\{\ell_{p-1}, \ell_p\} < \min\{u_p, u_{p+1}\}.$$

Choose ξ_p between the latter bounds and choose $\xi_{p-1} < \xi_p < \xi_{p+1}$ in their overlap intervals. By (20), both selected indices remain supercritical. $\square$

A.4. **Shared equilateral-enclosure optimization.** Let R denote counterclockwise rotation through $2\pi/3$. For a nonempty compact $K \subset \mathbb{R}^2$, put

$$h_K(n) := \max_{x\in K}\langle x, n\rangle.$$

Proposition A.4 (Equilateral enclosure gauge)**.** *The least side length $\Lambda(K)$ of a closed equilateral triangle containing K is*

$$\Lambda(K) = \frac{2}{\sqrt{3}} \min_{\|n\|=1} \sum_{j=0}^{2} h_K(\mathsf{R}^j n). \tag{22}$$

It is monotone under inclusion, translation invariant, and positively homogeneous. If K is a polygonal hull with at least two distinct points, a minimizing normal can be chosen perpendicular to a hull edge.

Proof. For fixed outward normals $n, \mathsf{R}n, \mathsf{R}^2 n$, the least containing triangle has the corresponding three support numbers, and its side is $2/\sqrt{3}$ times their sum. Translation invariance follows from

$$n + \mathsf{R}n + \mathsf{R}^2 n = 0,$$

while monotonicity and homogeneity are immediate. On an open angular cell with fixed support points, the support sum is $A\cos\theta + B\sin\theta$ and its second derivative is its negative. For a nonsingleton the support sum is positive, so an interior critical point is a maximum; a minimum occurs at a support tie, hence at a hull-edge normal. $\square$

A.4.1. *Finite caliper criterion.* Let J denote counterclockwise rotation through $\pi/2$. For a nonempty finite set F define

$$W_F(N) := \sum_{k=0}^{2} \max_{z\in F}\langle z, \mathsf{R}^k N\rangle. \tag{23}$$

Theorem A.5 (Finite caliper certificate)**.** *If F has at least two distinct points, then F is contained in a closed unit equilateral triangle if and only if there are distinct $p, q \in F$ and $\varepsilon \in \{-1, 1\}$ such that*

$$W_F\big(\varepsilon \mathsf{J}(q-p)\big) \le \frac{\sqrt{3}}{2}\|p-q\|. \tag{24}$$

A singleton is contained in equilateral triangles of arbitrarily small positive side length. For $F_x = \{O, A, B, x\}$, every clause is a finite polynomial clause of degree at most four in the coordinates of A, B, x; at most $12 \cdot 4^3$ clauses are needed.

Proof. Apply Proposition A.4 to conv$F$. A minimizing normal is perpendicular to a hull edge $[p, q]$ and, after cyclic rotation, equals $\varepsilon\mathsf{J}(q-p)/\|p-q\|$. Homogeneity gives (24); a nondegenerate segment follows by checking its two normals.

For the polynomial assertion fix (p, q, ε), put $N = \varepsilon\mathsf{J}(q-p)$ and $N_k = \mathsf{R}^k N$, and select support labels $s_k \in F_x$. Retain the strict condition

$$\|p-q\|^2 > 0.$$

On the cell

$$\langle s_k - y, N_k\rangle \ge 0 \qquad (y \in F_x,\ k = 0, 1, 2), \tag{25}$$

the test becomes

$$\sum_{k=0}^{2}\langle s_k, N_k\rangle \le \frac{\sqrt{3}}{2}\|p-q\|. \tag{26}$$

The left side is polynomial of degree at most two. Both sides are nonnegative on an active cell, so one squaring gives degree at most four. There are at most six unordered pairs, two normal signs for each pair, and 4^3 support triples for each pair-normal candidate. ☐

A.5. **Center and V triangle incidence feasibility.**

Lemma A.6 (Separation of nonadjacent boundary edges)**.** *Relative-interior points on two nonadjacent edges of H are more than unit distance apart.*

Proof. Up to symmetry take

$$x = (1-t)V_0 + tV_1, \qquad 0 < t < 1.$$

For $y = (1-s)V_2 + sV_3$, $0 < s < 1$, direct expansion gives

$$\|x-y\|^2 - 1 = (1-t)(2-t) + s(s+t) > 0.$$

For the other orbit, $y = (1-s)V_3 + sV_4$, put $u = t + s$ and obtain

$$\|x-y\|^2 - 1 = (u-1)^2 + 2 > 0.$$

These are the two nonadjacent-edge configurations. Since the diameter of a unit equilateral triangle is 1, the result supplies Proposition 2.3. ☐

Lemma A.7 (Local wedge calculation)**.** *The incidence conclusions of Lemma 2.4 hold.*

Proof. Normalize to $i = 0$ and use the affine chart

$$X = V_0 + x(V_1 - V_0) + y(V_5 - V_0).$$

Then $V_0 = (0,0)$,

$$\|(x,y)\|^2 = x^2 + y^2 - xy,$$

and, in the closed unit ball about V_0, membership in H is membership in

$$W := \{(x,y) : x \ge 0,\ y \ge 0\}.$$

Indeed,

$$x^2 + y^2 - xy = \left(y - \frac{x}{2}\right)^2 + \frac{3x^2}{4} = (x-y)^2 + xy.$$

The adjacent radial segments are

$$r_1 = \{(1,y) : 0 \le y \le 1\}, \qquad r_5 = \{(x,1) : 0 \le x \le 1\}.$$

Suppose $T_0 \cap r_1$ has positive length and let m be the largest x-coordinate of a vertex of T_0. If $m > 1$, a maximizing vertex (m, y) lies in the unit ball and has $0 < y < 1$, so it lies in H. If $m = 1$, positive-length contact with the support line $x = 1$ forces a side there, and both endpoints satisfy $1 + y^2 - y \le 1$ and lie in H. The argument for r_5 is symmetric. If both arms are met but only one vertex lies in H, that vertex would have $x > 1$ and $y > 1$, contradicting $(x-y)^2 + xy > 1$.

Finally, if all three vertices lay in convex H, then $T_0 \subset H$, contrary to the extreme point V_0 lying in $\mathrm{int}(T_0)$. Hence $o(T_0) \ge 1$, completing the calculation. ☐

A.6. **Exact-trace normalization.** Set

$$D_t := 1 - t + t^2, \qquad z := \sqrt{D_t}, \qquad 0 \le t \le 1.$$

After relabelling the vertices, every orientation belongs to one of the two families

	first side vector	second side vector
I	$z^{-1}(1,t)$	$z^{-1}(1-t,1)$
II	$z^{-1}(t,-(1-t))$	$z^{-1}(1,t)$.

In Type I write

$$U = \alpha + y - tx, \qquad V = \beta + x - (1-t)y, \qquad T = \{U \ge 0, V \ge 0, U+V \le z\}, \tag{27}$$

and in Type II write

$$U = \alpha + (1-t)x + ty, \qquad V = \beta + tx - y, \qquad T = \{U \ge 0, V \ge 0, U+V \le z\}. \tag{28}$$

In both charts

$$V_0 \in \operatorname{int}(T) \iff \alpha > 0, \quad \beta > 0, \quad \alpha + \beta < z. \tag{29}$$

The Type-II vertex in the wedge is

$$Q = \left(\frac{z - \alpha - t\beta}{D_t}, \frac{t(z-\alpha) + (1-t)\beta}{D_t} \right). \tag{30}$$

Lemma A.8 (Raw $(3,0)$ trace-normalization calculation)**.** *The translation asserted in Proposition 2.5 exists and preserves both the open and closed traces in H.*

Proof. Normalize to $i = 0$. A Type-II triangle cannot have $o = 3$, because the point Q in (30) lies in the wedge; together with the diameter-one condition, Lemma 2.4 puts it in H. Hence T has Type-I form. Its vertices are

$$Q_0 = \left(-\frac{(1-t)\alpha + \beta}{D_t}, -\frac{\alpha + t\beta}{D_t} \right),$$

$$\begin{aligned} Q_1 &= \left(\frac{z - (1-t)\alpha - \beta}{D_t}, \frac{tz - \alpha - t\beta}{D_t} \right), \\ Q_2 &= \left(\frac{(1-t)z - (1-t)\alpha - \beta}{D_t}, \frac{z - \alpha - t\beta}{D_t} \right). \end{aligned} \tag{31}$$

Thus $o = 3$ is equivalent to

$$\alpha + t\beta > tz, \qquad (1-t)\alpha + \beta > (1-t)z, \tag{32}$$

which forces $0 < t < 1$. Put $s = \alpha + \beta$ and $L = z - s > 0$. Then

$$\alpha > \frac{tL}{1-t}, \qquad \beta > \frac{(1-t)L}{t}. \tag{33}$$

For $(x,y) \in W$ satisfying

$$(1-t)x + ty \le L, \tag{34}$$

the minimum of U is attained at $(L/(1-t), 0)$ and that of V at $(0, L/t)$. Hence

$$T \cap W = \{(x,y) \in W : (1-t)x + ty \le L\}. \tag{35}$$

Keep t, s fixed and set

$$\widehat{\alpha} := \frac{tL}{1-t}, \qquad \widehat{\beta} := s - \widehat{\alpha}. \tag{36}$$

Then $0 < \widehat{\alpha} < \alpha$, $\widehat{\beta} > \beta > 0$, and $\widehat{\alpha} + \widehat{\beta} = s < z$. Changing these affine constants is a translation and preserves $V_0 \in \operatorname{int}(\widehat{T})$. Its third inequality remains (34); $\widehat{U} = 0$ only at the endpoint $(L/(1-t), 0)$ of that common boundary and $\widehat{V} > 0$. Therefore

$$\widehat{T} \cap W = T \cap W,$$

and their interiors have the same trace on W. The local wedge calculation identifies these with the closed and open traces in H.

Equation (36) places Q_1 on the boundary of H; Q_0 remains outside and the second strict inequality in (33) keeps Q_2 outside. Hence $(o(\widehat{T}), n(\widehat{T})) = (2, 0)$. Since the three reach segments lie in H, their open and closed traces, and therefore A_i, B_i, C_i and $A_i + B_i > 1$, are unchanged. □

A.7. **The local admissible set and radial envelope.** Let u, v be unit vectors meeting at $120°$ and define

$$K(a,b,c) := \operatorname{conv}\{0, au, bv, c(u+v)\}.$$

This is the translate of the body normalization at V_0, with $u = V_5 - V_0$ and $v = V_1 - V_0$. The exact local admissible set is

$$\mathcal{A} := \{(a,b,c) \in [0,1]^3 : \Lambda(K(a,b,c)) \le 1\}. \tag{37}$$

This set records the three reach lower bounds that a closed unit V triangle can realize.

Proposition A.9 (Exact local admissible set)**.** *Put*

$$s := a+b, \qquad d := a^2 + ab + b^2, \qquad m := \min\{a,b\}, \qquad M := \max\{a,b\},$$
$$q := s^4 - s^2 + ab,$$

and

$$\begin{aligned} F_L &:= c^4 - c^2 + mc - m^2, \\ F_T &:= (s^2-1)c^2 + Mc - M^2, \\ F_S &:= (m^2-1)c^2 + (2mM^2 + M)c + M^4 - M^2. \end{aligned}$$

Then

$$\mathcal{A} = \mathcal{A}_L \cup \mathcal{A}_T \cup \mathcal{A}_S, \tag{38}$$

where all variables lie in $[0,1]$ *and*

$$\begin{aligned} \mathcal{A}_L &:= \{d \le 1,\ s \le 1,\ q \le 0,\ F_L \le 0\}, \\ \mathcal{A}_T &:= \{d \le 1,\ s \le 1,\ q \ge 0,\ F_T \le 0,\ c \le 2M\}, \\ \mathcal{A}_S &:= \{d \le 1,\ s > 1,\ F_S \le 0,\ c \le 1/2\}. \end{aligned}$$

The set is coordinatewise down-closed. Define its maximal radial demand by

$$c_{\max}(a,b) := \max\{c \in [0,1] : (a,b,c) \in \mathcal{A}\}.$$

Then

$$c_{\max}(a,b) = \begin{cases} 1, & a = b = 0, \\ C_L(m), & s \le 1,\ q \le 0,\ (a,b) \neq (0,0), \\ \dfrac{2M}{1+\sqrt{4s^2-3}}, & s \le 1,\ q > 0, \\ \dfrac{M(2mM+1) - M\sqrt{4d-3}}{2(1-m^2)}, & s > 1, \end{cases} \tag{39}$$

where $C_L(m)$ is the unique root in $[\sqrt{3}/2, 1]$ of

$$z^4 - z^2 + mz - m^2 = 0.$$

Proof. First assume $s > 0$ and $cs \le ab$. When $a, b > 0$,

$$c(u+v) = \frac{c}{a}(au) + \frac{c}{b}(bv) + \left(1 - \frac{c}{a} - \frac{c}{b}\right) 0 \in \operatorname{conv}\{0, au, bv\};$$

the zero-coordinate cases follow by continuity. The distance from au to bv is $\sqrt{d}$. With $T = -bu - av$, the triangle $\operatorname{conv}\{au, bv, T\}$ is equilateral of side $\sqrt{d}$ and

$$0 = \frac{b^2}{d}(au) + \frac{a^2}{d}(bv) + \frac{ab}{d}T.$$

Thus the least enclosure side is $\sqrt{d}$. When $a = b = 0$, it is c.

Now assume $cs > ab$. The hull has cyclic vertices $0, bv, c(u+v), au$. Proposition A.4 reduces its least enclosure side to its four edge normals. For the edges through 0 one obtains

$$L_{0A} = b + \max\{a, c\}, \qquad L_{B0} = a + \max\{b, c\}.$$

For the edge from au to $c(u+v)$ put

$$R_a := \sqrt{a^2 - ac + c^2}, \qquad n_0 := \frac{1}{2R_a}(\sqrt{3}a, 2c - a).$$

The three support values in the rotated directions are

$$\frac{\sqrt{3}ac}{2R_a}, \qquad \frac{\sqrt{3}a(a-c)_+}{2R_a}, \qquad \frac{\sqrt{3}c\max\{b, (c-a)_+\}}{2R_a}.$$

Consequently

$$L_{AC} = \begin{cases} (a^2 + bc)/R_a, & a \ge c, \\ c(a+b)/R_a, & a < c \le a + b, \\ c^2/R_a, & c > a + b, \end{cases}$$

and L_{CB} is obtained by interchanging a, b. Hence

$$\Lambda(K(a,b,c)) = \min\{L_{0A}, L_{B0}, L_{AC}, L_{CB}\}.$$

Assume by symmetry that $a = m \le b = M$. Reading the active support points in these four formulas gives

range		active supports in cyclic order			unit frontier
s	q	first	second	third	equation
$s < 1$	$q < 0$	$\{0\}$	$\{c(u+v)\}$	$\{au, c(u+v)\}$	$F_L = 0$
$s < 1$	$q > 0$	$\{0\}$	$\{bv, c(u+v)\}$	$\{au\}$	$F_T = 0$
$s = 1$		$\{0, bv\}$	$\{bv, c(u+v)\}$	$\{au\}$	$c = M$
$s > 1$		$\{bv\}$	$\{bv, c(u+v)\}$	$\{au\}$	$F_S = 0$

The inactive-support inequalities reduce to $q \le 0$, $q \ge 0$, and $s > 1$. The positive unit-frontier equations square to $F_L = 0, F_T = 0, F_S = 0$.

It remains to select the geometric components introduced by squaring. In Cell T the roots are

$$c_- := \frac{2M}{1+\sqrt{4s^2-3}}, \qquad c_+ := \frac{2M}{1-\sqrt{4s^2-3}}.$$

The polynomial inequality also has a formal high component, while the geometric fiber is the component joined to $c = 0$. Since

$$c_- \le 2M \le c_+,$$

the exact selector is $c \le 2M$.

In Cell S,

$$F_S(a, b, 0) = M^2(M^2 - 1) < 0, \qquad F_S(a, b, M) = M^2(s^2 - 1) > 0.$$

Moreover $4F_S(a, b, 1/2)$ increases with m, and at $m = 1 - M$ equals $M^2(2M - 1)^2$. Because $s > 1$, one has $F_S(a, b, 1/2) > 0$. The quadratic opens downward, so $c \le 1/2$ selects exactly the smaller-root component containing 0.

In Cell L, the fiber from $c = 0$ reaches one root in $[\sqrt{3}/2, 1]$; uniqueness there follows from $4c^3 - 2c + m > 0$. At $q = 0$ the L and T formulas meet at $c = s$. This proves (38), including all equality and degenerate cases by continuity. Convexity of any containing triangle proves coordinatewise down-closedness. Solving the three selected frontier equations gives (39). □

For later variational use, the raw forward envelope retains its established notation

$$M_c(a) := \max\{b \in [0, 1] : (a, b, c) \in \mathcal{A}\}.$$

No closed-form evaluation of M_c is needed in the geometric body.

A.8. **Midpoint caliper optimization.**

Lemma A.10 (Four-point self-midpoint caliper)**.** *If $a + b > 1$, then*

$$\Lambda\left(\operatorname{conv}\left\{0, au, \frac{u+v}{2}, bv\right\}\right) > 1.$$

This supplies Lemma 2.12.

Proof. Put

$$K := \operatorname{conv}\left\{0, au, \frac{u+v}{2}, bv\right\}.$$

By Theorem A.5, it suffices to check a normal to each of the four hull edges. For the edge from 0 to au, the sum of the three supports is

$$S_{0A} = \max\left\{\frac{\sqrt{3}}{4}, \frac{\sqrt{3}a}{2}\right\} + \frac{\sqrt{3}b}{2} > \frac{\sqrt{3}}{2}.$$

If $a \ge 1/2$, this is $(\sqrt{3}/2)(a + b)$; if $a < 1/2$, then $b > 1/2$. The edge through bv is symmetric.

For the edge from au to $(u + v)/2$, put $D_a := \sqrt{4a^2 - 2a + 1}$. Projection gives

$$S_{AM} = \begin{cases} \dfrac{\sqrt{3}(a+b)}{2D_a}, & a \le 1/2, \\ \dfrac{\sqrt{3}(2a^2+b)}{2D_a}, & a > 1/2. \end{cases}$$

The first value exceeds $\sqrt{3}/2$ because $D_a \le 1$. In the second case, $b > 1 - a$ and

$$(2a^2 - a + 1)^2 - D_a^2 = a^2(2a-1)^2 \ge 0,$$

so it again exceeds $\sqrt{3}/2$. The remaining edge is symmetric. Formula (22) now gives $\Lambda(K) > 1$.

If $a + b = 1$ while all four points are contained with positive margin, both boundary reach demands can be increased slightly, contradicting the strict case just proved. □

A.9. **Signed-center optimization.** The CE1 and CE2 cases share one signed cell. The parameters below are used only in the appendices; the geometric body uses the exact C triangle type and midpoint interfaces.

Proposition A.11 (Signed CE1/CE2 center normal form)**.** *Let T_C be a closed unit equilateral triangle with $O \in \operatorname{int}(T_C)$ and a positive-length boundary trace. Normalize such a trace to $e_{0,1}$ and use affine coordinates*

$$X = V_0 + b(V_1 - V_0) + a(V_5 - V_0).$$

There are parameters

$$0 < R < 1, \qquad W := 1 - R, \qquad E := \sqrt{1 - RW},$$
$$\eta := 1 - E, \qquad P := E(1 - E),$$

and nonnegative slacks α, δ, with $k := \eta + \alpha + \delta$, such that

$$\begin{aligned} F_0 &= R + \alpha - a + Wb, \\ F_1 &= Rb + Wa - k, \qquad\qquad T_C = \{F_0, F_1, F_2 \ge 0\}. \\ F_2 &= W + \delta - b + Ra, \end{aligned} \tag{40}$$

Define

$$\Delta_R := P - \alpha - W\delta, \qquad \Delta_L := P - R\alpha - \delta. \tag{41}$$

Then

$$T_C \cap e_{0,1} = \left[\frac{k}{R}, W + \delta\right], \qquad \mathcal{H}^1(T_C \cap e_{0,1}) = \frac{\Delta_R}{R}, \tag{42}$$

and the possible companion trace is

$$T_C \cap e_{5,0} = \left[\frac{k}{W}, R + \alpha\right] \quad \textit{when } \Delta_L > 0, \qquad \mathcal{H}^1(T_C \cap e_{5,0}) = \frac{[\Delta_L]_+}{W}. \tag{43}$$

The normalized trace requires $\Delta_R > 0$, and

$$T_C \textit{ is } \mathrm{CE1} \iff \Delta_L \le 0, \qquad T_C \textit{ is } \mathrm{CE2} \iff \Delta_L > 0. \tag{44}$$

The six radial exits from O are

$$\begin{aligned} d_0^C &= E - \alpha - \delta, \qquad & d_1^C &= \frac{\delta}{R}, \quad & d_2^C &= \delta, \\ d_3^C &= \min\left\{\frac{\alpha}{R}, \frac{\delta}{W}\right\}, \quad & d_4^C &= \alpha, \quad & d_5^C &= \frac{\alpha}{W}. \end{aligned} \tag{45}$$

Moreover,

$$T_C \cap \{M_0, \dots, M_5\} = \{M_0\}. \tag{46}$$

For the closure of an original open C triangle, $\alpha, \delta > 0$.

Proof. Write $T_C \cap e_{0,1} = [s,t]$ with $0 \le s < t \le 1$. The two sides through the endpoints have a positive slope ratio. Reflect across the perpendicular bisector if necessary, call the ratio $R \le 1$, and put $W = 1 - R$. The three side equations are

$$F_1 = Rb + Wa - Rs, \qquad F_2 = -b + Ra + t, \qquad F_0 = Wb - a + E + Rs - t,$$

where $E = \sqrt{1 - RW}$ is the unit-side relation. If $R = 1$, then $F_0(O) = s - t < 0$, so $0 < R < 1$.

Set $\alpha := F_0(O)$ and $\delta := F_2(O)$. Then

$$t = W + \delta, \qquad Rs = \eta + \alpha + \delta = k,$$

which gives (40). The gradients are the three equilateral side-normal directions and $F_0 + F_1 + F_2 = E$.

On $e_{0,1}$, $a = 0$ and the active inequalities are $Rb \ge k$ and $b \le W + \delta$. Thus its length is

$$W + \delta - \frac{k}{R} = \frac{P - \alpha - W\delta}{R}.$$

On $e_{5,0}$, $b = 0$ and the active inequalities are $Wa \ge k$ and $a \le R + \alpha$, giving signed length

$$R + \alpha - \frac{k}{W} = \frac{P - R\alpha - \delta}{W}.$$

This proves the two trace formulas.

The positive right trace implies

$$\delta < \frac{P}{W} = \frac{ER}{1+E} < \frac{R}{2}.$$

On $e_{1,2}$, parametrized by $b = 1 + q, a = q$, one has $F_2 = \delta - R - Wq < 0$. The center classification leaves $e_{5,0}$ as the only possible companion edge, and its signed length proves (44); $\Delta_L = 0$ is only point contact.

Substituting the six ray parametrizations into (40) gives (45). Finally,

$$W(\alpha + \delta) \le \alpha + W\delta < P$$

implies

$$E - \alpha - \delta > E - \frac{P}{W} = \frac{E(E-R)}{W} > \frac{1}{2},$$

where the last inequality is equivalent to $(E - R)^2 > 0$. Also

$$\alpha < P < \min\left\{\frac{R}{2}, \frac{W}{2}\right\}, \qquad \delta < \frac{R}{2}.$$

Therefore $d_0^C > 1/2$ and $d_i^C < 1/2$ for $i = 1, \dots, 5$, proving the midpoint formula. Interior containment of O in the original open triangle makes all three side slacks positive, so $\alpha, \delta > 0$. $\square$

Lemma A.12 (CE2 total slack)**.** *In the CE2 sign range, with $T := \alpha + \delta$,*

$$T < \frac{\eta}{E}. \tag{47}$$

Proof. The CE2 inequalities are $\alpha + W\delta < P$ and $R\alpha + \delta < P$. Multiply them by W and R and add. Since $W + R^2 = W^2 + R = E^2$, this gives $E^2 T < P = E\eta$. $\square$

Corollary A.13 (Common center-boundary formula). *The full center-boundary contribution is*

$$L_{\partial H}(T_C) = \frac{\Delta_R}{R} + \frac{[\Delta_L]_+}{W}. \tag{48}$$

In CE1,

$$L_{\partial H}(T_C) \le P \le \frac{\sqrt{3}}{2} - \frac{3}{4},$$

and in CE2,

$$L_{\partial H}(T_C) = \frac{E}{1+E} - \frac{E^2(\alpha+\delta)}{RW} < \frac{1}{2}.$$

Proof. The first formula sums (42) and (43). In CE1, $R\alpha + \delta \ge P$; multiplication by W, together with $\alpha \ge RW\alpha$, gives $\alpha + W\delta \ge WP$. Thus $\Delta_R \le RP$ and the contribution is at most P. Since $E \ge \sqrt{3}/2$ and $E(1-E)$ decreases there, the stated cap follows.

In CE2, direct addition gives

$$L_{\partial H}(T_C) = \frac{P - E^2(\alpha+\delta)}{RW} = \frac{E}{1+E} - \frac{E^2(\alpha+\delta)}{RW} < \frac{1}{2}.$$

□

Lemma A.14 (Adjacent-edge diameter transfer). *For $0 \le q \le 1$, the zero-radial forward envelope is*

$$M_0(q) = \frac{-q + \sqrt{4-3q^2}}{2}.$$

Here M_0 without an argument remains the radial midpoint. If a unit equilateral triangle reaches parameters q, b on adjacent boundary edges, then $b \le M_0(q)$. The function is strictly decreasing and, for $q > 0$,

$$M_0(q) < 1 - \frac{q}{2}, \qquad 1 - M_0(q) > \frac{q}{2}.$$

Proof. The squared distance between the boundary points is $q^2 + qb + b^2 \le 1$. Solving the quadratic in b gives the formula. Differentiation gives

$$M_0'(q) = \frac{-1 - 3q/\sqrt{4-3q^2}}{2} < 0.$$

For $q > 0$, $\sqrt{4-3q^2} < 2$, which gives both strict comparisons. □

Appendix B. Trace-Length Optimization

This appendix evaluates the trace quantities whose exact interfaces appear in Section 3. The center and V triangle coordinates are used only here; the body retains the geometric hypotheses and final bounds.

B.1. **Conditional skeleton caps.**

Lemma B.1 (Center-skeleton calculation). *Under the hypotheses of Lemma 3.1, one has $L_S(T_C) < 3/2$.*

Proof. Normalize the unique midpoint as M_0 and a positive boundary trace to $e_{0,1}$. Use Proposition A.11; thus

$$W = 1 - R, \qquad E = \sqrt{1 - RW}, \qquad \eta = 1 - E, \qquad P = E(1 - E),$$

and the positive trace condition is

$$\alpha + W\delta < P. \tag{49}$$

The two possible boundary contributions are

$$\ell_{01} = W + \delta - \frac{\eta + \alpha + \delta}{R}, \qquad \ell_{50} = \frac{[P - (R\alpha + \delta)]_+}{W}.$$

The six radial contributions are bounded by

i	0	1	2	3	4	5
$\mathcal{H}^1(T_C \cap r_i)$	$E - \alpha - \delta$	δ/R	δ	$\min\{\alpha/R, \delta/W\}$	α	$\alpha/W.$

Hence

$$L_S(T_C) \le K_R + \mathcal{E}(\alpha, \delta), \qquad K_R := W + E - \frac{\eta}{R},$$

where

$$\mathcal{E}(\alpha, \delta) := -\frac{\alpha}{R} + \frac{\alpha}{W} + \delta + \min\left\{\frac{\alpha}{R}, \frac{\delta}{W}\right\} + \frac{[P - (R\alpha + \delta)]_+}{W}.$$

Splitting according to the minimum and positive part gives

$$\mathcal{E}(\alpha, \delta) \le \frac{P}{W}. \tag{50}$$

If $\alpha/R \le \delta/W$, the expression without the positive part is $\alpha/W + \delta$; when that part is active, the excess over P/W is $\alpha - R\delta/W \le 0$. The reflected order gives $\delta - W\alpha/R \le 0$. If the positive part vanishes, (49) gives the same bound strictly.

Put $A := 1 + R - R^2$. Then

$$\frac{3}{2} - K_R - \frac{P}{W} = \frac{1 + A - 2EA}{2RW}.$$

Both sides of $1 + A > 2EA$ are positive, and

$$(1 + A)^2 - 4A^2E^2 = R^2W^2(5 + 4R - 4R^2) > 0.$$

Therefore $K_R + P/W < 3/2$, as required. □

Lemma B.2 (No-support skeleton calculation)**.** *The hypotheses of Lemma 3.3 imply $L_S(T_i) \le 2$.*

Proof. The boundary contribution is $A_i + B_i \le 1$ by (9). If $P = sV_j \in r_j \setminus \{O\}$ and $j \notin \{i-1, i, i+1\}$, then

$$\|V_i - P\|^2 = 1 + s^2 - 2s\cos\frac{(j - i)\pi}{3} > 1.$$

The adjacent radial traces have zero length by hypothesis, so only r_i can add positive trace, of length at most one. □

For the remaining cap, reflect the local directions if necessary so that $0 \le A_i \le B_i \le 1$. The actual reach triple of a supercritical V triangle lies in the selected cell

$$\begin{aligned} A_i^2 + A_iB_i + B_i^2 &\le 1, \\ C_i &\le \frac{1}{2}, \\ (A_i^2 - 1)C_i^2 + (2A_iB_i^2 + B_i)C_i + (B_i^4 - B_i^2) &\le 0, \end{aligned} \tag{51}$$

where C_i belongs to the connected component containing 0 in the corresponding one-dimensional fiber. This is Cell S of Proposition A.9. The selector is essential: the quadratic is negative at $C_i = 0$, positive at $C_i = 1/2$, and opens downward.

Lemma B.3 (Supercritical skeleton calculation)**.** *Every V triangle with $A_i + B_i > 1$ has $L_S(T_i) < 3/2$.*

Proof. The corner normal form, Proposition B.5, makes Vd1 and Vd2 triangles nonsupercritical. Lemma 2.10 gives the same conclusion for T3-like triangles. Hence a supercritical triangle is Vd0 and has no positive adjacent-radial support. Reflect so that $A_i \le B_i$. Then

$$L_S(T_i) = A_i + B_i + C_i.$$

Set

$$s := A_i + B_i, \qquad \delta := B_i - A_i, \qquad \gamma_0 := \frac{3}{2} - s.$$

The diameter constraint gives

$$1 < s \le \frac{2}{\sqrt{3}}, \qquad 0 \le \delta \le \sqrt{4 - 3s^2}.$$

Let $P(\gamma)$ be the quadratic in (51). Substitution and multiplication by 16 yield

$$\begin{aligned} 16P(\gamma_0) = Q(s,\delta) = {} & \delta^4 + 8\delta^3 s - 6\delta^3 + 14\delta^2 s^2 - 18\delta^2 s + 5\delta^2 \\ & - 8\delta s^3 + 30\delta s^2 - 34\delta s + 12\delta \\ & + s^4 - 6s^3 - 19s^2 + 60s - 36. \end{aligned}$$

Its derivative factors as

$$\frac{\partial Q}{\partial \delta} = 2(2\delta + 4s - 3)\big(\delta^2 + \delta(4s-3) + (s-1)(2-s)\big) \ge 0.$$

Therefore

$$Q(s,\delta) \ge Q(s,0) = (s-1)(s^3 - 5s^2 - 24s + 36) > 0.$$

The cubic decreases on $[1, 2/\sqrt{3}]$ and at the right endpoint equals $88/3 - 136\sqrt{3}/9 > 0$. Hence $P(\gamma_0) > 0$. The selected smaller-root component in (51) forces $C_i < \gamma_0$, so $A_i + B_i + C_i < 3/2$. □

Lemma B.4 (Common skeleton-budget calculation)**.** *Under the hypotheses of Theorem 3.6, the total skeleton trace of the seven triangles is strictly less than* 12*.*

Proof. The center cap and the three V triangle caps give

$$\begin{aligned} L_S(T_C) + \sum_{i=0}^{5} L_S(T_i) &< \frac{3}{2} + \frac{3}{2}(N_+ + N_{\mathrm{sp}}) + 2(6 - N_+ - N_{\mathrm{sp}}) \\ &= \frac{27 - N_+ - N_{\mathrm{sp}}}{2} \le 12. \end{aligned}$$

The two classes counted by N_+ and N_{sp} are disjoint: every Vd1, Vd2, or T3-like triangle has positive adjacent-radial support and is nonsupercritical, while every supercritical triangle is Vd0. □

B.2. **Vd1 and Vd2 corner optimization.** Use temporary coordinates

$$X = V_0 + x(V_5 - V_0) + y(V_1 - V_0),$$

so $V_0 = (0,0)$, $V_5 = (1,0)$, $V_1 = (0,1)$, $O = (1,1)$, and $\|(x,y)\|^2 = x^2 + y^2 - xy$.

Proposition B.5 (Vd1/Vd2 corner normal form)**.** *Let T be the closure of an original* Vd1 *or* Vd2 *triangle at V_0, and let A_0, B_0 be its exact reaches toward V_5, V_1. There is a unique $t > 0$ such that, with $d = \sqrt{t^2+t+1}$,*

$$T = \left\{ (x,y) : \begin{array}{r} x - (t+1)y \le A_0, \\ ty - (t+1)x \le tB_0, \\ tx + y \le d - A_0 - tB_0 \end{array} \right\}. \tag{52}$$

The raw traces, parametrized from the indicated outer vertex, are

$$\begin{array}{r|c} r_0 & 0 \le s \le \dfrac{d - A_0 - tB_0}{t+1} \\ r_1 & \dfrac{t(1-B_0)}{t+1} \le s \le \dfrac{d - A_0 - tB_0 - 1}{t} \\ r_5 & \dfrac{1-A_0}{t+1} \le s \le d - A_0 - tB_0 - t. \end{array} \tag{53}$$

Intersecting with $[0,1]$ gives the actual traces, and

$$A_0 + B_0 < \frac{1}{2}. \tag{54}$$

Proof. The triangle has one vertex W outside H and two vertices U, Z in H. Because V_0 is an interior convex combination, both coordinates of W are negative. The axis endpoints $(A_0, 0), (0, B_0)$ lie on the two sides through W. Put $p = U - W$ and $q = Z - W$. Their order selects

$$q = (p_x - p_y, p_x), \qquad p_x > p_y > 0.$$

Writing $t = (p_x - p_y)/p_y$ gives uniquely

$$p = \frac{(t+1, 1)}{d}, \qquad q = \frac{(t, t+1)}{d},$$

which yields (52); substitution of $(s,s), (s,1), (1,s)$ yields (53).

If r_1 has positive trace, clearing its endpoint inequality gives

$$(t+1)A_0 + tB_0 < (t+1)(d-1) - t^2.$$

Since the left side is at least $t(A_0 + B_0)$,

$$A_0 + B_0 < \frac{(t+1)(d-1) - t^2}{t} < \frac{1}{2}.$$

The last inequality follows from

$$\left(t^2 + \frac{3t}{2} + 1\right)^2 - (t+1)^2(t^2+t+1) = \frac{t^2}{4} > 0.$$

Reflection treats support on r_5. $\square$

Lemma B.6 (Vd2 adjacent-midpoint cap)**.** *If an original open V triangle has* Vd2 *closure T_i and*

$$\{M_{i-1}, M_{i+1}\} \cap T_i \neq \varnothing,$$

then its exact boundary reaches satisfy

$$A_i + B_i < \frac{1}{3}.$$

Proof. In Proposition B.5, reflection sends

$$(t, A_0, B_0, M_1, M_5) \longmapsto (1/t, B_0, A_0, M_5, M_1).$$

Thus assume $t \geq 1$ and use $A_0 + B_0 \leq A_0 + tB_0$.

If $M_5 = (1, 1/2)$ lies in T, the third half-plane gives

$$A_0 + B_0 \leq A_0 + tB_0 \leq d - t - \frac{1}{2} \leq \sqrt{3} - \frac{3}{2} < \frac{1}{3}.$$

If $M_1 = (1/2, 1)$ lies in T, the second and third half-planes give

$$B_0 \geq \frac{t-1}{2t}, \qquad A_0 + B_0 \leq S(t) := d - t - \frac{1}{2t}.$$

Positive-length contact with r_5 gives

$$A_0 + B_0 < R(t) := \frac{2(t+1)d - 3t^2 - t - 2}{2t}.$$

Direct differentiation shows S increasing and R decreasing on $[1, \infty)$. For S', the sign reduces to

$$(2t+1)^2 t^4 - (t^2 + t + 1)(2t^2 - 1)^2 = (t+1)(t^3 + 3t^2 - 1) > 0,$$

while $R' < 0$ follows from $d > t + 1/2$. Splitting at $t = 4/3$ gives

$$\begin{aligned} 1 \leq t \leq \frac{4}{3}: &\quad A_0 + B_0 \leq \frac{8\sqrt{37} - 41}{24} < \frac{1}{3}, \\ t \geq \frac{4}{3}: &\quad A_0 + B_0 < \frac{7\sqrt{37} - 39}{12} < \frac{1}{3}. \end{aligned}$$

The final comparisons are $8\sqrt{37} < 49$ and $7\sqrt{37} < 43$. □

B.3. **Boundary contributions.**

Lemma B.7 (Boundary-trace table calculation)**.** *All identities and bounds in Theorem 3.7 hold.*

Proof. The two incident traces of a V triangle are initial intervals of lengths A_i, B_i. A relative-interior point of a nonincident edge is more than unit distance from V_i, while $V_i \in \operatorname{int}(T_i)$. Hence no other edge has positive-length trace and

$$L_{\partial H}(T_i) = A_i + B_i.$$

The $N_E(T_C) = 0$ row is the definition. The other C triangle rows are Corollary A.13. If a Vd0 triangle is nonsupercritical, its boundary trace is at most 1. If it is supercritical, its two incident endpoints have squared distance

$$A_i^2 + A_i B_i + B_i^2 \leq 1,$$

and therefore

$$\frac{3}{4}(A_i + B_i)^2 \leq 1.$$

This gives $A_i + B_i \leq 2/\sqrt{3}$. The Vd1/Vd2 row is (54). For an original T3-like triangle, the Type-II calculation in theorem A.2 gives $A_i + B_i = z - \alpha < z < 1$ because $0 < t < 1$ and $\alpha > 0$. □

B.4. **Common perimeter budgets.**

Lemma B.8 (Master perimeter deficit). *Put*

$$\theta := \frac{2}{\sqrt{3}} - 1.$$

Suppose a branch has center contribution $L_C := L_{\partial H}(T_C)$, exactly n supercritical Vd0 *triangles, and $m \geq 1$ distinct nonsupercritical V triangles $T_{i_1}, \dots, T_{i_m}$ such that*

$$L_{\partial H}(T_{i_j}) < q_j < 1 \qquad (1 \leq j \leq m).$$

Suppose also that each of the remaining $6 - n - m$ V triangles contributes at most 1. If

$$L_C + n\theta \leq \sum_{j=1}^{m} (1 - q_j), \tag{55}$$

then the seven triangles do not cover ∂H.

Proof. The supercritical Vd0 cap is $2/\sqrt{3}$, so the total contribution is strictly less than

$$\begin{aligned} L_C + n\frac{2}{\sqrt{3}} + \sum_{j=1}^{m} q_j + (6 - n - m) &= 6 + L_C + n\theta - \sum_{j=1}^{m}(1 - q_j) \\ &\leq 6. \end{aligned}$$

The strictness comes from $m \geq 1$ and the distinguished strict caps, in contradiction with perimeter subadditivity. $\square$

Lemma B.9 (Small center slacks in the one-Vd branch). *Assume there is exactly one supercritical* Vd0 *triangle, exactly one* Vd1 *or* Vd2 *triangle, and four nonsupercritical* Vd0 *triangles. Every candidate surviving the perimeter budget is CE2 and satisfies*

$$\alpha + \delta < \frac{1}{24}, \qquad \alpha + \delta < \frac{\min\{R, W\}}{6}. \tag{56}$$

Proof. The exceptional V triangle has strict cap 1/2. Perimeter coverage and Lemma B.8 force

$$L_C + \theta > \frac{1}{2}.$$

Put $\kappa := 1/2 - \theta = 3/2 - 2/\sqrt{3}$. The CE1 cap gives

$$L_C + \theta \leq \left(\frac{\sqrt{3}}{2} - \frac{3}{4}\right) + \left(\frac{2}{\sqrt{3}} - 1\right) < \frac{1}{2},$$

so every survivor is CE2.

Set $T := \alpha + \delta$. The CE2 formula gives

$$T < M(E) := \frac{RW}{E^2}\left(\frac{E}{1+E} - \kappa\right) = \frac{1-E}{E^2}\big(E - \kappa(1+E)\big).$$

Since

$$M'(E) = -\frac{E - 2\kappa}{E^3} < 0 \qquad \left(\frac{\sqrt{3}}{2} \leq E < 1\right),$$

where $2\kappa < \sqrt{3}/2$, one has

$$T < M(\sqrt{3}/2) = \frac{8\sqrt{3}}{9} - \frac{3}{2} < \frac{1}{24}.$$

The last comparison follows from $64\sqrt{3} < 111$.

Also $E/(1+E) < 1/2$ gives

$$T < \frac{RW}{E^2}\theta.$$

Because $E^2 = W + R^2 = R + W^2$, the factor RW/E^2 is at most both R and W; and $\theta < 1/6$. This proves (56). □

Lemma B.10 (CE2 endpoints dominate total slack)**.** *Assume* $\Delta_R > 0$ *and* $\Delta_L > 0$. *Put*

$$T := \alpha + \delta, \qquad x := \frac{\eta + T}{W}, \qquad y := \frac{\eta + T}{R}.$$

Then

$$T < \frac{1}{2}\min\{x, y\}. \tag{57}$$

Proof. Equation (47) gives $T < \eta/E$. Since

$$E^2 - (2R-1)^2 = 3RW > 0,$$

one has $|2R-1|T < \eta$. Therefore

$$x - 2T = \frac{\eta - (1-2R)T}{W} > 0, \qquad y - 2T = \frac{\eta - (2R-1)T}{R} > 0.$$

□

Lemma B.11 (Perimeter-budget substitutions)**.** *Items (1)–(5) of Corollary 3.8 are impossible.*

Proof. For item (1), use $L_C = 0$, one supercritical cap $2/\sqrt{3}$, and one strict Vd1/Vd2 cap $1/2$ in Lemma B.8; the required comparison is

$$\frac{2}{\sqrt{3}} - 1 < \frac{1}{2}.$$

For item (2), $L_C < 1/2$, the distinguished V triangle has strict cap $1/2$, and the remaining five contribute at most 1.

For item (3), the CE1 cap, the exceptional cap, the supercritical cap, and four unit caps give

$$\left(\frac{\sqrt{3}}{2} - \frac{3}{4}\right) + \frac{1}{2} + \frac{2}{\sqrt{3}} + 4 < 6.$$

For item (4), the CE2 sum is strictly less than

$$\frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{2}{\sqrt{3}} + 3 < 6.$$

For item (5), Lemma B.6 gives the strict $1/3$ cap, and the total is strictly less than

$$\frac{1}{2} + \frac{1}{3} + \frac{2}{\sqrt{3}} + 4 < 6.$$

Each inequality contradicts the length-6 perimeter requirement. □

Appendix C. Area-Loss Optimization

This appendix evaluates the three area optimizations used in section 4. The body fixes the feasible triangles, the loss functional, and the strict cyclic handoffs; here we introduce coordinates only to solve those problems.

C.1. **Wedge chart.**

Lemma C.1 (Local wedge reduction)**.** *After rotating indices to $i = 0$, write*

$$X = V_0 + x(V_1 - V_0) + y(V_5 - V_0), \qquad W = \{(x, y) : x \geq 0,\ y \geq 0\}.$$

If a closed unit equilateral triangle T contains V_0, then

$$T \cap H = T \cap W.$$

Normalized Euclidean area is twice the corresponding (x, y)-coordinate area.

Proof. The two basis vectors have length one and angle $120°$, so

$$\|(x, y)\|^2 = x^2 + y^2 - xy, \qquad |\det(V_1 - V_0, V_5 - V_0)| = \frac{\sqrt{3}}{2}.$$

The hexagon is

$$H = \{0 \leq x, y \leq 2,\ |x - y| \leq 1\} \subset W.$$

Conversely, if $(x, y) \in T \cap W$, then the diameter-one condition gives $x^2 + y^2 - xy \leq 1$. Hence

$$|x - y|^2 \leq x^2 + y^2 - xy \leq 1, \qquad \frac{3}{4}x^2 \leq x^2 + y^2 - xy,$$

and the same estimate holds for y. Thus $|x - y| \leq 1$ and $0 \leq x, y \leq 2/\sqrt{3} < 2$, proving $(x, y) \in H$. The determinant identity proves the area assertion. □

C.2. **Local square-loss optimization.**

Lemma C.2 (Solved local square-loss optimization)**.** *For the feasible family $\mathcal{T}(a, b)$ of section 4, the minimum loss satisfies equations (10) and (11).*

Proof. In the coordinates of Lemma C.1, physical rotation through $60°$ is $\mathsf{R}(x, y) = (x - y, x)$. Put

$$0 \leq t \leq 1, \qquad D = 1 - t + t^2, \qquad z = \sqrt{D}.$$

After choosing one of the six directed sides, every orientation has one of the following two forms:

$$\begin{array}{c|cc} & d & \mathsf{R}d \\ \hline \mathrm{I} & z^{-1}(1, t) & z^{-1}(1 - t, 1) \\ \mathrm{II} & z^{-1}(t, -(1 - t)) & z^{-1}(1, t). \end{array} \tag{58}$$

Indeed these forms cover, respectively, the physical direction sectors $[120°, 180°]$ and $[60°, 120°]$; their endpoints cover the common boundary orientations.

For Type I introduce

$$U = \alpha + y - tx, \qquad V = \beta + x - (1 - t)y.$$

Then T is the simplex $U, V \geq 0$, $U + V \leq z$. Containment of $(0, 0), (0, a), (b, 0)$ gives

$$\alpha \geq tb, \quad \beta \geq (1 - t)a, \quad \alpha + \beta + (1 - t)b \leq z, \quad \alpha + \beta + ta \leq z. \tag{59}$$

The two inequalities defining W become

$$(1 - t)U + V \geq (1 - t)\alpha + \beta, \qquad U + tV \geq \alpha + t\beta.$$

For fixed t, lowering α or β therefore enlarges $T \cap W$. The outside loss is minimized at $\alpha = tb$, $\beta = (1 - t)a$. At that placement the portion outside W has vertices, in the (U, V)-plane,

$$(0, 0), \quad (a + tb, 0), \quad (tb, (1 - t)a), \quad (0, b + (1 - t)a).$$

Since the Jacobian of $(x, y) \mapsto (U, V)$ has absolute value D,

$$\mathcal{L}(T) \geq \mathcal{L}_{\mathrm{I}}(a, b; t) = \frac{(1-t)a^2 + tb^2 + 2t(1-t)ab}{D}. \tag{60}$$

If $a \leq b$, then

$$D(\mathcal{L}_{\mathrm{I}} - a^2) = t\{b^2 - a^2 + (1-t)(a^2 + 2ab)\} \geq 0;$$

the reflected factorization proves $\mathcal{L}_{\mathrm{I}} \geq b^2$ when $b \leq a$. This proves (10) in Type I.

Suppose now that $a + b > 1$ and $a \leq b$. Set

$$r = \frac{a}{b}, \qquad t_0 = \frac{1 - r^2}{1 + 2r}.$$

Since $b > 1/(1 + r)$, feasibility in (60) and (59) excludes $0 \leq t < t_0$: for $0 < t < t_0$,

$$\left(\frac{1 + r(1-t)}{1 + r}\right)^2 - D = \frac{t(1 + 2r)(t_0 - t)}{(1 + r)^2} > 0,$$

so $b + (1-t)a > z$, while $t = 0$ would give $a + b \leq 1$. Hence $t \geq t_0$, and

$$D(\mathcal{L}_{\mathrm{I}} - b^2) = (1-t)b^2\{r^2 - 1 + t(2r + 1)\} \geq 0.$$

If $b \leq a$, put $r = b/a$ and $t_1 = (r^2 + 2r)/(1 + 2r)$. The identical reflected calculation excludes $t > t_1$ and gives

$$D(\mathcal{L}_{\mathrm{I}} - a^2) = ta^2\{r^2 + 2r - t(1 + 2r)\} \geq 0.$$

Thus (11) holds in Type I.

For Type II put

$$U = \alpha + (1-t)x + ty, \qquad V = \beta + tx - y.$$

Again T is the simplex $U, V \geq 0$, $U + V \leq z$. Its exact reaches on the positive y- and x-axes are

$$A = \beta, \qquad B = z - \alpha - \beta, \qquad A + B \leq z \leq 1. \tag{61}$$

In particular Type II is impossible when $a + b > 1$. Its wedge intersection is the quadrilateral with vertices

$$(0, 0), \quad (B, 0), \quad \left(\frac{B + (1-t)A}{D}, \frac{A + tB}{D}\right), \quad (0, A).$$

Hence

$$\mathcal{L}(T) = \mathcal{L}_{\mathrm{II}}(A, B; t) = 1 - \frac{(1-t)A^2 + tB^2 + 2AB}{D}. \tag{62}$$

Let $S = A + B$. If $A \leq B$, write $A = rS$, $B = (1-r)S$ with $0 \leq r \leq 1/2$. Since $S^2 \leq D$, and with

$$C = (1-t)r^2 + t(1-r)^2 + 2r(1-r),$$

we have $\mathcal{L}_{\mathrm{II}} \geq 1 - C$. The exact factorization

$$1 - C - r^2 D = (1-t)(1 - 2r + tr^2) \geq 0$$

gives $\mathcal{L}_{\mathrm{II}} \geq r^2 D \geq A^2$. Reflection gives $\mathcal{L}_{\mathrm{II}} \geq B^2$ when $B \leq A$. Since $A \geq a$ and $B \geq b$, this proves (10). The two orientation types are exhaustive, so the theorem follows. $\square$

C.3. T3-like **limiting optimization.**

Lemma C.3 (Solved T3-like loss optimization)**.** *Under the hypotheses of theorem 4.2, the actual V triangle satisfies equation (13).*

Proof. The first assertion is Lemma 2.10. For the loss bound apply the closed-trace domination in Proposition 2.7. It gives a translated Type-II triangle $\widehat{T}$ of the same area with

$$T \cap H \subset \widehat{T} \cap H, \qquad \mathcal{L}(T) \geq \mathcal{L}(\widehat{T}).$$

In the notation of that proposition, $0 < t < 1$, $D = 1 - t + t^2$, $z = \sqrt{D}$, and after reflection the translated majorant has exact reaches

$$A = \beta, \qquad B = z - \beta$$

and unique wedge vertex

$$Q = \left(\frac{B + (1-t)A}{D}, \frac{A + tB}{D} \right), \qquad Q_x > 1, \quad Q_y \leq 1.$$

Thus $z - tA > D$, so

$$A < L := \frac{z - D}{t} = \frac{z(1-z)}{t}. \tag{63}$$

The exact normalized loss is

$$\mathcal{L}(\widehat{T}) = \mathcal{L}(z, A) = \frac{tA^2 + (1-t)(z-A)^2}{D}.$$

This decreases for $A < L < (1-t)z$, and therefore

$$\mathcal{L}(T) \geq \mathcal{L}(\widehat{T}) > \mathcal{L}(z, L). \tag{64}$$

Set $r = (1-z)/t$. Then $0 < r < 1/2$, and eliminating t, z gives

$$t = \frac{1-2r}{1-r^2}, \qquad z = \frac{r^2 - r + 1}{1 - r^2},$$

$$L = \alpha_{\lim}(r) := \frac{r(r^2 - r + 1)}{1 - r^2}, \qquad z - L = \beta_{\lim}(r) := \frac{r^2 - r + 1}{1 + r}.$$

The limiting normalized inside area and loss are

$$F_0(r) = \frac{(r^2 - r + 1)^2}{1 - r^2}, \qquad \mathcal{L}_0(r) = 1 - F_0(r).$$

Since the exact reach A is at least the required reach lower bound $a \geq m$, (63) gives $\alpha_{\lim}(r) > m$. Direct simplification yields

$$\mathcal{L}_0(r) - \{2\alpha_{\lim}(r) - 4\alpha_{\lim}(r)^2\} = \frac{r^2(5r^4 - 8r^3 + 13r^2 - 8r + 2)}{(1-r^2)^2} \geq 0,$$

because the last polynomial is at least $9r^2 - 8r + 2 \geq 2/9$ on $[0, 1/2]$. Also $\alpha_{\lim}(r) \leq r$, and

$$\mathcal{L}_0(r) - \frac{1}{4} = \frac{(1-2r)(2r^3 - 3r^2 + 6r - 1)}{4(1-r^2)} \geq 0 \quad \text{when } \alpha_{\lim}(r) \geq \frac{1}{4};$$

the cubic is increasing and has value $11/32$ at $r = 1/4$.

Finally $\psi(u) = 2u - 4u^2$ is increasing on $[0, 1/4]$ and never exceeds $1/4$ on $[0, 1/2]$. If $\alpha_{\lim}(r) \leq 1/4$, then $\mathcal{L}_0(r) \geq \psi(\alpha_{\lim}(r)) \geq \psi(m)$; if $\alpha_{\lim}(r) \geq 1/4$, then $\mathcal{L}_0(r) \geq 1/4 \geq \psi(m)$. Together with (64), this proves (13). The reflected crossing is identical. ∎

C.4. **Cyclic loss optimization.** The variables ξ_i, a_i, b_i are the strict handoffs already fixed in section 4; in particular, no endpoint selection is relaxed here.

Lemma C.4 (Solved cyclic area-loss optimization)**.** *Both implications in theorem 4.3 hold.*

Proof. Put $m = \min_i \xi_i$ and $M = \max_i \xi_i$. The reflection $y_i = 1 - \xi_{-i-1}$ swaps each V triangle's two coordinates and preserves both alternatives and the total loss. Hence assume $m \le 1 - M$. Every selected V triangle coordinate is then at least m, so (10) gives loss at least m^2 for every V triangle.

In case (1), choose an ascent out of a minimum plateau, $\xi_{p-1} = m < \xi_p$. Its V triangle contains the coordinate $1 - m$ and loses at least $(1-m)^2$ by (11). A second ascent has one coordinate strictly larger than $1/2$ and hence loses strictly more than $1/4$. The other four V triangles lose at least m^2 each, so the total loss is strictly larger than

$$4m^2 + (1-m)^2 + \frac{1}{4} = 5\left(m - \frac{1}{5}\right)^2 + \frac{21}{20} > 1.$$

In case (2), rotate so that the unique ascent leaves the minimum: $\xi_5 = m < \xi_0 = M$. The supercritical V triangle loses at least $(1-m)^2$. Lemma 2.10 shows that a T3-like V triangle is different from it, and Theorem 4.2 gives that V triangle loss at least $2m - 4m^2$. The remaining four V triangles lose at least m^2 each. Their total loss is at least

$$(1-m)^2 + (2m - 4m^2) + 4m^2 = 1 + m^2 > 1.$$

□

Appendix D. Nonzero-Gap Finite-Enclosure Optimization

This appendix solves the point, radial, disk, and interval calculations used by the nonzero-gap rows of table 2. The geometric witnesses and their case predicates remain in the body.

D.1. **Exact gap endpoints.**

Lemma D.1 (Actual open-trace endpoints)**.** *For every actual gap J_i,*

$$J_i = X_i([B_i, 1 - A_{i+1}]).$$

The formula includes the singleton case $B_i = 1 - A_{i+1}$.

Proof. The definitions of A_{i+1} and B_i, together with convexity, give the closed traces $X_i([0, B_i])$ and $X_i([1 - A_{i+1}, 1])$. Because U_i, U_{i+1} are open, their edge traces are respectively $X_i([0, B_i))$ and $X_i((1 - A_{i+1}, 1])$. Taking the set difference in the definition of J_i gives the displayed closed interval, including both endpoints. □

D.2. **Common-pair reach adapters.**

Lemma D.2 (Solved one-gap common-pair adapter)**.** *The reaches defined from row A in theorem 5.5 satisfy its displayed coordinatewise alternatives.*

Proof. On the five gap-free edges choose

$$x_i \in (1 - A_{i+1}, B_i) \qquad (1 \le i \le 5).$$

Since $N_+ = 0$, one has $B_i \le 1 - A_i$ for every i. The exact gap endpoints and the five handoffs therefore give

$$\ell \le x_5 \le x_4 \le x_3 \le x_2 \le x_1 \le r.$$

With $p = 1 - r = A_1$ and $q = \ell = B_0$, this yields

$$p \le A_i, \qquad q \le B_i \qquad (i \in \mathbb{Z}/6\mathbb{Z}).$$

This is stronger than the alternative asserted in the body and retains the singleton case $\ell = r$. $\square$

Lemma D.3 (Solved two-gap common-pair adapter)**.** *The reaches defined from row B in theorem 5.7 satisfy its displayed coordinatewise alternatives.*

Proof. In the signed CE2 chart of Appendix A, the complementary far endpoints of the two positive C traces are

$$p = 1 - (R + \alpha) = W - \alpha, \qquad q = 1 - (W + \delta) = R - \delta.$$

The row predicates leave exactly

$$(N_+, t) \in \{(0,0), (0,1), (0,2), (1,0)\}.$$

Normalize the gap edges to $e_{5,0}$ and $e_{0,1}$. On the four intervening edges choose

$$x_i \in (1 - A_{i+1}, B_i) \qquad (1 \le i \le 4).$$

If $N_+ = 0$, then $T_1, \ldots, T_5$ are nonsupercritical immediately. In the remaining case $(N_+, t) = (1, 0)$, let T_s be supercritical. If $s \ne 0$, then theorem 2.12 and the unique-center-midpoint identity give

$$M_s \notin T_s \cup T_C.$$

Any other open V triangle covering M_s would be adjacent and would have positive-length trace on r_s, contrary to $d = t = 0$. Thus $s = 0$, and $T_1, \ldots, T_5$ are again nonsupercritical. Gap containment and the displayed handoffs now give

$$\begin{aligned} p < B_5 \le 1 - A_5 < x_4, \\ x_4 < B_4 \le 1 - A_4 < x_3, \\ x_3 < B_3 \le 1 - A_3 < x_2, \\ x_2 < B_2 \le 1 - A_2 < x_1, \\ x_1 < B_1 \le 1 - A_1 < 1 - q. \end{aligned}$$

Thus

$$p < x_4 < x_3 < x_2 < x_1 < 1 - q,$$

and, including the two roles incident to the gap edges,

$$q \le A_i, \qquad p \le B_i \qquad (1 \le i \le 5).$$

Finally, if T_j is T3-like and supports r_{j-1} or r_{j+1}, then $t \ge 1$. The row-B identity $N_+ t = 0$ gives $N_+ = 0$, and hence

$$A_j + B_j \le 1.$$

Coordinatewise antitonicity and reflection give

$$C_\pm(A_j, B_j) \le \max\{C_+(p,q), C_-(p,q)\} \le c_{\max}(p,q)$$

by theorem D.8. This proves both assertions. $\square$

D.3. **Vd1 two-chart replacement.** The replacement below uses two different vertex charts. It preserves the five skeleton pieces that can change, then recomputes the output gap rank from the new open traces.

Lemma D.4 (Half-square admissibility)**.** *If $(x, y, z) \in \mathcal{A}$ satisfies*

$$x \geq \frac{1}{2}, \qquad 0 \leq z \leq \frac{1}{2},$$

then

$$y \leq 1 - z.$$

Proof. Suppose $y > 1 - z$. Then $x + y > 1$, so the selected supercritical cell of theorem A.9 applies. In the ordered half $x \leq y$ its necessary polynomial is

$$F(x, y, z) = (x^2 - 1)z^2 + (2xy^2 + y)z + y^4 - y^2 \leq 0.$$

It is nondecreasing in x, its derivative in y is positive for $y \geq 1 - z$, and

$$F\left(\frac{1}{2}, 1 - z, z\right) = \frac{z^2(2z - 5)(2z - 1)}{4} \geq 0.$$

This is a contradiction. In the reflected half $y < x$, apply the same argument to $F(y, x, z)$. At the joining corner,

$$F(1 - z, 1 - z, z) = z(1 - 2z) \geq 0,$$

so no selected feasible point is lost at equality. □

Proposition D.5 (Vd1 two-chart replacement and router)**.** *Assume the normalized replacement hypotheses in the proof of theorem 5.15: T_0 is Vd1, T_1 is the unique supercritical Vd0 role, $M_1 \in T_0$, and only T_0, T_1 have positive trace on r_0 or r_1. Then there are two open unit equilateral triangles U_0', U_1' that preserve the covered skeleton, make all six V roles nonsupercritical Vd0, and route by the recomputed $N'_{\mathrm{gap}} \in \{0, 1, 2\}$.*

Proof. Let A_0, B_0, C_0 be the actual incoming, outgoing, and own-radial reaches of the Vd1 role. Its corner normal form supplies $\vartheta \geq 1$ and

$$\omega := \sqrt{\vartheta^2 + \vartheta + 1}$$

such that

$$C_0 = \frac{\omega - A_0 - \vartheta B_0}{\vartheta + 1}, \qquad \lambda = \frac{\vartheta(1 - B_0)}{\vartheta + 1},$$

$$u_{\mathrm{adj}} = \frac{\omega - A_0 - \vartheta B_0 - 1}{\vartheta}.$$

The strict midpoint and one-sided-support conditions give

$$A_0 + C_0 < 1, \qquad A_0 < \lambda \leq \frac{1}{2},$$

$$B_0 < \frac{1}{2}, \qquad u_{\mathrm{adj}} < 1 - A_0.$$

For the first inequality, evaluation on the closed midpoint face reduces strictness to

$$(2\vartheta^2 + 4\vartheta + 1)^2 - 4(\vartheta + 1)^2(\vartheta^2 + \vartheta + 1) = 4\vartheta^3 + 4\vartheta^2 - 4\vartheta - 3 > 0.$$

The remaining inequalities follow directly from the displayed endpoint formulas and the strict midpoint inequalities.

The center-free shared edge and the supported interval give

$$A_1 \geq 1 - B_0 > \frac{1}{2}, \qquad C_1 \geq \lambda$$

for the actual reaches of T_1. The selected supercritical component has $C_1 \leq 1/2$, so theorem D.4 yields

$$B_1 \leq 1 - C_1 \leq 1 - \lambda.$$

Since $A_0 < \lambda$,

$$A_0 + B_1 < 1.$$

Define the C-triangle reach on r_1, measured from O, by

$$d_1^C := \max\{x \in [0,1] : xV_1 \in T_C\}, \qquad c_1^{\mathrm{req}} := 1 - d_1^C.$$

Only U_C, U_1, and the adjacent trace of U_0 can contribute a positive interval to r_1. Open skeleton coverage gives

$$c_1^{\mathrm{req}} < \max\{C_1, u_{\mathrm{adj}}\} < \max\{1 - B_1, 1 - A_0\}.$$

Put $L := 1 - B_1$. Since $A_0 < L$, choose

$$A_0 < p_1 < p_2 < 1 - B_1$$

so that

$$p_1 < \frac{1}{2}, \qquad 1 - p_1 > C_0,$$
$$\max\{p_2, 1 - p_2\} > c_1^{\mathrm{req}}.$$

This is possible because the supremum of $p \mapsto \max\{p, 1-p\}$ on (A_0, L) is $\max\{L, 1-A_0\}$. Choose

$$0 < \varepsilon < \min\left\{p_1 - A_0,\, p_2 - p_1,\, 1 - B_1 - p_2,\, 1 - p_1 - C_0,\, \max\{p_2, 1 - p_2\} - c_1^{\mathrm{req}}\right\}.$$

Every margin is strictly positive.

Use the distinct charts

$$X_0(x, y) = V_0 + x(V_5 - V_0) + y(V_1 - V_0),$$
$$X_1(x, y) = V_1 + x(V_0 - V_1) + y(V_2 - V_1).$$

Both have metric $x^2 + y^2 - xy$. For $0 \leq p \leq 1/2$, set

$$\Delta_p^- = \mathrm{conv}\{(0, 1-p), (1, 1-p), (0, -p)\},$$

and for $1/2 < p \leq 1$, set

$$\Delta_p^+ = \mathrm{conv}\{(p, 0), (p, 1), (p-1, 0)\}.$$

For $0 < \varepsilon < p$, the shifted open minus template

$$D_{p,\varepsilon}^- := \mathrm{int}(\Delta_p^-) + (-\varepsilon, 0)$$

has actual reaches

$$(p - \varepsilon,\, 1 - p,\, 1 - p)$$

on the two positive axes and the radial diagonal. For $0 < \varepsilon < 1 - p$, the shifted open plus template

$$D_{p,\varepsilon}^+ := \mathrm{int}(\Delta_p^+) + (0, -\varepsilon)$$

has reaches

$$(p,\, 1 - p - \varepsilon,\, p).$$

Substitution in their three side half-planes also shows that both contain the chart origin and have no positive neighboring support.

Now define

$$U_0' := X_0(D^-_{p_1,\varepsilon}),$$

$$U_1' := \begin{cases} X_1(D^-_{p_2,\varepsilon}), & p_2 \le 1/2, \\ X_1(D^+_{p_2,\varepsilon}), & p_2 > 1/2. \end{cases}$$

These are the two required charts; the split at $p_2 = 1/2$ is part of the construction. Each new boundary-reach sum is $1-\varepsilon < 1$, and neither triangle has positive neighboring support. Hence both outputs are nonsupercritical Vd0.

Only

$$e_{5,0}, \quad e_{0,1}, \quad e_{1,2}, \quad r_0, \quad r_1$$

can change. On $e_{5,0}$, the new reach $p_1 - \varepsilon > A_0$ contains the former trace. On r_0, $1 - p_1 > C_0$. On the shared edge,

$$(1-p_1) + (p_2 - \varepsilon) = 1 + (p_2 - p_1 - \varepsilon) > 1,$$

so the two new open traces overlap. On $e_{1,2}$, the V_1-chart reach is greater than B_1 in both the minus and plus branches. On r_1, its reach $\max\{p_2, 1-p_2\}$ is greater than c_1^{req}. Thus all five affected pieces, and hence the full skeleton, remain covered.

Set $U_i' := U_i$ for $2 \le i \le 5$ and $T_i' := \overline{U_i'}$ for $0 \le i \le 5$. Then

$$A(T_i') := \max\{x \in [0,1] : V_i + x(V_{i-1} - V_i) \in T_i'\},$$
$$B(T_i') := \max\{x \in [0,1] : V_i + x(V_{i+1} - V_i) \in T_i'\},$$
$$C(T_i') := \max\{x \in [0,1] : V_i + x(O - V_i) \in T_i'\}.$$
$$(o(T_i'), n(T_i')) \in \{(1,0),(2,0)\} \qquad (0 \le i \le 5),$$
$$A(T_i') + B(T_i') \le 1 \qquad (0 \le i \le 5),$$
$$A(T_0') + B(T_0') = A(T_1') + B(T_1') = 1 - \varepsilon < 1.$$
$$|\{i \in \mathbb{Z}/6\mathbb{Z} : A(T_i') + B(T_i') > 1\}| = 0.$$

For $2 \le i \le 5$, the original actual triples are unchanged:

$$\big(A(T_i'), B(T_i'), C(T_i')\big) = (A_i, B_i, C_i).$$

Finally recompute the output gap rank from the modified open traces:

$$N'_{\mathrm{gap}} := \left|\left\{i \in \mathbb{Z}/6\mathbb{Z} : e_{i,i+1} \setminus \big((U_i' \cap e_{i,i+1}) \cup (U_{i+1}' \cap e_{i,i+1})\big) \ne \varnothing\right\}\right|.$$

Then $N'_{\mathrm{gap}} \in \{0,1,2\}$. No equality with the input gap rank is used. If $N'_{\mathrm{gap}} = 0$, apply the boundary-complete length row of theorem 3.10. If $N'_{\mathrm{gap}} = 1$, the primed actual reaches satisfy theorem 5.5, so the row-A branch of theorem 5.9 applies. If $N'_{\mathrm{gap}} = 2$, the primed actual reaches satisfy theorem 5.7, so the row-B branch of the same proposition applies. These alternatives are exhaustive. □

D.4. **Neighboring-arm optimization.** Normalize at V_0. Let $A_a \in e_{5,0}$ and $B_b \in e_{0,1}$ have reaches a, b, and put

$$X_c = (1-c)V_1 \in r_1.$$

Thus $C_+(a,b)$ is the maximal c for which V_0, A_a, B_b, X_c lie in a closed unit equilateral triangle, and $C_-(a,b) = C_+(b,a)$.

For $0 \le a \le 1/2$, let $p(a)$ be the unique root in $[a,1]$ of

$$p^3 - (a+2)p^2 + 2(a+1)p - 1 = 0, \tag{65}$$

and define

$$\sigma(a) = 1 - p(a), \qquad \tau(a) = 1 - a - (p(a) - a)(1 - p(a)).$$

Proposition D.6 (Neighboring-ray formula)**.**

$$C_+(a,b) = \begin{cases} \textit{undefined}, & a+b > 1, \\ 1-b, & a+b \le 1,\ a > 1/2, \\ 1-b, & 0 \le a \le 1/2,\ 0 \le b \le \sigma(a), \\ p(a), & 0 \le a \le 1/2,\ \sigma(a) \le b \le \tau(a), \\ a + \dfrac{1}{2} - \sqrt{(a+b)^2 - \dfrac{3}{4}}, & 0 \le a \le 1/2,\ \tau(a) < b \le 1-a. \end{cases} \tag{66}$$

At $b = \tau(a)$ the plateau value $p(a)$ is selected.

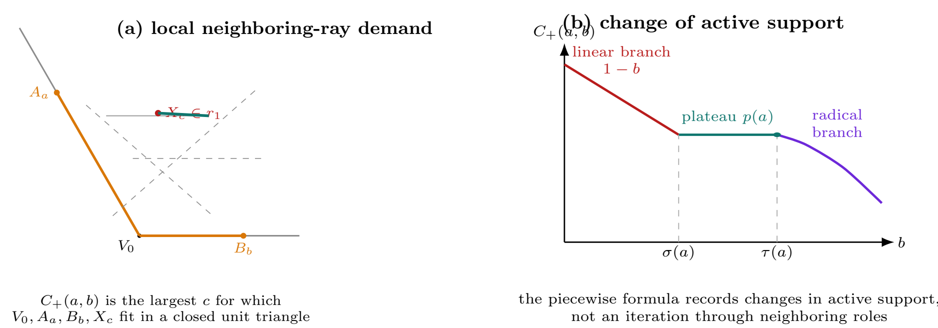


FIGURE 20. Geometry behind the neighboring-ray capacity $C_+(a,b)$. The local problem asks how far a triangle anchored at V_0 and the two boundary points A_a, B_b can reach on the neighboring arm r_1. As b varies with a fixed, the active support pattern changes from the linear branch to a plateau and then to the radical branch, with transition points $\sigma(a)$ and $\tau(a)$. The graph is schematic; the transition order, endpoint selection, and branch labels are exact.

Proof. Use the cone basis

$$E_- = V_5 - V_0, \qquad E_+ = V_1 - V_0.$$

In these coordinates

$$V_0 = (0,0), \quad A = (a,0), \quad B = (0,b), \quad X = (c,1),$$

and the metric is $u^2 + v^2 - uv$.

Let R, S satisfy $R^2 + S^2 - RS = 1$, and define the support coordinates

$$U(u,v) = (R-S)u + Sv,$$
$$V(u,v) = Su - Rv, \qquad W(u,v) = Ru + (S-R)v.$$

For this orientation, the least enclosing equilateral side is

$$\max_Y W(Y) - \min_Y U(Y) - \min_Y V(Y),$$

where Y ranges over V_0, A, B, X. At a maximal feasible c, one support side contains a hull edge. Put $\mathcal{Y} := \{V_0, A, B, X\}$. The non-dominated possibilities are

support predicate	candidate
$\{V_0, A\} \subseteq \operatorname{argmin}_{Y\in\mathcal{Y}} U(Y)$ $\{B, X\} \subseteq \operatorname{argmin}_{Y\in\mathcal{Y}} V(Y)$	$c = 1 - b$
$\{A, X\} \subseteq \operatorname{argmin}_{Y\in\mathcal{Y}} V(Y)$ $B \notin \operatorname{argmin}_{Y\in\mathcal{Y}} U(Y)$	$c = p(a)$
$\{A, X\} \subseteq \operatorname{argmin}_{Y\in\mathcal{Y}} V(Y)$ $B \in \operatorname{argmin}_{Y\in\mathcal{Y}} U(Y)$	$c = a + \frac{1}{2} - \sqrt{(a+b)^2 - \frac{3}{4}}$

We verify the three patterns and their selectors.

For the first pattern take $R = S = 1$. The containing triangle is

$$U \geq 0, \qquad V \geq -b, \qquad W \leq c,$$

with side $b + c$. It contains A exactly when $a \leq c$. Hence the unit value is $c = 1 - b$, feasible precisely when $a + b \leq 1$.

For the second pattern put $d = c - a$. The side through A, X gives $R = Sd$, and the unit condition gives $S^2(d^2 - d + 1) = 1$. With B inactive, the opposite support is $W(V_0) = 0$ and $W(X) = -1$. Eliminating S gives

$$(c-a)\big(c^3 - (a+2)c^2 + 2(a+1)c - 1\big) = 0.$$

The nondegenerate root is $c = p(a)$. For $0 \leq a \leq 1/2$ it is unique because the cubic is strictly increasing on $[a, 1]$. The remaining containment condition is

$$b \leq 1 - p(a)(1 + a - p(a)) = \tau(a).$$

This branch dominates $1 - b$ exactly when $b \geq \sigma(a)$.

For the third pattern the active supports give

$$S = -\frac{1}{a+b}, \qquad d^2 - d + 1 = (a+b)^2.$$

The selected root $0 \leq d \leq 1/2$ is

$$d = \frac{1 - \sqrt{4(a+b)^2 - 3}}{2},$$

which gives the radical branch of equation (66). The inactive support inequalities reduce to

$$0 \leq a \leq \frac{1}{2}, \qquad \tau(a) \leq b \leq 1 - a.$$

On this interval the radical candidate dominates $1-b$ because, with $s=a+b\le 1$,

$$s-\frac{1}{2}\ge\sqrt{s^2-\frac{3}{4}}.$$

The three selected intervals are ordered since

$$\tau(a)-\sigma(a)=(p(a)-a)p(a)\ge 0,\qquad \tau(a)\le 1-a.$$

If $a+b>1$, none of the three support patterns is feasible, including $c=0$, so the capacity is undefined. Reflection gives C_-. □

A consequence used repeatedly below is

$$\boxed{C_+(p,q),C_-(p,q)\le 1-\min\{p,q\}}\qquad (p+q\le 1).$$

Indeed this is immediate on the linear branch; on the plateau branch the cubic gives $p(a)\le 1-a$, and the radical branch is no larger than its selected plateau at the transition.

Lemma D.7 (Type-aware radial calculation)**.** *The points $D_i=(1-\Gamma_i)V_i$ of theorem 5.2 are missed by every open V role.*

Proof. Put $D_i=dV_i$, so $1-d=\Gamma_i$. If $D_i\in U_i$, openness of the radial trace makes the actual radial reach strictly larger than $c_{\max}(a_i,b_i)$, contradicting the defining maximum. If $D_i\in U_{i-1}$ or $D_i\in U_{i+1}$, that role must have positive trace on r_i; the applicable term C_+ or C_- is therefore present in Γ_i, and the same open-endpoint argument is a contradiction. The actual-type condition is essential here: no neighboring capacity is used when the corresponding positive trace is forbidden.

For the remaining roles, rotation reduces the distances to

$$\|dV_i-V_{i\pm 2}\|^2=1+d+d^2>1,\qquad \|dV_i-V_{i+3}\|=1+d>1.$$

Their diameter is one, so they cannot contain D_i. □

Lemma D.8 (Solved common-pair comparison)**.** *The inequalities in theorem 5.3 hold.*

Proof. Assume first that $q=\min\{p,q\}$. The demand $(p,q,1-q)$ is admissible: in the edge-normal realization the active side has length

$$q+\max\{p,1-q\}=1.$$

Thus $c_{\max}(p,q)\ge 1-q$. In the three selected branches of theorem D.6, the linear value is $1-q$; the selected cubic value is at most $1-q$; and the radical branch decreases from that cubic transition. Hence $C_+(p,q)\le 1-q$. Reflection gives $C_-(p,q)\le 1-q$ and treats $p\le q$. Coordinatewise antitonicity follows because increasing either anchor replaces the feasible family by a subfamily. □

D.5. **Disk and complementary-gap optimization.** Put $\mathcal{D}_\eta:=\mathbb{B}(O,\eta)$.

Lemma D.9 (Disk-plus-point formula)**.** *If $\|G\|=\rho$, then*

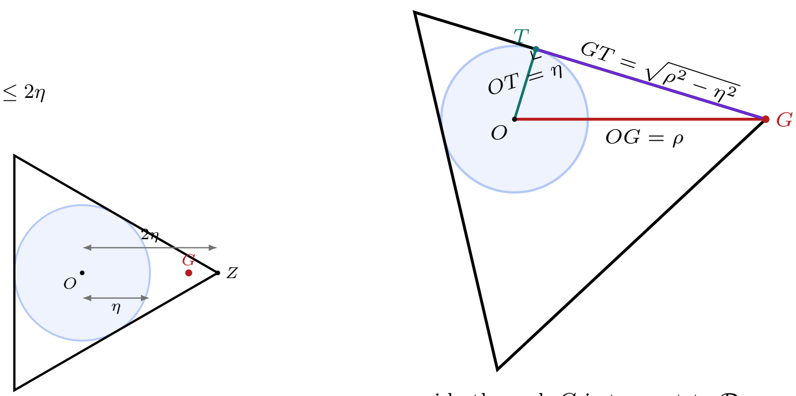
$$\Lambda(\mathcal{D}_\eta\cup\{G\})=\begin{cases}3\eta/h, & \rho\le 2\eta,\\[4pt] \dfrac{3\eta+\sqrt{3(\rho^2-\eta^2)}}{2h}, & \rho>2\eta.\end{cases}$$

Proof. Apply theorem D.18. In the disk-only cell the support sum is 3η. Otherwise a point–disk tie gives the second active projection

$$\frac{-\eta + \sqrt{3(\rho^2 - \eta^2)}}{2},$$

which yields the second branch. □

(a) $\rho \le 2\eta$ **(b)** $\rho > 2\eta$



$G \in [O, Z]$ and the disk-only minimizer still contains G
$\Lambda = \dfrac{3\eta}{h}$

one side through G is tangent to $\mathcal{D}_\eta$
$\Lambda = \dfrac{3\eta + \sqrt{3(\rho^2 - \eta^2)}}{2h}$

FIGURE 21. The two minimizing configurations in the disk-plus-point formula. For $\rho \le 2\eta$, the centered equilateral triangle tangent to $\mathcal{D}_\eta$ already contains G. For $\rho > 2\eta$, the point G becomes a vertex of the minimizing triangle and one incident side is tangent to the disk. The tangent length $GT = \sqrt{\rho^2 - \eta^2}$ is the geometric source of the radical term.

Lemma D.10 (Solved complementary-gap optimization)**.** *The set $\mathcal{D}(p, q) \cup J(p, q)$ of theorem 5.6 has enclosure number at least one.*

Proof. Put $c = c_{\max}(p, q)$, $m = \min\{p, q\}$, and $d = 1 - c$. The farther endpoint G of $J(p, q)$ satisfies

$$\|G\|^2 = 1 - m + m^2.$$

By theorem D.9, the nontrivial branch reduces to

$$3c(1 - c) \ge m(1 - m).$$

Common-pair domination gives $c \ge 1 - m \ge 1/2$. For $c \le h$, the left side is at least $3h(1 - h) > 1/4$. For $c > h$, the exact local support inequalities from theorem A.9 give

$$m \le c\left(\frac{1}{2} - \sqrt{c^2 - \frac{3}{4}}\right).$$

At this upper bound,

$$3c(1-c) - c\left(\frac{1}{2} - \sqrt{c^2 - \frac{3}{4}}\right)\left[1 - c\left(\frac{1}{2} - \sqrt{c^2 - \frac{3}{4}}\right)\right]$$
$$= \frac{c(1-c)}{2}\left(5 - 2c - 2c^2 + 2\sqrt{c^2 - \frac{3}{4}}\right) \geq 0.$$

The whole interval $J(p,q)$ is contained in every convex enclosure of its farther endpoint and the centered disk. This proves the claim, including the singleton endpoint. □

D.6. **CE2 short-ray optimization.**

Lemma D.11 (Solved CE2 short-ray optimization)**.** *Under the hypotheses of theorem 5.8,*

$$c_{\max}(p,q) < 1 - e, \qquad e := \min\{\alpha, \delta\},$$

and consequently $\{D_2, D_4\} \not\subseteq T_C$.

Proof. In the signed CE2 chart of Appendix A,

$$p = W - \alpha, \qquad q = R - \delta, \qquad M = \max\{R, W\}.$$

The two strict CE2 inequalities imply $P > (1+M)e$. Since $E^2 = 1 - M + M^2$,

$$(1+M)^2 - (3M-1)^2E^2 = 3M(1-M)(3M^2 - 2M + 3) > 0.$$

Hence $\eta = P/E > (3M-1)e$. The identity $RW = \eta + P$ gives

$$Wq > \eta + e > 3Me, \qquad Rp > \eta + e > 3Me.$$

Thus

$$3e < p, q < 1 - 5e, \qquad e < \frac{P}{1+M} \leq \frac{2\sqrt{3}-3}{6} < \frac{1}{12}.$$

Set $c = 1 - e$ and $f_c(t) = c^4 - c^2 + ct - t^2$. This function is concave, and

$$f_c(3e) = e(1 - 7e - 4e^2 + e^3) > 0,$$
$$f_c(1-5e) = e(2 - 15e - 4e^2 + e^3) > 0.$$

Therefore $f_c(p), f_c(q) > 0$. Since $c > p + q$ and $p + c, q + c > 1$, all four exact support-contact side lengths at radial demand c exceed one. Hence $c_{\max}(p,q) < 1 - e$.

If $e = \delta$, the forced point on r_2 lies strictly beyond the C exit; if $e = \alpha$, the forced point on r_4 does. Endpoint strictness therefore gives the asserted noncontainment. □

D.7. **High-radial and selected-chord calculations.** The zero-radial diameter envelope is

$$M_0(x) = \frac{-x + \sqrt{4 - 3x^2}}{2}. \tag{67}$$

For

$$0 < e < 1 - \frac{\sqrt{3}}{2},$$

define the selected low root

$$\epsilon(e) = \frac{1-e}{2}\left(1 - \sqrt{4(1-e)^2 - 3}\right). \tag{68}$$

It satisfies

$$e < \epsilon(e), \qquad \epsilon(e) < \frac{2e}{1-2e}, \tag{69}$$

and, for $e \le 1/8$,

$$\epsilon(e) \le 2e + 5e^2. \tag{70}$$

These follow by substituting the proposed upper bounds in the selected quadratic and checking its sign.

Lemma D.12 (Direct high-radial threshold)**.** *Let a nonsupercritical V triangle have incoming boundary reach A and outgoing boundary reach B. If*

$$C \ge 1-e, \qquad A > \epsilon(e),$$

then

$$\boxed{B \le \epsilon(e).}$$

Proof. The condition $C > 1/2$ forces $A+B \le 1$. In the exact cells of Proposition A.9, the only selected component with $A > \epsilon(e)$ has $B \le \epsilon(e)$; equivalently the two roots of the selected T-cell quadratic separate the feasible fiber. The selector $c \le 2\max\{A, B\}$ removes the formal high root. □

For $0 \le c \le 1/2$, the strict-supercritical outgoing supremum is

$$M_c^{\sup} = \frac{c + \sqrt{c^2 - 8c + 4}}{2}.$$

It is not attained. Thus an actual strict-supercritical V triangle with own-radial reach at least c has outgoing boundary reach strictly below $M_c^{\sup}$.

Lemma D.13 (Selected-arc chord bounds)**.** *Let $0 < e < 1/2$. On the selected Q_+ component for radial demand $1-e$, let p be the incoming boundary reach and let q be the following incoming reach forced across a center-free edge. Then*

$$q - p = 1 - \sqrt{1 - x + x^2},$$

where

$$q = e + (1-e)x, \qquad p = e + (1-e)x - 1 + \sqrt{1 - x + x^2}.$$

The selected arc is increasing and concave. In particular, on every interval used below,

$$q \ge p + \lambda(p - e)$$

for the chord slope λ joining the appropriate selected endpoints. The CE1 applications may take

$$\lambda > 1 - 4e \quad (e = \alpha), \qquad \lambda > 1 - 5e \quad (e = \alpha/R).$$

Proof. The selected T-cell equation reduces, after putting $x = (q-e)/(1-e)$, to

$$x(1-x) = (q-p)(2-q+p).$$

The selected component takes the smaller root, giving the displayed parameterization. Strict concavity follows by differentiation. A concave graph lies above its chords. For the first slope, equation (69) gives

$$\frac{e}{\epsilon(e) - e} > 1 - 4e.$$

For the second, use equation (70) when $e \le 1/8$ and $\epsilon(e) < (1-e)/2$ otherwise; the resulting slope is greater than $1 - 5e$ throughout $0 < e < 1 - \sqrt{3}/2$. □

D.8. **Exact direct output formulas.** The preceding proofs refer to three local inequalities. We derive them here from Proposition A.9. This replaces the former presentation by iterated reach maps.

For a nonsupercritical triangle with incoming reach at least a and radial reach at least c, let

$$\mathcal{B}_c(a) = \max\{b : (a, b, c) \text{ is admissible and } a + b \leq 1\}.$$

This is an outgoing capacity for one triangle; it is never composed in the proof.

For $1/2 < c \leq \sqrt{3}/2$, put

$$h(c) = \frac{2c}{1 + \sqrt{16c^2 - 3}}.$$

For $\sqrt{3}/2 < c < 1$, put

$$\ell(c) = \frac{c}{2}\left(1 - \sqrt{4c^2 - 3}\right), \qquad r(c) = c - \ell(c).$$

Also set

$$Q_-(a, c) = \frac{\sqrt{a^2 - ac + c^2}}{c} - a,$$
$$Q_+(a, c) = \frac{2(1 - a^2)c^2}{2ac^2 + c + \sqrt{(2ac^2 + c)^2 - 4(1 - c^2)(1 - a^2)c^2}}.$$

Proposition D.14 (Four direct high-radial outputs)**.** *For $1/2 < c \leq \sqrt{3}/2$,*

$$\mathcal{B}_c(a) = \begin{cases} 1 - a, & 0 \leq a \leq 1 - c, \\ Q_+(a, c), & 1 - c < a < h(c), \\ h(c), & a = h(c), \\ Q_-(a, c), & h(c) < a < c, \\ 1 - a, & c \leq a \leq 1. \end{cases}$$

For $\sqrt{3}/2 < c < 1$,

$$\mathcal{B}_c(a) = \begin{cases} 1 - a, & 0 \leq a \leq 1 - c, \\ Q_+(a, c), & 1 - c < a \leq \ell(c), \\ \ell(c), & \ell(c) < a < r(c), \\ Q_-(a, c), & r(c) \leq a < c, \\ 1 - a, & c \leq a \leq 1. \end{cases}$$

At $a = \ell(c)$, the selected value is $r(c)$; immediately to the right it is $\ell(c)$.

Proof. Since $c > 1/2$, any containing V triangle contains the radial midpoint. Therefore $a + b \leq 1$, and the supercritical cell of Proposition A.9 is absent. On the cap $b = 1 - a$, the T-cell inequality becomes

$$M(c - M) \leq 0,$$

where $M = \max\{a, b\}$. Thus the linear cap is feasible exactly when $a \leq 1 - c$ or $a \geq c$.

Off the linear cap, a maximal point lies on $F_L = 0$ or $F_T = 0$. The selected L-cell equation has the two ordered roots $\ell(c), r(c)$; the selector $q \leq 0$ retains the smaller coordinate. The ordered halves of $F_T = 0$ give Q_+ and Q_-. Their equality boundary $a = b$ is $h(c)$ in the low-radial range, while their $q = 0$ boundaries are

$(\ell(c), r(c))$ and $(r(c), \ell(c))$ in the high range. These conditions produce the two displayed formulas.

The formal other root in the Q_+ quadratic violates the geometric selector $c \leq 2\max\{a, b\}$ and is discarded. At the high-range junction, the selected T-cell point has value $r(c)$, whereas the adjacent selected L-cell point has value $\ell(c)$; this is the genuine downward jump stated above. □

D.8.1. *Vd corner margins.* In the exact Vd1/Vd2 corner form, let the boundary reaches be a, b, let $t > 0$, and put $d = \sqrt{t^2 + t + 1}$. The own-radial reach is

$$c_0 = \frac{d - a - tb}{t + 1}.$$

If the role has positive support on the forward adjacent arm, its far endpoint there is

$$u_+ = \frac{d - a - tb - 1}{t}.$$

Since $d < t + 1$,

$$c_0 < 1 - \frac{a + tb}{t + 1} \leq 1 - \min\{a, b\},$$

and

$$u_+ < 1 - b - \frac{a}{t} < 1 - b.$$

Thus lower boundary demands $a \geq A$, $b \geq H$ imply

$$c_0 < 1 - \min\{A, H\}, \qquad u_+ < 1 - H.$$

Reflection supplies the backward adjacent-arm statements. These two one-line inequalities are exactly the radial separations used in the one-Vd placement proof.

D.9. **CE1 reverse-path calculation.** The CE1 part of the seven-point enclosure theorem uses five boundary handoffs and the three transverse radial reaches at T_4, T_3, T_2.

Proposition D.15 (CE1 three-transverse return certificate)**.** *Use the signed CE1 center variables*

$$0 < R < 1, \qquad w = 1 - R, \qquad E = \sqrt{1 - Rw},$$
$$\eta = 1 - E, \qquad P = E(1 - E),$$

and write

$$A = \alpha, \qquad D = \delta.$$

Assume

$$A > 0, \qquad D > 0, \qquad A + wD < P, \qquad D + RA \geq P. \tag{71}$$

Put

$$X = R - D, \qquad m = \frac{A}{R}, \qquad h_0 = \frac{\eta + A + D}{2R}. \tag{72}$$

Let T_0 be the unique supercritical role and let $T_1, \ldots, T_5$ be nonsupercritical Vd0 *roles. Assume*

$$A_1 \geq X,$$
$$B_i + A_{i+1} \geq 1 \quad (1 \leq i \leq 4), \qquad B_5 + A_0 \geq 1,$$

and

$$C_4 \geq 1 - A, \qquad C_3 \geq 1 - m, \qquad C_2 \geq 1 - D. \tag{73}$$

Then

(74) $$\boxed{A_0 > 1 - h_0.}$$

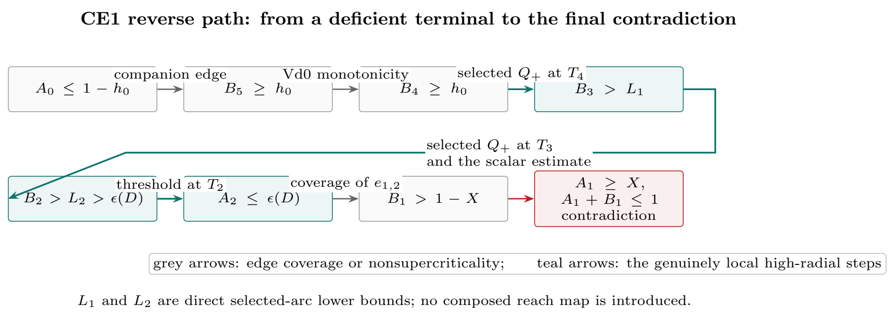


FIGURE 22. Logical structure of the CE1 direct reverse-path certificate. The proof starts from the negation of the desired terminal bound at T_0, moves through the actual boundary reaches, uses two selected-arc estimates and one high-radial threshold, and returns to T_1 with incompatible incoming and outgoing demands.

Proof. The sign difference in equation (71) gives

$$RD > wA,$$

so the center exit on r_3 is $m = A/R$. Moreover

(75) $$0 < A < P < \frac{1}{8}, \qquad 0 < D < \frac{R}{2}, \qquad 0 < m < \frac{w}{2} < \frac{1}{2}.$$

We first record

(76) $$\epsilon(A) < X, \qquad h_0 > A.$$

Indeed, $A + wD < P$ and $D = R - X$ give $A < wX - \eta$. Convexity of

$$wX - \frac{X}{2(1+X)}$$

on $R/2 < X < R$ gives

$$A < \frac{X}{2(1+X)}.$$

The selected-root bound equation (69) yields $\epsilon(A) < X$. Also

$$2R(h_0 - A) = \eta + D + (1 - 2R)A > 0;$$

when $R > 1/2$, use $A < P = \eta E$ for the last sign.

Suppose, contrary to equation (74), that

(77) $$A_0 \le 1 - h_0.$$

The assumed final handoff gives $B_5 \ge h_0$. Nonsupercriticality of T_5 and the handoff $B_4 + A_5 \ge 1$ imply

(78) $$\boxed{B_4 \ge B_5 \ge h_0.}$$

The assumed transverse radial reach at T_4 is at least $1 - A$. Apply the exact local catalogue to the reflected boundary pair (B_4, A_4). If the local point is not on the selected Q_+ component, the constant, Q_-, and linear components together with equation (76) give

$$1 - A_4 \geq 1 - \epsilon(A) > 1 - X.$$

The center-free edges then carry this lower bound backward to $B_1 > 1 - X$, contradicting $A_1 \geq X$ and $A_1 + B_1 \leq 1$.

Hence the T_4 state is selected Q_+. The selected-chord estimate theorem D.13 and equation (78) give

$$B_3 > L_1, \qquad L_1 = (2 - 4A)h_0 - (1 - 4A)A. \tag{79}$$

The selected-state condition also forces

$$\boxed{X > \frac{1}{2}.} \tag{80}$$

To see this, selectedness gives

$$h_0 \leq \epsilon(A) < \frac{2A}{1 - 2A}.$$

If $X \leq 1/2$, the lower CE1 inequality gives

$$2Rh_0 = \eta + A + D \geq w(R + A).$$

Consequently

$$f_R(A) < 0, \qquad f_R(z) = w(R + z)(1 - 2z) - 4Rz.$$

The quadratic is strictly concave and $f_R(0) > 0$. If $R \leq 1/2$, then

$$f_R(P) \geq \frac{1 - E}{2}(6E^3 - 2E^2 - 5E + 2) > 0.$$

If $R > 1/2$, then $A < A_* = w/2 - \eta$ and

$$f_R(A_*) = \frac{1 - E}{2}(E + 11R - 3ER - 5) > 0.$$

Concavity contradicts $f_R(A) < 0$ in both cases, proving equation (80). Thus

$$m < \frac{w}{2} < \frac{1}{4}. \tag{81}$$

If $B_3 \geq 1/2$, then $B_3 > 1 - X$ and the preceding immediate contradiction applies. Assume $B_3 < 1/2$. At T_3 the reflected incoming reach is $B_3 > h_0 > m$, while its assumed transverse radial reach is at least $1 - m$. If this state is not selected Q_+, the exact local catalogue gives

$$B_2 \geq 1 - B_3 > \frac{1}{2} > 1 - X,$$

and the remaining center-free edge again contradicts T_1.

Thus both T_4 and T_3 are selected Q_+. A second use of theorem D.13 gives

$$B_2 > (2 - 5m)B_3 - (1 - 5m)m.$$

Since $2 - 5m > 0$, equation (79) yields

$$\boxed{B_2 > L_2, \qquad L_2 = (2 - 5m)L_1 - (1 - 5m)m.} \tag{82}$$

We now prove the scalar estimate

$$\boxed{D < \frac{1}{10}, \qquad L_2 > 2D + 3D^2 > \epsilon(D), \qquad \epsilon(D) < X.} \tag{83}$$

The T_4 selected-state inequality and equation (70) give

$$D \le D_h := A(4R-1) + 10RA^2 - \eta. \tag{84}$$

If $D \ge 1/10$, then $A + wD < P$ gives

$$A < \overline{A} = \frac{w(5R-1)}{10}.$$

The right side of equation (84) is increasing in A, while $\eta > Rw/2$, so

$$D < U(R) := \overline{A}(4R-1) + 10R\overline{A}^2 - \frac{Rw}{2}.$$

Direct expansion gives

$$\frac{1}{10} - U(R) = -\frac{R}{10}P_4(R),$$

where

$$P_4(R) = 25R^4 - 60R^3 + 26R^2 + 22R - 14.$$

For $y = 2R - 1 \in (0,1)$,

$$16P_4(R) = 25y^4 - 20y^3 - 106y^2 + 124y - 39 \le -101y^2 + 124y - 39 < 0.$$

Hence $U(R) < 1/10$, a contradiction. This proves $D < 1/10$.

Put

$$\Psi(D) = L_2 - 2D - 3D^2, \qquad J(D) = L_2 - \frac{23}{10}D.$$

Since $D < 1/10$,

$$\Psi(D) - J(D) = 3D\left(\frac{1}{10} - D\right) > 0.$$

For fixed R, A, differentiation of equation (82) gives

$$J'(D) = \frac{S(R,A)}{10R^2},$$

where

$$S(R,A) = 100A^2 - (40R+50)A + R(20-23R).$$

The nonempty interval $P - RA \le D \le D_h$ implies

$$A > \frac{4Rw}{25R-4} =: A_*.$$

On the relevant interval $\partial S/\partial A < -45$, and

$$S(R,A) < S(R,A_*) = -\frac{Rq_3(R)}{(25R-4)^2},$$

where

$$q_3(R) = 8775R^3 - 14260R^2 + 7928R - 1120.$$

For $y = 2R - 1$,

$$8q_3(R) = 8775y^3 - 2195y^2 + 997y + 3007 > 0.$$

Thus J decreases with D, so $J(D) \ge J(D_h)$.

At $D = D_h$, one has $h_0 = 2A + 5A^2$. If $L_{2,h}$ is the corresponding value, then

$$L_{2,h} - A(1+4R) = \frac{A}{R^2}Q(R,A),$$

where

$$Q(R,A) = 100A^3R - 40A^2R^2 - 30A^2R + 12AR^2 - 15AR + 5A \\ - 4R^3 + 5R^2 - R.$$

This polynomial is concave in A on $0 \le A \le Rw/2$. Its endpoint values are

$$Q(R,0) = Rw(4R-1) > 0$$

and

$$Q\left(R, \frac{Rw}{2}\right) = \frac{Rw}{2}(25R^5 - 30R^4 + 20R^3 - 3R^2 - 7R + 3) > 0.$$

The last positivity follows after putting $y = 2R - 1$:

$$32p(R) = 25y^5 + 65y^4 + 90y^3 + 106y^2 - 35y + 5 > 0.$$

Hence $L_{2,h} > A(1+4R)$.

Finally $A < P = \eta E$, $A < Rw/2$, and $1/E > 1 + Rw/2$ give

$$\frac{D_h}{A} < C_* := 4R - 2 + 5R^2w - \frac{Rw}{2}.$$

Moreover

$$1 + 4R - \frac{23}{10}C_* = \frac{230R^3 - 253R^2 - 81R + 112}{20} > 0.$$

Thus $J(D_h) > 0$, so

$$L_2 > 2D + 3D^2.$$

Substitution in the selected-root quadratic gives $\epsilon(D) < 2D + 3D^2$, and the argument proving $\epsilon(A) < X$ also gives $\epsilon(D) < X$. This proves equation (83).

The assumed transverse radial reach at T_2 is at least $1 - D$. Since $B_2 > \epsilon(D)$, theorem D.12, applied after reflection, gives

$$A_2 \le \epsilon(D).$$

Coverage of $e_{1,2}$ therefore forces

$$B_1 \ge 1 - A_2 \ge 1 - \epsilon(D) > 1 - X,$$

contradicting $A_1 \ge X$ and $A_1 + B_1 \le 1$. Hence equation (77) is impossible. □

D.10. **Transverse seven-point calculation.**

Lemma D.16 (Endpoint upper squeeze)**.** *If an open unit triangle U' contains K_{tr}, use its local signed-center variables and put*

$$X = R - \delta, \qquad Q = \frac{\eta + \alpha + \delta}{2R}.$$

Then its incoming reach at V_0 satisfies

$$A_0(U') < 1 - Q.$$

Proof. The actual gap endpoints lie strictly inside the edge trace of U'. Writing its two corresponding closed reaches locally as $\widehat{B}_0, \widehat{A}_1$, one has

$$\widehat{B}_0 = \ell > 2Q, \qquad \widehat{A}_1 = 1 - r > X.$$

The diameter inequality for the two boundary endpoints gives

$$A_0(U') \le M_0(\widehat{B}_0) < M_0(2Q) = -Q + \sqrt{1 - 3Q^2} < 1 - Q.$$

No radial point on r_0 occurs in this calculation. □

Lemma D.17 (Solved CE2 transverse threshold)**.** *For an open unit equilateral triangle U_C, put $T_C := \overline{U_C}$. Then*

$$N_E(T_C) = 2 \Longrightarrow K_{\mathrm{tr}} \nsubseteq U_C.$$

Proof. Nonsupercriticality and the five gap-free edges give

$$A_1 < A_2 < A_3 < A_4 < A_5 < A_0$$

for the hypothetical containing triangle. The endpoint squeeze therefore implies $B_2, B_4 > Q$. Containment of the actual radial endpoints P_2, P_4 gives

$$C_2 > 1 - \delta, \qquad C_4 > 1 - \alpha.$$

The signed CE2 threshold calculations give

$$R - \delta > \epsilon(\alpha), \qquad \min\{\epsilon(\alpha), \epsilon(\delta)\} < Q.$$

If $\epsilon(\alpha) < Q$, theorem D.12 at T_4 gives $B_4 < Q$; otherwise the same lemma at T_2 gives $B_2 < Q$. Both alternatives contradict the preceding strict lower bounds. $\square$

D.11. **Disk, one-third, and supported-tail optimizations.**

Proposition D.18 (Disk–finite-set calipers)**.** *Let $P = \{p_1, \dots, p_m\}$, and let J denote rotation by $\pi/2$. Unless the disk is the active support in all three directions at a minimizing orientation, a minimizing normal belongs to the finite list*

$$\pm \frac{\mathsf{J}(p_k - p_i)}{\|p_k - p_i\|} \quad (p_i \neq p_k)$$

or

$$\frac{\eta p_i \pm \sqrt{\|p_i\|^2 - \eta^2}\, \mathsf{J} p_i}{\|p_i\|^2} \quad (\|p_i\| \geq \eta).$$

Thus one side of a least enclosing equilateral triangle either contains two points, or is tangent to the disk and contains one point.

Proof. The tie directions partition the normal circle into cells with fixed support sources. On a cell with q disk supports,

$$\sum_{j=0}^{2} h_{\mathcal{D}_\eta \cup P}(\mathsf{R}^j n(\theta)) = q\eta + \langle v, n(\theta) \rangle.$$

If any point is active, the last inner product is positive and the second derivative is its negative. Hence an interior critical point is a maximum. A minimum occurs at a point–point or point–disk tie. On a disk-only cell the support sum is 3η, already the global minimum. $\square$

Lemma D.19 (Half-edge one-third radial envelope)**.** *If*

$$M \geq \frac{1}{2}, \qquad 0 < m \leq M, \qquad M + m < 1,$$

then

$$\boxed{c_{\max}(M, m) < 1 - \frac{m}{3}.} \tag{85}$$

Proof. Put $s = M + m$ and $c_0 = 1 - m/3$. Only the L and T cells of theorem A.9 can occur.

For $m \le 3/8$, substitution into the L polynomial gives

$$F_L(c_0) = \frac{m}{81}(m^3 - 12m^2 - 63m + 27) > 0;$$

the cubic decreases on $[0, 3/8]$ and equals $891/512$ at $3/8$. Thus the selected L root is below c_0. At the L/T transition the two roots agree. On the T cell,

$$c_T(s) = \frac{2(s-m)}{1+\sqrt{4s^2-3}},$$

and, writing $r = \sqrt{4s^2 - 3}$,

$$c_T'(s) = \frac{2(r + 4ms - 3)}{r(1+r)^2} < 0$$

because $m \le s/2$ and $r + 2s^2 - 3 < 0$ for $s < 1$. Hence the T root remains below c_0.

For $3/8 \le m < 1/2$, one has

$$s^4 - s^2 + mM \ge \left(\frac{7}{8}\right)^4 - \left(\frac{7}{8}\right)^2 + \frac{3}{16} = \frac{33}{4096} > 0,$$

so only the T cell occurs. It decreases with M; at $M = 1/2$, $s \ge 7/8$ and its value is at most $4/5 < 5/6 < c_0$. □

Lemma D.20 (Rescuer endpoint and boundary-tail budget)**.** *Suppose a special role at V_0 has boundary endpoint a on $e_{5,0}$ and supported interval $[c, u] \subset r_1$, measured from V_1 toward O. Put*

$$\varepsilon = 1 - u, \qquad M = M_c^{\sup}.$$

Assume

$$\varepsilon V_1 \in U_C, \qquad a + \varepsilon \le 1,$$

$$a \le 1 - M, \qquad \frac{a}{a+\varepsilon} \le 1 - M. \tag{86}$$

If T_1 is uniquely supercritical with own-radial reach at least c and T_2, T_3, T_4, T_5 are nonsupercritical on the center-free four-edge path, then the skeleton is not covered.

Proof. Let h_T be the far boundary demand on T_5. If the companion C trace does not hide a, then $h_T \ge 1 - a \ge M$.

In the hiding configuration, $k/W \le a$, while $\varepsilon V_1 \in U_C$ gives $\delta/R \ge \varepsilon$. Therefore

$$Wa \ge k = \eta + \alpha + \delta > \alpha + R\varepsilon,$$

so

$$a - R(a + \varepsilon) > \alpha.$$

Also $Wa > \delta \ge R\varepsilon$, so $R < a/(a + \varepsilon)$. Since $a + \varepsilon \le 1$,

$$R + \alpha < a + R(1 - a - \varepsilon) < \frac{a}{a+\varepsilon} \le 1 - M.$$

Thus again $h_T \ge M$.

The strict-supercritical envelope gives $B_1 < M$. Adding

$$A_2 \ge 1 - B_1, \quad B_2 + A_3 \ge 1, \quad B_3 + A_4 \ge 1, \quad B_4 + A_5 \ge 1, \quad B_5 \ge h_T$$

gives

$$\sum_{i=2}^{5}(A_i + B_i) > 4,$$

contrary to nonsupercriticality. □

D.12. **T3-like supported-tail adapter.**

Lemma D.21 (Solved T3-like supported-tail adapter)**.** *The exact hypotheses of theorem 5.14 imply the rescuer-tail hypotheses in theorem D.20.*

Proof. Normalize the center midpoint to M_0 and reflect so that T_0 is T3-like, $M_1 \in T_0$, and T_1 is uniquely supercritical. Write the T3-like interval on r_1, measured from V_1 toward O, as $[c, u]$, with $c \le 1/2 \le u$. Its O-side endpoint is

$$P_T = (1-u)V_1.$$

The open endpoint and the V-type restrictions exclude every V role from P_T, so $P_T \in U_C$.

Let A_0 be the actual boundary reach of T_0 on $e_{5,0}$. The translated T3-like normal form has

$$R_0 = \frac{D + \sqrt{4 - 3D^2}}{2}, \qquad x = \frac{A_0 D}{R_0}, \qquad \theta = \frac{D-1}{R_0},$$

$$c = x + \theta, \qquad \frac{1 - 4\theta + \theta^2}{2(1-2\theta)} \le x \le \frac{1}{2} - \theta.$$

Let $M = M_c^{\sup}$. The equation $s(1-M) = c$, where $s(z) = z(2-z)/(1+z)$, and

$$2x^2 + (\theta - 1)x + \theta \ge 0$$

give

$$A_0 \le x \le 1 - M, \qquad \frac{A_0}{A_0 + 1 - u} = x \le 1 - M.$$

Finally $A_0 + 1 - u = R_0/D \le 1$. These are exactly the hypotheses of theorem D.20. □

D.13. **One-Vd radial-separation calculations.**

Lemma D.22 (Solved adjacent Vd separation)**.** *Under the adjacent placement*

$$\sigma = 0, \qquad \tau \in \{1, 5\},$$

the radial-separation inequality $c_{\mathrm{loc}} < 1 - d_2^C$ holds on the routed arm r_2.

Proof. In the adjacent placement, the residuals satisfy

$$\rho_R + \rho_L < \frac{1}{2}.$$

The signed CE2 domain gives

$$\alpha + \delta < \frac{1}{6}\min\{R, W\}.$$

If $\rho_L = W - \alpha$, then

$$4\delta + \alpha \le 4(\alpha + \delta) < \frac{2W}{3} < W,$$

so $4\delta < \rho_L$.

Otherwise $\rho_L = 1 - A_0$. The opposite residual must be

$$\rho_R = 1 - (W + \delta), \qquad B_0 \geq \frac{k}{R}.$$

The boundary-endpoint diameter inequality gives

$$\rho_L > \frac{k}{2R} > \frac{\eta}{2R}.$$

Combining this with the preceding residual-sum inequality gives

$$\delta > R - \frac{1}{2} + \frac{\eta}{2R} =: J(R).$$

The function J is increasing, $J(3/8) = 1/24$, and the displayed center-slack inequality gives $\delta < 1/24$. Thus $R < 3/8$. Then

$$(2 - 3R)^2 - E^2 = (3 - 8R)(1 - R) > 0,$$

so $\eta/R > 1/3$, and the preceding endpoint-distance inequality yields

$$\rho_L > \frac{1}{6} > 4\delta.$$

This proves the residual estimate used by the adjacent Vd terminal. The new one-third radial envelope then gives

$$C_2 < 1 - \frac{\rho_L}{3} < 1 - \delta.$$

□

Lemma D.23 (Solved nonadjacent Vd separation)**.** *Under the nonadjacent placement*

$$\sigma = 0, \qquad \tau \in \{2, 3, 4\},$$

the point $D_\tau = \min\{\rho_R, \rho_L\} V_\tau$ lies beyond both the C-triangle exit and the actual Vd radial endpoint.

Proof. Let $x = k/W$ and $y = k/R$ be the two center near endpoints, and let B, U be the farthest already-covered extents. The endpoint-distance inequality and monotonicity of M_0 give

$$B \leq M_0(x), \qquad U \leq M_0(y).$$

Since

$$1 - M_0(z) > \frac{z}{2} \qquad (z > 0),$$

the residual tails satisfy

$$\rho_R > \frac{x}{2}, \qquad \rho_L > \frac{y}{2}.$$

The signed center-slack estimate

$$\alpha + \delta < \frac{1}{2} \min\{x, y\}$$

therefore gives

$$\alpha + \delta < \min\{\rho_R, \rho_L\}.$$

The Vd own-radial margin then places $\min\{\rho_R, \rho_L\} V_\tau$ beyond the Vd trace, while the last strict residual inequality places it beyond the C exit. □

Lemma D.24 (Solved Vd1 supported-tail adapter)**.** *When the unique Vd1 role is based at T_0, its actual corner data imply the two strict-supercritical envelope inequalities required by theorem D.20.*

Proof. Use the Vd1 corner parameter $t \geq 1$, let $d = \sqrt{t^2+t+1}$, and write the supported interval as $[c, u]$. Then

$$c = \frac{t(1-b)}{t+1}, \qquad u = \frac{d-a-tb-1}{t}.$$

The midpoint condition gives

$$a + tb \leq d - 1 - \frac{t}{2}.$$

Eliminating $tb = t - c(t+1)$ gives

$$a \leq F(t,c) := d - 1 - \frac{3t}{2} + c(t+1).$$

For fixed $c \leq 1/2$, the function F decreases with t, so

$$a \leq F(1,c) = \sqrt{3} - \frac{5}{2} + 2c =: L(c).$$

The inequality $2L(c) \leq 1 - M_c^{\sup}$ reduces to

$$20c^2 + (18\sqrt{3} - 52)c + 47 - 24\sqrt{3} \geq 0,$$

which holds on $[0, 1/2]$. Hence

$$a \leq 1 - M_c^{\sup}.$$

Put $\varepsilon = 1 - u$. Direct substitution gives

$$\varepsilon = \varepsilon_0 + \frac{a}{t}, \qquad \varepsilon_0 = \frac{2t+1-c(t+1)-d}{t} > 0.$$

The map

$$z \longmapsto \frac{z}{z + \varepsilon_0 + z/t}$$

is increasing, while $\varepsilon_0 + F(t,c)/t = 1/2$. Therefore

$$\frac{a}{a+\varepsilon} \leq \frac{F(t,c)}{F(t,c)+1/2} \leq 2L(c) \leq 1 - M_c^{\sup}.$$

The last two estimates are the local hypotheses of the common rescuer-tail lemma. □

Appendix E. Zero-Gap Nine-Point Optimization

This appendix supplies the exact frontier, radial, pointwise, Newton, and support-arc calculations for the zero-gap nine-point obstruction. It uses the public parameters and witness $K_{\mathrm{wit}}(a,b)$ from section 5.7. The symbols A, B, C introduced below are appendix-only Newton inner points required by the exact certificate; they do not replace the public frontier points Q_-, Q_0, Q_+.

The finite caliper criterion used here is the shared result theorem A.5 in Appendix A. It is not rederived in this appendix.

E.1. **Corner-clipped AB-unions.** Apply the rigid motion that sends the supercritical corner V_4 to the local origin and its two incident edge directions to e_A, e_B. Within this calculation O, A, B, C denote the local origin, anchors, and corner cone; the feasible union keeps the public symbol $\mathcal{R}_{AB}(a, b)$.

Let e_A, e_B be unit vectors with angle $120°$, put

$$O = 0, \qquad A = ae_A, \qquad B = be_B, \qquad C = \{ue_A + ve_B : u, v \geq 0\},$$

and define

$$\mathcal{R}_{AB}(a, b) = C \cap \bigcup \{T : T \text{ is a closed unit equilateral triangle and } O, A, B \in T\}. \tag{87}$$

The finite description in Theorem A.5 applies to the four-point set $\{O, A, B, x\}$ and makes membership in (87) an exact finite semialgebraic condition of degree at most four.

The strict supercritical branch admits a simpler closed form. In the next statement write $x = ue_A + ve_B$ and $\|x\|^2 = u^2 + v^2 - uv$.

Theorem E.1 (Strict supercritical AB-union)**.** *Assume*

$$a + b > 1, \qquad \rho = a^2 + ab + b^2 < 1,$$

put $h = \sqrt{3}/2$, $D = \sqrt{4\rho - 3}$, *and define*

$$\alpha = \frac{h(a + 2b - aD)}{2\rho}, \qquad \beta = \frac{h(a - b + (a + b)D)}{2\rho},$$
$$\gamma = \frac{h(-a + b + (a + b)D)}{2\rho}, \qquad \delta = \frac{h(2a + b - bD)}{2\rho}.$$

Then all four coefficients are positive and

$$\mathcal{R}_{AB}(a, b) = (C \cap \mathcal{D}_A \cap \mathcal{H}_A) \cup (C \cap \mathcal{D}_B \cap \mathcal{H}_B), \tag{88}$$

where

$$\mathcal{D}_A = \{(u, v) : (u - a)^2 + v^2 - (u - a)v \leq 1\},$$
$$\mathcal{D}_B = \{(u, v) : u^2 + (v - b)^2 - u(v - b) \leq 1\},$$
$$\mathcal{H}_A = \{(u, v) : \alpha(u - a) + \beta v \leq 0\},$$
$$\mathcal{H}_B = \{(u, v) : \gamma u + \delta(v - b) \leq 0\}.$$

Its non-axis frontier consists, in counterclockwise order, of the A-circle, the A-line, the B-line, and the B-circle.

Proof. The identities

$$4\rho - 3(a + b)^2 = (a - b)^2, \qquad (a + b)^2 D^2 - (a - b)^2 = 4\rho((a + b)^2 - 1)$$

give $0 < D < 1$ and positivity of the four coefficients.

We apply the caliper criterion of Theorem A.5. For a point $x = (u, v)$ in the relative interior of C but outside $\triangle OAB$, the four hull vertices occur in the order O, B, x, A. The two cone-axis hull edges cannot certify a unit triangle: their support sums, divided by h, are respectively

$$\max\{a, u\} + \max\{b, v - u\}, \qquad \max\{b, v\} + \max\{a, u - v\},$$

and are at least $a + b > 1$. It remains to test the Ax and Bx calipers.

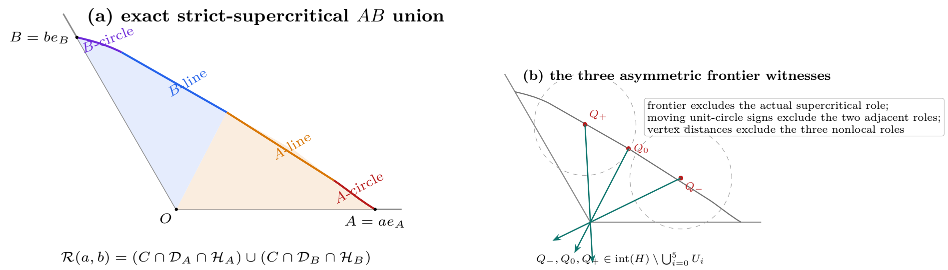


FIGURE 23. The strict-supercritical AB frontier and the three asymmetric witnesses. The source-conditioned union has, in counterclockwise order, an A-circle arc, an A-line segment, a B-line segment, and a B-circle arc. The points Q_-, Q_0, Q_+ are selected on this exact frontier and are then excluded from the six actual V roles by frontier, moving-circle, and vertex-distance arguments. The drawing is schematic; the labelled frontier order and witness roles are exact.

For the Ax caliper put $p = u - a$ and $r^2 = p^2 + v^2 - pv$. Exact evaluation of the three supports gives

$$\frac{G_A}{h} = \frac{\max\{r^2, -ap + (a+b)v\} + N_A}{r}, \qquad N_A = \begin{cases} 0, & p \le 0, \\ ap, & 0 \le p \le v, \\ (a+b)p - bv, & p \ge v. \end{cases} \tag{89}$$

No support label is omitted in these three sectors. If $p \le 0$, $G_A \le h$ is equivalent to

$$r \le 1, \qquad -ap + (a+b)v \le r.$$

The second inequality is the half-plane condition in (88), because

$$r^2 - (-ap + (a+b)v)^2 = \frac{(cp + dv)(c'p + d'v)}{4\rho},$$

where

$$c = a + 2b - aD, \quad d = a - b + (a+b)D, \quad c' = a + 2b + aD, \quad d' = a - b - (a+b)D,$$

and $c, c', d > 0 > d'$. Thus the low sector is exactly $\mathcal{D}_A \cap \mathcal{H}_A$. In the middle sector $0 < p \le v$ one has $r \le v$, whereas the numerator in (89) is at least $(a+b)v > r$, so no fit is possible.

In the high sector $p \ge v$, the inequalities $G_A \le h$ imply

$$r^2 + (a+b)p - bv \le r, \qquad bp + av \le r. \tag{90}$$

Write $x - A = ry(\theta)$, $0 \le \theta \le 60°$, with cone coordinates

$$y(\theta) = te_A + we_B, \quad t = \frac{2}{\sqrt{3}}\cos(\theta - 30°), \quad w = \frac{2}{\sqrt{3}}\sin\theta.$$

Then (90) gives

$$r \le f(\theta) = 1 - (a+b)t + bw, \qquad S(\theta) = bt + aw \le 1.$$

Let n_B be the unit vector with dual coordinates (γ, δ). The identities

$$b\gamma + (a+b)\delta = h, \quad (a+b)\gamma + a\delta = \frac{h}{2}(1+D), \quad b\delta - a\gamma = \frac{h}{2}(1-D)$$

show that the unique angle θ_B of this normal satisfies $S(\theta_B) = 1$ and $f(\theta_B) = (1-D)/2$. The function S has at most one critical point, a maximum, so $S(\theta) \le 1$ implies $\theta \le \theta_B$. On the feasible part $f, f' \ge 0$; hence $f(\theta)\cos(\theta_B - \theta)$ is increasing. Consequently

$$\langle n_B, x - B\rangle \le a\gamma - b\delta + \frac{h}{2}(1-D) = 0.$$

The diameter bound also gives $\|x - B\| \le 1$. Thus every high sector fit belongs to $\mathcal{D}_B \cap \mathcal{H}_B$. Reflection exchanges a, u with b, v and supplies the corresponding result for the Bx caliper. This proves (88) in the cone interior; the axes follow by taking limits, since both sides are closed.

For completeness, the line–circle junctions are obtained by solving the two displayed line and circle equations. The line normals satisfy

$$\alpha^2 + \alpha\beta + \beta^2 = h^2, \qquad \gamma^2 + \gamma\delta + \delta^2 = h^2,$$

and the line determinant is

$$\alpha\delta - \gamma\beta = \frac{3(1-D)(D+3)}{16\rho} > 0.$$

The successive cone slopes are

$$+\infty, \quad \frac{2}{1-D}, \quad \frac{a(1-D)+2b}{2a+b(1-D)}, \quad \frac{1-D}{2}, \quad 0;$$

the two middle comparisons reduce to $a(4-(1-D)^2) > 0$ and $b(4-(1-D)^2) > 0$. This proves the asserted four-piece order. □

E.2. **Exact-one handoffs and common reach bounds.**

Lemma E.2 (Exact-one handoff chain)**.** *Assume the six V triangles cover ∂H and exactly one actual V triangle is supercritical. Rotate its index to 4. The strict handoffs supplied by Proposition 2.11 satisfy*

$$\xi_3 < \xi_2 < \xi_1 < \xi_0 < \xi_5 < \xi_4.$$

With

$$a = 1 - \xi_3, \qquad b = \xi_4, \qquad p = 1 - b, \qquad q = 1 - a,$$

one has

$$0 < a, b < 1, \qquad a + b > 1, \qquad a^2 + ab + b^2 < 1, \tag{91}$$

and every selected V triangle obeys

$$a_i \ge p, \qquad b_i \ge q.$$

Proof. Every selected V triangle other than 4 is strictly nonsupercritical, so $\xi_i < \xi_{i-1}$ for $i \ne 4$, which is precisely (E.2). In particular ξ_3 is the global minimum and ξ_4 the global maximum. The two anchors of V triangle 4 lie in its open triangle; their mutual squared distance is $a^2 + ab + b^2$, strictly below the diameter squared of the closed unit triangle. This proves (91). Finally every $\xi_j \in [\xi_3, \xi_4]$, so $1 - \xi_{i-1} \ge 1 - \xi_4 = p$ and $\xi_i \ge \xi_3 = q$. □

E.3. **Six directly forced radial points.** Put $h = \sqrt{3}/2$ and

$$s = p + q, \qquad m = \min(p, q), \qquad M = \max(p, q), \qquad \chi = s^4 - s^2 + pq.$$

The strict domain (91) gives

$$pq > (1-s)(2-s), \qquad m > 1-s, \qquad s > 1-m. \tag{92}$$

Let $c_* = c_{\max}(p, q)$ be the exact radial envelope of Proposition A.9. On the present subcritical domain, equation (39) becomes

$$c_* = \begin{cases} C_L(m), & \chi \le 0, \\ \dfrac{2M}{1 + \sqrt{4s^2 - 3}}, & \chi > 0, \end{cases} \tag{93}$$

where $C_L(m)$ is the unique root in $[\sqrt{3}/2, 1]$ of

$$z^4 - z^2 + mz - m^2 = 0.$$

At $\chi = 0$ the two expressions agree and equal s.

Lemma E.3 (Direct radial forcing)**.** *One has*

$$0 < c_* < 1, \qquad c_* > 1 - m. \tag{94}$$

Assume that every $T_i = \overline{U_i}$ is Vd0 *and that $U_C, U_0, \ldots, U_5$ cover H. Then, for every i,*

$$D_i = (1 - c_*)V_i \in U_C.$$

Consequently U_C contains the centered closed disk

$$\mathcal{D}_\eta = \{x : \|x\| \le \eta\}, \qquad \eta = h(1 - c_*). \tag{95}$$

Proof. On the C_L branch, $C_L(m) < 1$ because the defining polynomial is positive at 1; if $s < \sqrt{3}/2$, then $c_* \ge \sqrt{3}/2 > s > 1 - m$, while if $s \ge \sqrt{3}/2$ its value at s is $\chi \le 0$, so $c_* \ge s > 1 - m$. On the other branch put $E = \sqrt{4s^2 - 3}$. The selector gives $sE > M - m$, whence $c_* < s < 1$. Moreover, with

$$A_* = \frac{2M}{1 - m} - 1,$$

one has

$$A_*^2 - E^2 = \frac{4(1-s)\{(1-m)(1-2m) + m(2-m)(1-s)\}}{(1-m)^2} > 0.$$

Thus $A_* > E$ and $c_* > 1 - m$. This proves (94).

Fix i. If $D_i \in U_i$, openness supplies $\varepsilon > 0$ with $c_* + \varepsilon < 1$ such that $(1 - c_* - \varepsilon)V_i \in U_i$. The same triangle contains its two selected handoff anchors. Since their incoming and forward-reach lower bounds dominate p and q, convexity and passage to the closure give a closed unit triangle realizing $(p, q, c_* + \varepsilon)$. This contradicts the definition of $c_* = c_{\max}(p, q)$. Hence $D_i \notin U_i$.

If D_i belonged to U_{i-1} or U_{i+1}, a small plane ball inside that open triangle would meet r_i in a positive-length interval. This contradicts the Vd0 definition. For the remaining triangles put $r = 1 - c_* > 0$. Directly,

$$\|rV_i - V_{i\pm 2}\|^2 = 1 + r + r^2 > 1, \qquad \|rV_i - V_{i+3}\| = 1 + r > 1.$$

Since $V_j \in U_j$ and two points of an open unit triangle are at distance strictly below 1, these inequalities exclude D_i from the other three V triangles. Thus no V triangle contains D_i. The point lies in $\operatorname{int}(H)$ by (94), so the assumed cover forces $D_i \in U_C$.

The six points D_i form a regular hexagon of circumradius $1-c_*$. Their convex hull contains its centered incircle of radius $h(1-c_*)$, and convexity of U_C gives (95). $\square$

We next define the three asymmetric witnesses. At V_4 use local coordinates

$$P = V_4 + u(V_5 - V_4) + v(V_3 - V_4), \qquad \mathsf{q}(u,v) = u^2 + v^2 - uv.$$

The forward-reach lower bound is b and the backward-reach lower bound is a. Accordingly, set

$$\alpha = \frac{h(b+2a-bD)}{2\rho}, \qquad \beta = \frac{h(b-a+(a+b)D)}{2\rho},$$
$$\gamma = \frac{h(-b+a+(a+b)D)}{2\rho}, \qquad \delta = \frac{h(2b+a-aD)}{2\rho},$$

where now $\rho = a^2+ab+b^2$ and $D = \sqrt{4\rho-3}$. The two line pieces of Theorem E.1 are

$$L_-(\lambda) = (b-\beta\lambda, \alpha\lambda), \qquad L_+(\mu) = (\delta\mu, a-\gamma\mu).$$

They meet at $\lambda_* = \mu_* = 8h\rho/(3(D+3))$ in

$$J = \left(\frac{2b+a-aD}{D+3}, \frac{b+2a-bD}{D+3}\right). \tag{96}$$

Put

$$E_- = (1-b, 2-b), \qquad E_+ = (2-a, 1-a).$$

The actual adjacent handoffs have local centers

$$Z_5(t) = (1+t, t), \qquad 1-a \le t \le b, \quad \text{and} \quad Z_3(y) = (y, 1+y), \qquad 1-b \le y \le a.$$

The restrictions of the two unit-circle equations are

$$g_-(\lambda) = \frac{3}{4}\lambda^2 - 3(\alpha+b\beta)\lambda + 3b^2 - 3b + 2,$$
$$g_+(\mu) = \frac{3}{4}\mu^2 - 3(\delta+a\gamma)\mu + 3a^2 - 3a + 2. \tag{97}$$

Let λ_- and μ_+ be their first roots; explicitly,

$$\lambda_- = 2(\alpha+b\beta) - \frac{2}{3}\sqrt{9(\alpha+b\beta)^2 - 9b^2 + 9b - 6},$$
$$\mu_+ = 2(\delta+a\gamma) - \frac{2}{3}\sqrt{9(\delta+a\gamma)^2 - 9a^2 + 9a - 6}.$$

Lemma E.4 (Moving adjacent-triangle signs)**.** *Throughout* (91)*,*

$$\lambda_* < \lambda_- < h^{-1}, \qquad \mu_* < \mu_+ < h^{-1}. \tag{98}$$

Moreover $J_u, J_v < 1/2$. The point $L_+(\mu_+)$ satisfies

$$(L_+(\mu_+))_u + (L_+(\mu_+))_v < 3 - 2a,$$

and the reflected bound holds for $L_-(\lambda_-)$. Writing $F(P,t) = \mathsf{q}(P - Z_5(t)) - 1$, one has, for every $t \in [1-a, b]$,

$$F(L_+(\mu_+), t) \ge 0, \qquad F(J,t) > 0, \qquad F(L_-(\lambda_-), t) > 0. \tag{99}$$

Under the reflection $(u,v,a,b) \leftrightarrow (v,u,b,a)$, the analogous signs hold for every $y \in [1-b, a]$: the first-root point $L_-(\lambda_-)$ has nonnegative sign against $Z_3(y)$, and the other two witnesses have strictly positive sign.

Proof. Put

$$U = a + b - 1, \qquad z = a - b, \qquad D^2 = z^2 + 6U + 3U^2.$$

Then $0 < U < 1/6$ and $z^2 < \Omega := 1 - 6U - 3U^2$. We give the fixed-line calculation for the circle centered at $E_+ = (2 - a, 1 - a)$; reflection gives the calculation for E_-. Put

$$F(P, t) = \mathsf{q}(P - (1 + t, t)) - 1, \qquad t_0 = 1 - a,$$

so this circle is $F(P, t_0) = 0$. At the junction, exact expansion gives

$$(D + 3)^2 F(J, t_0) = 3D(z^2 + Uz + 1 - 3U) + P_J,$$

where

$$P_J = z^4 + 5z^2 + 6Uz^2 + 3U^2z^2 + 9Uz + 3U + 15U^2 > 0;$$

indeed $z^2 + Uz + 1 - 3U \geq 1 - 3U - U^2/4 > 0$, and $5z^2 + 9Uz + 15U^2$ is positive definite. Thus $F(J, t_0) > 0$. Also

$$J_u + J_v = \frac{(1 + U)(3 - D)}{3 + D} < 1.$$

Indeed this inequality is $3U < D(2 + U)$, which follows from $D^2 \geq 3U(2 + U)$ and $(2 + U)^3 > 3U$. Since

$$\frac{\partial F}{\partial t}(P, t) = 1 + 2t - P_u - P_v,$$

$F(J, t)$ is strictly increasing for $t \geq 0$, and therefore is positive throughout $[1 - a, b]$.

On the opposite line L_-, the derivative at the junction, after multiplication by a positive factor, has at $t = 1 - a$ the numerator

$$N_0 = D(9 + 3z^2 + 6Uz - 9U^2) + P_0,$$

where

$$P_0 = 18U^3 + 6U^2z^2 + 63U^2 + 18Uz^2 + 18Uz + 54U + 2z^4 + 9z^2 + 9.$$

The coefficient of D is at least $9 - 6U - 9U^2 > 0$, while $P_0 \geq 9 + 54U - 18U > 0$. Hence $N_0 > 0$. At the other endpoint $t = b$, the corresponding numerator is

$$N_1 = D(9 + 3z^2 + 6U - 3U^2) + P_1,$$

where

$$P_1 = 9U^4 - 3U^3z + 45U^3 + 9U^2z^2 - 6U^2z + 72U^2 - Uz^3 + 21Uz^2 + 9Uz + 45U + 2z^4 + 9z^2 + 9.$$

Its D-coefficient is positive and, using $|z| < 1$,

$$P_1 \geq 9 + 45U - 9U - U - 6U^2 - 3U^3 > 0.$$

The junction derivative is affine in t, so it is positive throughout $[1 - a, b]$. Since the restriction of F to L_- is a convex quadratic, is positive at the junction, and is increasing there, it has no intersection on the opposite line side for any actual adjacent handoff.

On the correct line L_+, the value at its outer endpoint h^{-1} has, after multiplication by a positive factor, numerator

$$A_* = P_2 - 4DU(1 + U + z),$$

where

$$P_2 = 3U^4 + 6U^3z + 6U^3 + 4U^2z^2 + 12U^2z + 12U^2 + 2Uz^3 + 6Uz^2 + 2Uz + 6U + z^4 + 2z^2 - 3.$$

For fixed U, replace the odd powers of z by $x = |z|$. The resulting polynomial $P_2^+(U,x)$ satisfies

$$\partial_x P_2^+ = 2(3U^3 + 4U^2x + 6U^2 + 3Ux^2 + 6Ux + U + 2x^3 + 2x) > 0,$$

and therefore its supremum for $z^2 < \Omega$ occurs at $x = \sqrt{\Omega}$. There

$$P_2^+(U,\sqrt{\Omega}) = 4U(U + \sqrt{\Omega} - 3) < 0.$$

As $1 + U + z = 2a > 0$, this gives $A_* < 0$. The positive junction value and negative endpoint value force the first root strictly between them, proving the L_+ half of (98); reflection proves the other half.

For the coordinate bounds, the exact identities

$$D^2 - (3U - z)^2 = 6U(1 + z - U), \qquad D^2 - (3U + z)^2 = 6U(1 - z - U)$$

give $J_u, J_v < 1/2$. It remains to control the coordinate sum along L_+. For $E = L_+(h^{-1})$, the sign of $3 - 2a - E_u - E_v$ is the sign of

$$D(6U + 2z + 6) + B(U,z),$$

where

$$B(U,z) = -9U^3 - 9U^2z - 9U^2 - 3Uz^2 - 18Uz + 3U - 3z^3 + 3z^2 - 3z + 3.$$

Here

$$B_z = -3(3z^2 + 2Uz + 6U - 2z + 3U^2 + 1) < 0,$$

because the quadratic in parentheses has discriminant $-32U^2 - 80U - 8 < 0$. Hence its minimum on $z^2 < \Omega$ is at $z = \sqrt{\Omega}$, where $B(U,\sqrt{\Omega}) = 6(1 - 3U - \sqrt{\Omega}) > 0$ because $(1-3U)^2 - \Omega = 12U^2$. Thus $E_u + E_v < 3 - 2a$. Also $J_u + J_v < 1 < 3 - 2a$, and the coordinate sum is affine on the line segment, so the same strict bound holds at the first root. Now

$$\frac{\partial F}{\partial t}(P,t) = 1 + 2t - P_u - P_v.$$

Since $F(L_+(\mu_+), 1-a) = 0$, the coordinate-sum bound makes its t-derivative strictly positive for $t \geq 1 - a$. Together with the no-opposite-line result, this proves (99). Reflection gives both the bound on L_- and the complete $Z_3(y)$ statement. $\square$

Define the local and global witnesses by

$$Q_-^{\text{loc}} = L_-(\lambda_-), \qquad Q_0^{\text{loc}} = J, \qquad Q_+^{\text{loc}} = L_+(\mu_+), \tag{100}$$

and by the displayed conversion from (u,v) to the plane.

Lemma E.5 (Direct asymmetric forcing)**.** *Assume that $U_C, U_0, \ldots, U_5$ cover H. Then*

$$Q_-, Q_0, Q_+ \in \operatorname{int}(H) \setminus \bigcup_{i=0}^{5} U_i,$$

and consequently all three witnesses belong to U_C.

Proof. The root order and positivity of the line coordinates give $0 < u, v < 1$ for all three points. Each belongs to $\mathcal{R}_{AB}(b,a)$ and hence lies in a closed unit triangle containing V_4, so $\mathsf{q}(u,v) \le 1$; hence $|u-v| < 1$ and all three points lie in $\operatorname{int}(H)$.

The closure of U_4 is one of the triangles represented by $\mathcal{R}_{AB}(b,a)$, since it contains V_4, X_3, X_4. All three witnesses lie on its exact non-axis frontier. If one belonged to the open triangle U_4, a sufficiently small plane ball about it would lie both in U_4 and in the interior of the corner cone, making the witness an interior point of $\mathcal{R}_{AB}(b,a)$. This contradiction excludes U_4.

For the triangles U_0, U_1, U_2, their contained vertices have local coordinates

i	0	1	2
V_i	$(2,1)$	$(2,2)$	$(1,2)$.

For $P = (u,v)$,

$$\|P-(2,1)\|^2 - 1 = \mathsf{q}(u,v) + 2 - 3u, \quad \|P-(1,2)\|^2 - 1 = \mathsf{q}(u,v) + 2 - 3v.$$

The inequalities $J_u, J_v < 1/2$ and the line order give distance greater than one from V_0 for Q_-, Q_0 and from V_2 for Q_0, Q_+. For the remaining pair, the first-root circle and the sum bound in Lemma E.4 give

$$\|Q_+ - V_0\|^2 - 1 = a(3 - a - (Q_+)_u - (Q_+)_v) > a^2,$$

and, by reflection, $\|Q_- - V_2\|^2 - 1 > b^2$. All three points are farther than one from $V_1 = (2,2)$, because writing $u = 1-\xi$, $v = 1-\eta$ gives

$$\|P - V_1\|^2 = 1 + \xi + \eta + \xi^2 + \eta^2 - \xi\eta > 1.$$

Since two points in one open unit equilateral triangle have distance strictly less than one, these inequalities exclude U_0, U_1, U_2.

The actual handoff $X_5 = Z_5(\xi_5)$ belongs to U_5, and $1 - a < \xi_5 < b$ by (E.2). The three signs in (99) therefore put every witness at distance at least one from X_5, excluding U_5. Similarly, if $y = 1 - \xi_2$, then $1 - b < y < a$, $X_2 = Z_3(y) \in U_3$, and the reflected signs exclude U_3. Thus no V triangle contains any witness. The assumed cover now forces all three interior witnesses into U_C. ☐

The nine directly forced points are $D_0, \dots, D_5, Q_-, Q_0, Q_+$. Under a putative cover, Lemmas E.3 and E.5 therefore force the compact set

$$K_{\mathrm{wit}}(a,b) = \mathcal{D}_\eta \cup \{Q_-, Q_0, Q_+\} \subset U_C. \tag{101}$$

E.4. **Tangent reduction and exact mixed overlaps.** For the nontrivial regime $c_* > 2/3$, we first remove the square radicals in the first-root witnesses. Normalize $\bar{\alpha} = \alpha/h$, and similarly for the other three coefficients, and put

$$\xi = \lambda/h, \quad \zeta = \mu/h, \quad \xi_* = \zeta_* = \frac{8\rho}{3(D+3)}.$$

Let g_-, g_+ be the rescaled forms of (97), and take one Newton step at the junction:

$$\widehat{\xi}_- = \xi_* - \frac{g_-(\xi_*)}{g_-'(\xi_*)}, \qquad \widehat{\zeta}_+ = \zeta_* - \frac{g_+(\zeta_*)}{g_+'(\zeta_*)}.$$

Define, in local coordinates,

$$\begin{aligned} A &= \left(b - \frac{3}{4}\bar{\beta}\widehat{\xi}_-, \frac{3}{4}\bar{\alpha}\widehat{\xi}_-\right), \\ B &= J, \\ C &= \left(\frac{3}{4}\bar{\delta}\widehat{\zeta}_+, a - \frac{3}{4}\bar{\gamma}\widehat{\zeta}_+\right), \end{aligned}$$

and convert them to global vectors.

Lemma E.6 (Newton inner reduction)**.** *The points above have coordinates rational in a, b, D and satisfy*

$$A \in (Q_0, Q_-), \qquad C \in (Q_0, Q_+).$$

Consequently

$$\widehat{K} = \operatorname{conv}(\mathcal{D}_\eta \cup \{A, B, C\}),$$

obeys $\widehat{K} \subseteq \operatorname{conv}(K_{\mathrm{wit}})$.

Proof. Each quadratic g_-, g_+ is strictly convex, positive, and decreasing at the junction. Its tangent zero therefore lies strictly between the junction and its first root. This gives the two open-segment inclusions. All displayed coefficients and Newton quotients are rational in a, b, D, and convexity gives the asserted containment. □

Put $W = C + RA$. For a disk $\mathbb{D}(X, r)$ let

$$I(X, r) = \{n : \|n\| = 1,\ \langle X, n\rangle + r \geq h\}.$$

E.5. **Ray order and four overlaps.**

Proposition E.7 (Technical four-overlap verification)**.** *Assume* (91) *and $c_* > 2/3$. Then*

$$\|A\|, \|C\| > \eta,$$

the rays A, B, C, W, RA occur in that cyclic order, and the four consecutive cap pairs

$$I(A, 2\eta) \leftrightarrow I(B, 2\eta) \leftrightarrow I(C, 2\eta) \leftrightarrow I(W, \eta) \leftrightarrow I(RA, 2\eta)$$

overlap across their intervening short sectors.

Proof. Put $\tau = h - 2\eta = h(2c_* - 1)$. The rays of

$$A,\ B,\ C,\ W,\ RA \tag{102}$$

occur in this cyclic order from A to RA. Indeed the two line distances give

$$\frac{\operatorname{cross}(A, B)}{\|B - A\|} = \alpha(1 - b) + \beta > 0, \qquad \frac{\operatorname{cross}(B, C)}{\|C - B\|} = \gamma + \delta(1 - a) > 0.$$

If $A = (-X, -Y)$ and $C = (-P, -Q)$ in the centered cone basis, then $0 < X, Y, P, Q < 1$, $X, Q > 1/2$, and

$$\frac{\operatorname{cross}(C, RA)}{h} = PX + (Q - P)Y > 0.$$

The last sign is immediate unless $Q < P$ and $X < Y$, in which case it is larger than $Q - P/2 > 0$. The ray of $W = C + RA$ lies between those of C and RA. The same coordinates give

$$\|A\|, \|C\| > h/2 > \eta. \tag{103}$$

For the two equal-radius overlaps it is enough to prove

$$\frac{\operatorname{cross}(A,B)}{\|B-A\|} \ge \tau, \qquad \frac{\operatorname{cross}(B,C)}{\|C-B\|} \ge \tau. \tag{104}$$

We now give the exact analytic verification. By reflection take $a \le b$ and set $m = 1-b$, $M = 1-a$, $U = a+b-1$, $s = m+M$, and $w = M-m$. The smaller of the two normalized left sides in (104) is

$$d = \frac{\mathcal{A} + U(2m-1) + D(\mathcal{A}+U)}{2\rho}, \qquad \mathcal{A} = 1 - m + m^2. \tag{105}$$

On the C_L sheet, write $F_L(z) = z^4 - z^2 + mz - m^2$. This defining polynomial evaluated at $r = 1 - m/2 + m^2/2$ is

$$F_L(r) = \frac{m^2(m-1)}{16}(m^5 - 3m^4 + 11m^3 - 17m^2 + 28m - 12) > 0.$$

Indeed the polynomial in parentheses is increasing on $(0, 1/2)$ and has value $-33/32$ at $1/2$; its derivative is $28 - 34m + m^2(5m^2 - 12m + 33) > 0$. Since $F_L(h) = -(m-h/2)^2 \le 0$ and the selected root is unique, $2c_* - 1 \le \mathcal{A}$. Put

$$X_0 = \mathcal{A} + U, \qquad Y_0 = 2\rho\mathcal{A} - \mathcal{A} - U(2m-1).$$

Then

$$d - \mathcal{A} = \frac{D(\mathcal{A}+U) - Y_0}{2\rho},$$

and exact expansion gives

$$\begin{aligned} Y_0 &= 2(m^2 - m + 1)U^2 + (2m^3 - 2m + 3)U \\ &\quad + (m^2 - m + 1)(2m^2 - 2m + 1) > 0. \end{aligned}$$

Furthermore

$$D^2 X_0^2 - Y_0^2 = 4m\rho\Phi(U, m),$$

where

$$\begin{aligned} \Phi(U,m) &= (1-m)(m^2 - m + 2)U^2 \\ &\quad + (2 - m^2)(m^2 - m + 1)U \\ &\quad - m(1-m)^2(m^2 - m + 1). \end{aligned}$$

In these variables

$$\chi = U^4 - 4U^3 + 5U^2 - (m+2)U + m(1-m).$$

Since $U^2(U^2 - 4U + 5) > 0$, the selector $\chi \le 0$ implies $U > m(1-m)/(m+2)$; Φ is increasing in U and at that lower endpoint is

$$\frac{m^2(1-m)(m^4 - 2m^3 + 6m^2 - 6m + 4)}{(m+2)^2} > 0.$$

The last quartic equals $(m^2 - m)^2 + 5m^2 - 6m + 4 > 0$. Thus $d > \mathcal{A} \ge 2c_* - 1$.

On the other radial sheet let $E = \sqrt{4s^2 - 3}$. Then $0 \le w < sE$ and $c_* = (s+w)/(1+E)$. At fixed U, differentiation of (105) can be checked without an implicit estimate. Namely,

$$d = \frac{1+D}{2} - UB(D, w), \qquad B(D, w) = \frac{(1+U)(3+D) + w(1-D)}{D^2 + 3}.$$

Here $D^2 = w^2 + 6U + 3U^2$, so $0 \le D' = w/D \le 1$, and

$$B_w = \frac{1-D}{D^2+3} \ge 0.$$

Writing $N = (1+U)(3+D) + w(1-D)$, direct differentiation gives

$$B_D = \frac{(1+U-w)(D^2+3) - 2DN}{(D^2+3)^2} > -\frac{34}{27}.$$

Indeed the first term in the numerator is positive, while $1 + U - w = 2a > 0$, $N < (7/6)4 + 1 = 17/3$, $D < 1$, and $D^2 + 3 \ge 3$. If $B_D < 0$, then $B_D D' \ge B_D$; otherwise $B_D D' \ge 0$. Hence $B' = B_w + B_D D' > -34/27$, and $U < 1/6$ yields

$$d' < \frac{1}{2} + \frac{1}{6}\frac{34}{27} = \frac{115}{162} < 1,$$

whereas $(2c_* - 1)' = 2/(1+E) \ge 1$. Hence their difference decreases with w. At the boundary $w = sE$, where $\chi = 0$ and $c_* = s$, the preceding sheet applies: this is an admissible point because $h < s < 1$ and $D^2 = 4s^4 - 12s + 9 < 1$. Indeed $s^4 - 3s + 2 = (s-1)(s^3 + s^2 + s - 2) < 0$ on this interval; the second factor is increasing and already equals $(7h-5)/4 > 0$ at $s = h$. The preceding sheet gives $d \ge 2s - 1$. Therefore (104) holds on both sheets, and reflection supplies the other line.

It remains to check the two mixed overlaps. The exact rational envelopes, Gram reductions, and global positive-basis identities are recorded once in Appendix F. Theorem F.3 gives precisely the overlaps for $\mathbb{D}(C, 2\eta)$ with $\mathbb{D}(W, \eta)$ and for $\mathbb{D}(W, \eta)$ with $\mathbb{D}(RA, 2\eta)$.

Thus the four consecutive pairs of caps centered on the rays (102) overlap. $\square$

E.6. **Newton reduction in the body notation.** The containment established by theorem E.6 is recorded under the public routing label

(106)
$$\widehat{K} = \operatorname{conv}\big(\mathbb{B}(O,\eta) \cup \{A, B, C\}\big) \subseteq \operatorname{conv}(K_{\mathrm{wit}}(a,b)), \qquad \Lambda(K_{\mathrm{wit}}(a,b)) \ge \Lambda(\widehat{K}).$$

E.7. **Support arcs and the enclosure conclusion.** Let R denote counterclockwise rotation through $2\pi/3$. Retain the points A, B, C, the vector $W = C + \mathsf{R}A$, the forced disk $\mathcal{D}_\eta$, and the exact witness set K_{wit} constructed above. For uniform notation put

$$P_- = A, \qquad P_0 = B, \qquad P_+ = C, \qquad P_* = W, \qquad r_* = \eta,$$

and

$$\widehat{K} = \operatorname{conv}\big(\mathcal{D}_\eta \cup \{A, B, C\}\big).$$

For $X \in \mathbb{R}^2$ and $r \ge 0$, write

$$\mathbb{B}(X, r) = \{Y : \|Y - X\| \le r\},$$

and define the level-h, $h = \sqrt{3}/2$, support arc

$$\operatorname{Cap}(X, r) = \{n \in \mathbb{S}^1 : \langle X, n\rangle + r \ge h\}.$$

Lemma E.8 (Cyclic support-arc covering)**.** *Let K be compact and put*

$$\Sigma_K = K + \mathsf{R}K + \mathsf{R}^2 K.$$

Suppose Σ_K contains the full R-orbits of

$$\mathbb{B}(P_-, 2r_*), \quad \mathbb{B}(P_0, 2r_*), \quad \mathbb{B}(P_+, 2r_*), \quad \mathbb{B}(P_*, r_*).$$

Assume that the rays

$$P_-, P_0, P_+, P_*, \mathsf{R}P_-$$

occur in this cyclic order, that every corresponding support arc is connected with half-width less than $\pi/2$, and that each consecutive pair of arcs intersects across the intervening short angular sector. Then

$$\Lambda(K) \geq 1.$$

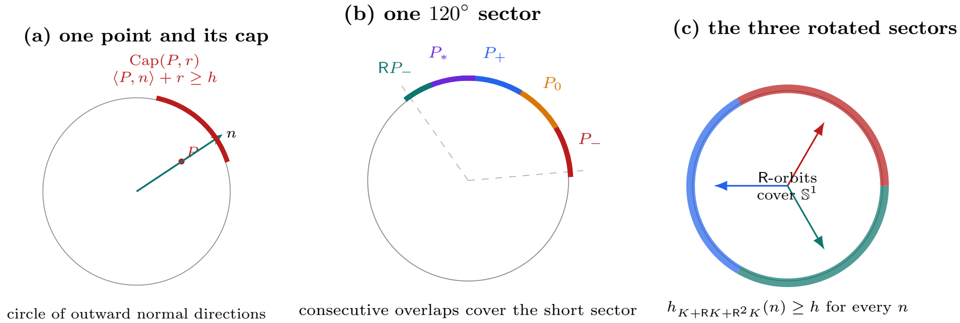


FIGURE 24. The support-cap conclusion for the zero-gap witness. A point and a disk radius determine an arc of normal directions in which their support is at least h. The four consecutive overlap checks cover the sector from the ray of P_- to that of $\mathsf{R}P_-$, and the three 120° rotations cover the full normal circle. The arc widths are schematic; the cyclic order and overlap pattern are the exact hypotheses of the support-arc lemma.

Proof. The four consecutive intersections and the cyclic order make the five arcs cover the sector from the ray of P_- to the ray of $\mathsf{R}P_-$. Their R-orbits cover $\mathbb{S}^1$. Hence for every unit vector n one of the displayed disks has support at least h in direction n, and therefore

$$h_{\Sigma_K}(n) \geq h.$$

Since

$$h_{\Sigma_K}(n) = \sum_{j=0}^{2} h_K(\mathsf{R}^j n),$$

Proposition A.4 gives $\Lambda(K) \geq 1$. $\square$

Lemma E.9 (Solved witness-enclosure optimization)**.** *For every parameter pair in equation (16),*

$$\Lambda(K_{\mathrm{wit}}(a, b)) \geq 1.$$

Proof. The centered disk alone gives

$$\Lambda(K_{\mathrm{wit}}(a, b)) \geq 3(1 - c_*),$$

which proves the claim when $c_* \leq 2/3$. Assume $c_* > 2/3$. By equation (106), it is enough to enclose $\widehat{K}$. The Minkowski sum

$$\widehat{K} + \mathsf{R}\widehat{K} + \mathsf{R}^2\widehat{K}$$

contains the full rotation orbits of the four disks in theorem E.8. The ray order, radius bounds, and equal-radius intersections are theorem E.7; the two mixed intersections are theorem F.3. Therefore theorem E.8 gives $\Lambda(\widehat{K}) \geq 1$, and monotonicity gives the result. □

Appendix F. Exact Unequal-Radius Support-Arc Intersections

The two mixed cap overlaps in the zero-gap nine-point theorem are certified by exact symbolic arithmetic. This section gives the mathematical reduction and verification algorithms; the repository provenance manifests identify the accompanying code and sparse integer data. No floating-point arithmetic, interval arithmetic, adaptive subdivision, branch-and-bound, or unrecorded numerical tolerance is used.

F.1. **Certificate manifest.** The machine-readable provenance manifest identifies the six sparse-data shards and both exact verifiers by full Git blob and records their roles. They decode to one canonical sparse-polynomial transcript with SHA–256 digest

`dc46aaf263655d5159ecd3a81db72ee82477951d06172f4743b248df37209485`

The derivation verifier reconstructs the transcript from the witness formulas and compares every coefficient with the authenticated object. The positivity verifier derives the Bernstein coefficients from that object and checks their signs exactly.

F.2. **Rational radial upper envelopes.** Retain the notation of Section E. Thus

$$p = 1 - b, \qquad q = 1 - a, \qquad s = p + q, \qquad m = \min\{p, q\}, \qquad M = \max\{p, q\},$$

$$\chi = s^4 - s^2 + mM, \qquad h = \frac{\sqrt{3}}{2}, \qquad c_* = c_{\max}(p, q).$$

The strict witness domain gives

$$mM > (1 - s)(2 - s), \qquad m > 1 - s, \qquad s > \frac{2}{3}. \tag{107}$$

Put

$$F_m(x) = x^4 - x^2 + mx - m^2, \qquad \kappa = F'_m(s) = 4s^3 - 2s + m.$$

Then

$$\kappa > 4s^3 - 3s + 1 = (s + 1)(2s - 1)^2 > 0.$$

Define

$$\bar{c}_L = s - \frac{\chi}{\kappa} \quad (\chi \leq 0), \qquad \bar{c}_T = s - \frac{\chi}{1 - m} \quad (\chi \geq 0). \tag{108}$$

Lemma F.1 (Branchwise radial upper bounds)**.** *On the corresponding radial cells,*

$$c_* \leq \bar{c}_L < 1, \qquad c_* \leq \bar{c}_T < 1.$$

At $\chi = 0$ both upper bounds equal $s = c_*$.

Proof. On the L cell, $F_m(s) = \chi \leq 0$ and $F_m(c_*) = 0$. For $x \geq s > 2/3$, $F''_m(x) > 0$, so convexity gives

$$0 \geq F_m(s) + F'_m(s)(c_* - s),$$

and hence $c_* \leq \bar{c}_L$. If $U = 1 - s$, then

$$\chi + U\kappa > U\{m + 1 - 4U + 8U^2 - 3U^3\} > 0$$

by (107); therefore $\bar{c}_L < 1$.

On the T cell put

$$E = \sqrt{4s^2 - 3}, \qquad m_- = \frac{s(1-E)}{2}, \qquad m_+ = \frac{s(1+E)}{2}.$$

Then

$$\chi = (m - m_-)(m_+ - m), \qquad s - c_* = \frac{2(m - m_-)}{1+E}.$$

Consequently

$$\frac{\chi}{s - c_*} = \frac{1+E}{2}(m_+ - m) \le 1 - m,$$

which gives $c_* \le \bar{c}_T$. The strict inequality $\chi > 0$ gives $\bar{c}_T < s < 1$. □

Reflect so that $z = a - b \ge 0$ and put $U = a + b - 1$. Then

$$s = 1 - U, \qquad m = \frac{1 - U - z}{2}, \qquad D^2 = z^2 + 6U + 3U^2 =: K(U, z),$$

and

$$\begin{aligned} \Xi(U,z) = 4\chi = 1 - 10U + 21U^2 - 16U^3 + 4U^4 - z^2, \\ G(U,z) = 2\kappa = 5 - 21U + 24U^2 - 8U^3 - z. \end{aligned}$$

Thus

$$\bar{c}_L = 1 - U - \frac{\Xi(U,z)}{2G(U,z)}, \qquad \bar{c}_T = 1 - U - \frac{\Xi(U,z)}{2(1+U+z)}. \tag{109}$$

F.3. **Gram reduction and the geometric implication.** On the active cell let $\bar{c}$ denote the corresponding expression in (109), and put

$$\bar{\eta} = h(1 - \bar{c}), \qquad e = 1 - \bar{c}, \qquad t = 2\bar{c} - 1.$$

Lemma F.1 gives $0 < \bar{\eta} \le \eta = h(1 - c_*)$. It therefore suffices to prove the mixed intersections for the smaller disks with radii $2\bar{\eta}, \bar{\eta}, 2\bar{\eta}$.

Represent a global vector as (x, hy) and put $C' = \mathsf{R}^{-1}C$. Define

$$N_A = \|A\|^2, \qquad N_C = \|C\|^2, \qquad \nu = \langle A, C' \rangle,$$

$$\Delta_0 = x_A y_{C'} - y_A x_{C'} > 0.$$

Thus the Euclidean cross product is $h\Delta_0$, and the Gram identity is

$$h^2 \Delta_0^2 = N_A N_C - \nu^2. \tag{110}$$

Define

$$Q_A = \Delta_0^2 - \|tA - eC'\|^2, \qquad Q_C = \Delta_0^2 - \|tC' - eA\|^2. \tag{111}$$

Lemma F.2 (Residual signs imply the mixed cap intersections)**.** *If $Q_A, Q_C \ge 0$, then*

$$I(C, 2\bar{\eta}) \cap I(C + \mathsf{R}A, \bar{\eta}) \ne \varnothing,$$

and

$$I(C + \mathsf{R}A, \bar{\eta}) \cap I(\mathsf{R}A, 2\bar{\eta}) \ne \varnothing.$$

Consequently the same intersections hold with η in place of $\bar{\eta}$.

Proof. Rotate the first pair by R^{-1}. Its centers become C' and $A+C'$, so their difference is A. An outer common tangent normal n_A is chosen with

$$\langle A, n_A\rangle = \bar\eta.$$

Since $\|A\| > \bar\eta$ and $\Delta_0 > 0$, choose the tangent side for which

$$\langle C', n_A\rangle = \frac{\bar\eta\nu + \sqrt{N_A - \bar\eta^2}\, h\Delta_0}{N_A}.$$

The common support of the two disks is therefore

$$2\bar\eta + \frac{\bar\eta\nu + \sqrt{N_A - \bar\eta^2}\, h\Delta_0}{N_A}.$$

Let $\bar\tau = h - 2\bar\eta = ht$. Expanding (111) and using (110) gives

$$\bar P_A := (N_A - \bar\eta^2)h^2\Delta_0^2 - (\bar\tau N_A - \bar\eta\nu)^2 = h^2 N_A Q_A. \tag{112}$$

Thus $Q_A \ge 0$ implies

$$\sqrt{N_A - \bar\eta^2}\, h\Delta_0 \ge \bar\tau N_A - \bar\eta\nu,$$

and the common support is at least $2\bar\eta + \bar\tau = h$. Hence the two level-h caps intersect. The same calculation with A and C' exchanged gives

$$\bar P_C := (N_C - \bar\eta^2)h^2\Delta_0^2 - (\bar\tau N_C - \bar\eta\nu)^2 = h^2 N_C Q_C, \tag{113}$$

and proves the second intersection. Enlarging the disk radii from $\bar\eta$ to η only enlarges the caps. $\square$

F.4. **Eight exact integer-polynomial signs.** The Newton witnesses A, C are rational functions of U, z, D. Substituting (109) into (111), clearing denominators, and reducing by

$$D^2 - K(U,z) = 0$$

gives, for $B \in \{L, T\}$ and $X \in \{A, C\}$,

$$\operatorname{num}(Q_{B,X}) = 32(K+3)^2\{A_{B,X}(U,z) + DB_{B,X}(U,z)\}, \tag{114}$$

where all eight core polynomials lie in $\mathbb{Z}[U,z]$. Every omitted denominator is strictly positive on the strict domain: its factors are positive constants, $(D+3)^4$, $G(U,z)^2$ or $(1+U+z)^2$, and squares of the two Newton-derivative numerators. Here $G = 2\kappa > 0$, and the fixed-line sign theorem proves both Newton derivatives strictly negative.

The derivation verifier constructs A, C, both rational envelopes, and all four residuals directly in the exact field $\mathbb{Q}(U,z,D)$. It performs the reduction modulo $D^2 - K$, removes the common positive factor, primitive-normalizes the eight integer polynomials, and compares their sparse term lists coefficient by coefficient with the authenticated transcript. Thus the data are checked against the geometric formulas rather than trusted as an independent precomputation.

F.5. **Three global positive-basis certificates.** Set

$$\mathcal{G}(U) = 1 - 6U - 3U^2, \qquad \mathcal{P}(U) = 1 - 10U + 21U^2 - 16U^3 + 4U^4,$$

$$U_\triangle = 1 - \frac{\sqrt3}{2}, \qquad U_{\max} = \frac{2}{\sqrt3} - 1.$$

The complete reflected domain is the union of the three closed cells

cell	U-range	z^2-range
T	$0 \le U \le U_\triangle$	$0 \le z^2 \le \mathcal{P}(U)$
L_0	$0 \le U \le U_\triangle$	$\mathcal{P}(U) \le z^2 \le \mathcal{G}(U)$
L_1	$U_\triangle \le U \le U_{\max}$	$0 \le z^2 \le \mathcal{G}(U)$

For the T cell use

$$U = \frac{(x-1)(3-x)}{4x}, \qquad z = y\frac{9-x^4}{8x^2}, \qquad 1 \le x \le \sqrt{3}, \quad 0 \le y \le 1,$$

and then $x = 1 + (\sqrt{3}-1)r$. This maps $[0,1]^2$ onto the complete cell. After multiplication by positive powers of 2 and x, the four transformed core polynomials have global tensor Bernstein expansions with data

	bidegree	coefficient count	zero count
$A_{T,A}$	$(88,22)$	2047	2
$B_{T,A}$	$(80,20)$	1701	0
$A_{T,C}$	$(88,22)$	2047	2
$B_{T,C}$	$(80,20)$	1701	0.

Every nonzero coefficient is strictly positive.

For an L-cell polynomial write

$$F(U,z) = E_F(U,w) + zO_F(U,w), \qquad w = z^2,$$

and

$$R_F(U,w) = E_F(U,w)^2 - wO_F(U,w)^2.$$

The implications $E_F \ge 0$ and $R_F \ge 0$ give $E_F \ge \sqrt{w}\,|O_F|$, hence $F \ge 0$ for $z \ge 0$. Use the exact charts

$$U = U_\triangle r, \qquad w = \mathcal{P}(U) + s\{\mathcal{G}(U) - \mathcal{P}(U)\}$$

for L_0, and

$$U = U_\triangle + (U_{\max} - U_\triangle)r, \qquad w = s\mathcal{G}(U)$$

for L_1. The positivity verifier forms one global Bernstein expansion for each of E_F, R_F on each square. The authenticated transcript records the bidegrees and coefficient counts of all sixteen expansions; every coefficient is strictly positive.

The verifier implements polynomial addition, multiplication, substitution, and power-to-Bernstein conversion over exact integers. Coefficients in $\mathbb{Q}(\sqrt{3})$ are represented as rational pairs and signed by exact rational comparisons. If $P(r,s) = \sum a_{ij} r^i s^j$ has bidegree (m,n), its exact tensor Bernstein coefficients are

$$b_{k\ell} = \sum_{i \le k} \sum_{j \le \ell} a_{ij} \frac{\binom{k}{i}}{\binom{m}{i}} \frac{\binom{\ell}{j}}{\binom{n}{j}}.$$

This is precisely the conversion implemented by the verifier.

Theorem F.3 (Exact mixed-overlap certificate)**.** *On the complete strict witness domain with $c_* > 2/3$, all eight polynomials in* (114) *are nonnegative. Hence $Q_A, Q_C \ge 0$, and the two mixed cap intersections required by Proposition E.7 hold.*

Proof. The T chart proves the four T-cell core signs by nonnegative Bernstein coefficients. On each L chart the positive Bernstein certificates for E_F and R_F prove the four L-cell signs. Since $D \ge 0$, (114) gives $Q_A, Q_C \ge 0$ for $z \ge 0$; reflection supplies $z \le 0$. Lemma F.2 then gives the two cap intersections. □

Remark F.4 (Reproduction). The authenticated data and the two verifiers constitute the exact electronic certificate used by the proof. Reproduce it from its package directory by executing

```
verify_mixed_overlap_core_derivation.py
verify_global_core_positivity.py
```

Both commands use exact arithmetic and must reproduce the transcript digest above and the asserted sign checks.